

1) Protecting Your Computing Devices

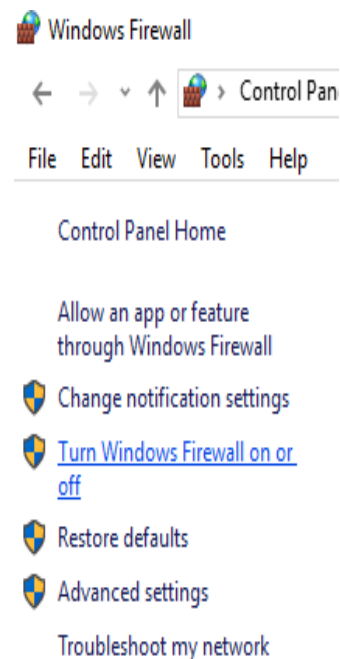
1. Turn on the Firewall

Windows Firewall is a Microsoft Windows application that filters information coming to your system from the Internet and blocking potentially harmful programs. It is necessary to turn on because Windows Defender Firewall helps prevent hackers and malicious software from gaining access to your PC through the internet or a network.

Steps to Turn ON Firewall:

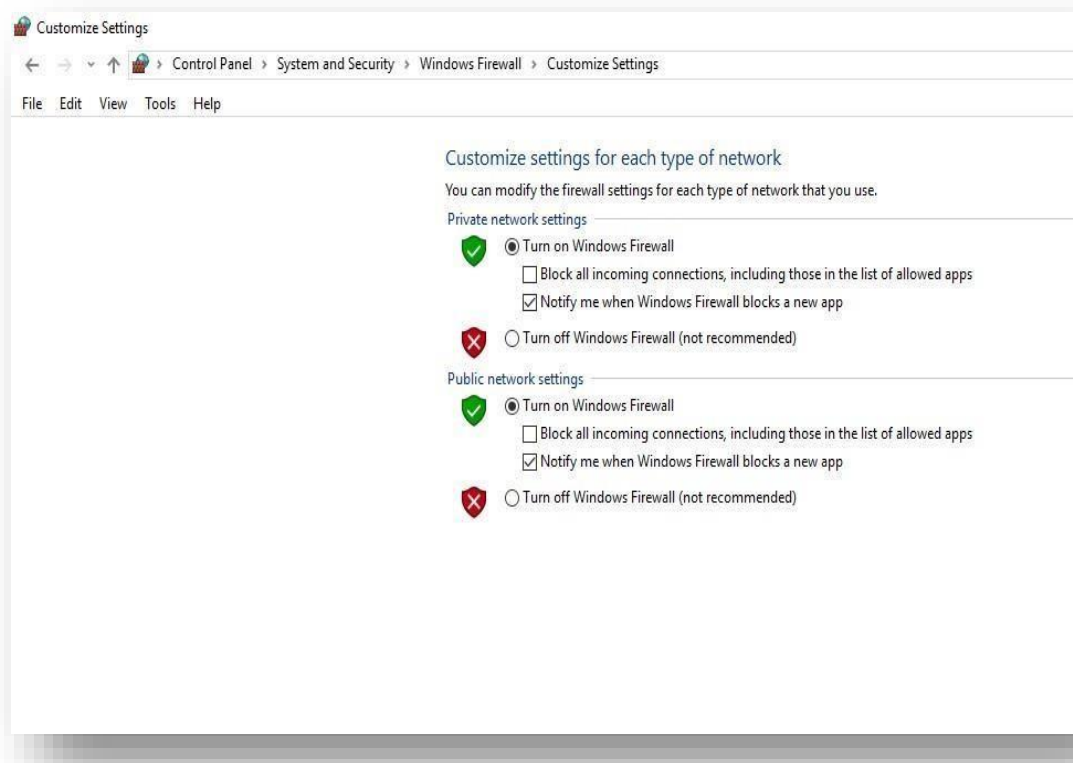
Step 1: Go to Start and open Control Panel.

Step 2: Select System and Security > Windows Defender Firewall.



Step 3: Choose Turn Windows Firewall on or off.

Step 4: Choose Turn Windows Firewall on or off. Select Turn on Windows Firewall for domain, private, and public network settings.



2. Install antivirus and antispyware

Antivirus: Antivirus software is a type of program designed and developed to protect Operating system from malware like viruses, computer worms, botnets, rootkits, keyloggers and etc.

Antispyware: Anti-spyware software is a type of program designed to prevent and detect unwanted spyware program installations and to remove those programs if installed. It is also used to manage browsers.

Steps to Install Antivirus:

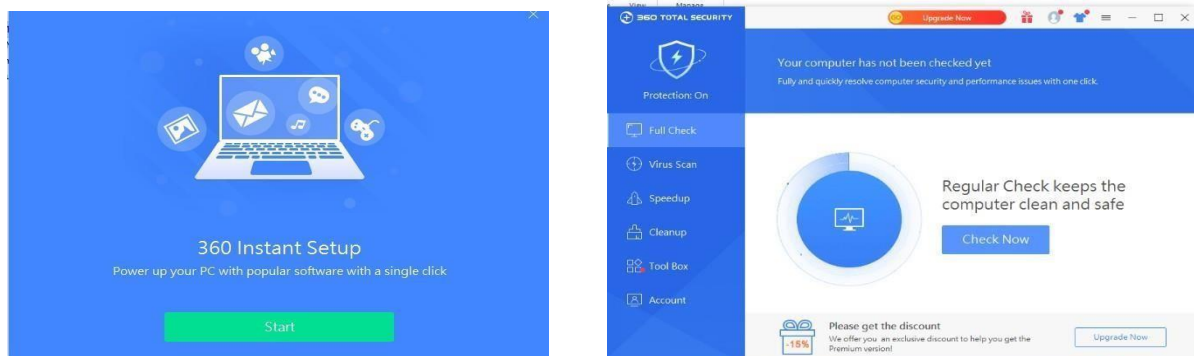
Step 1: Go to browser, search for total 360 security antivirus software or any other antivirus software.



Step 2: Click on the download.

Step 3: Open the Downloads folder and double click the downloaded file and install it.

Step 4: Click on start > go to Full Check and Check Now.



Steps to Install Antispyware (Spywareblaster):

Anti-spyware is a type of software that is designed to detect and remove unwanted spyware programs. Spyware is a type of malware that is installed on a computer without the user's knowledge in order to collect information about them. This can pose a security risk to the user, but more frequently spyware degrades system performance by taking up processing power, installing additional software, or redirecting users' browser activity.

Anti-spyware software detects spyware through rules-based methods or based on downloaded definition files that identify common spyware programs. Anti-spyware software can be used to find and remove spyware that has already been installed on the user's computer, or it can act much like an anti-virus program by providing real-time protection and preventing spyware from being downloaded in the first place.

Some antivirus software vendors include McAfee, AVG, Trend Micro, etc. Some anti-spyware software vendors include Microsoft, Webroot, McAfee, etc.

Step 1: Go to browser and search for antispyware blaster.



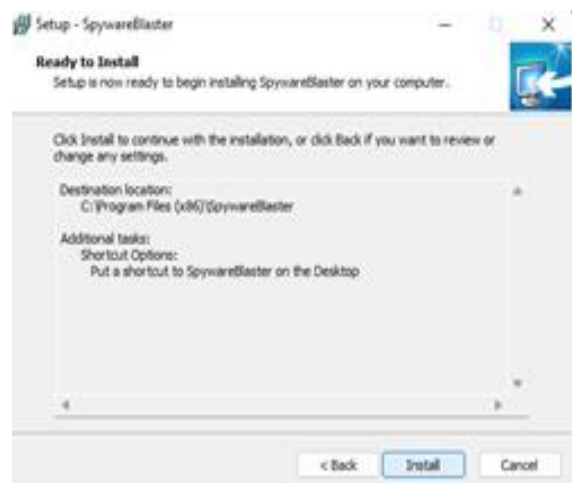
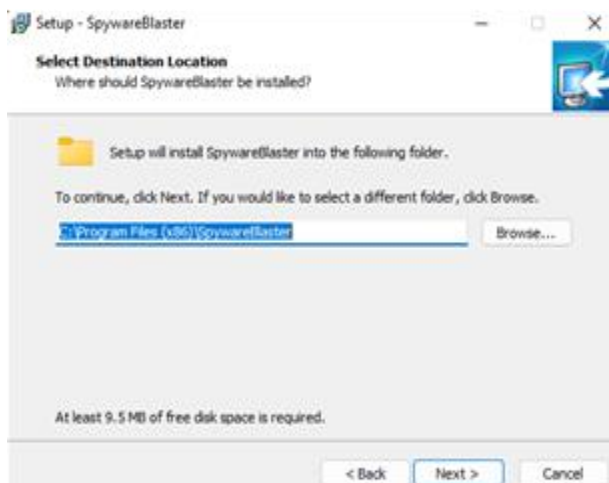
Step 2: Select downloads and click on spywareBlaster.



Step 3: Select Download SpywareBlaster 6.0

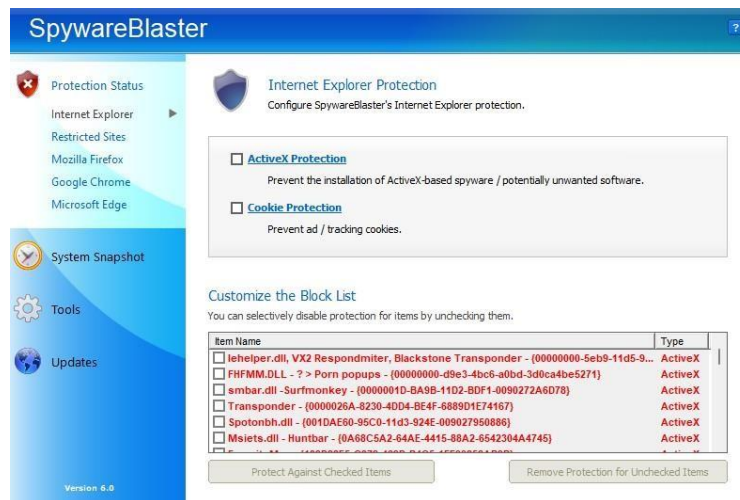
Step 4: Open the spywareblastersetup60 from the downloads and install it

Step 5: Double-click on setup file > Agree to agreement > Browse the location > click next > Install.



Step 6: Open the application > click next > select automatic updating and click next.

Step 7: Select the browser you to protect > select activate protection.



3. Manage your operating system and browser

GOOGLE CHROME

- **Setting the default browser** – Open Google chrome web browser→click on 3 dots→ select settings→select Default browser→ Make Google Chrome the default browser button.
- **Automatic download** – Open Google chrome web browser→click on 3 dots→ select settings→choose “privacy and security”→select “site settings” and choose “Additional permissions” and select the “Automatic downloads” and check “Sites can ask to automatically download multiple files”.
- **Handling Cookies** – Open Google chrome web browser→click on 3 dots→ select settings→choose “privacy and security”→select Cookies and other site data → check the “Block third-party cookies in Incognito”.
- **Do not save Passwords** – Open Google chrome web browser→click on 3 dots→ select settings→select “Auto fill” and select the “password” and uncheck the “Offer to save passwords”.

- **Microphone Access** – Open Google chrome web browser→click on 3 dots→ select settings→choose “privacy and security”→select “site settings” and choose “permission” and select the “Microphone” and check “Sites can ask to use your microphone”.
- **Camera Access** – Open Google chrome web browser→click on 3 dots→ select settings→choose “privacy and security”→select “site settings” and choose “permission” and select the “Camera” and check “Sites can ask to use camera”.

INTERNET EXPLORER

- **Block unwanted pop-ups** – Open Internet explorer→Go to Tools menu→ select Internet options→select Privacy tab→check the “turn on pop-up blocker” box.
- **Block unwanted plugins** - Open Internet explorer→Go to Tools menu→ select Internet options→select Advanced tab and scroll down to Multimedia. Uncheck “Play animations” and “Play sounds” in webpages if they are checked.
- **Set your browser to not set passwords** - Open Internet explorer→Go to Tools menu→ select Internet options→select Content tab and click the AutoComplete Settings button and uncheck the “user names and passwords on forms” box
- **Handling cookies** - Open Internet explorer→Go to Tools menu→ select Internet options→ select Privacy tab and click the “Advanced” button. Check the “Override” box and the “Accept” button for First-party cookies and “Prompt” button for Third-party cookies. The “Always allow...” button should not be checked. Click OK. When done, click the Apply button.

Browser security is an important part in keeping your information safe.

- Your browser is the window to the internet and also the first line of defence against malware threats. Some small tweaks to your browser security settings are all that you need to make your time online that much safer.

Browser features and their security vulnerabilities

- Browsers use many tools for various tasks, such as Java, Flash Player, ActiveX, etc. But these often come with security flaws, which cybercriminals exploit to get access to your PC. A quick rundown of these tools will help you figure out if you need them or not.

Deactivate ActiveX.

- A browser add-on that comes preinstalled on Internet Explorer or Microsoft Edge and only works with these browsers. ActiveX acts as a middle man between your PC and Java/Flash based interactions in certain sites.
- This creates security problems by giving malicious websites a window into your PC. What's more, ActiveX is rarely used nowadays, so be on your guard if a site asks you to install it and accept the installation only if you are 150% sure that site is trustworthy.

Try to disable JavaScript.

- JavaScript is a programming language used by websites to run various programs and features. Sites such as YouTube or Google Docs need it to function, but so do advertising, pop-up software and a whole host of other spammy elements from the internet.
- Cybercriminals use JavaScript in malicious ways in order to infect your device with malware and other harmful software. If you disable JavaScript altogether you will get a much quicker and simplified browser experience, with little to no ads, pop-ups, and greatly improved page load times and generally a cleaner Internet experience at the cost of specialized tools such as Google Docs or YouTube.
- This doesn't need to be as drastic as it sounds, since browsers do allow you to white list certain sites which can run JavaScript.

Delete Cookies.

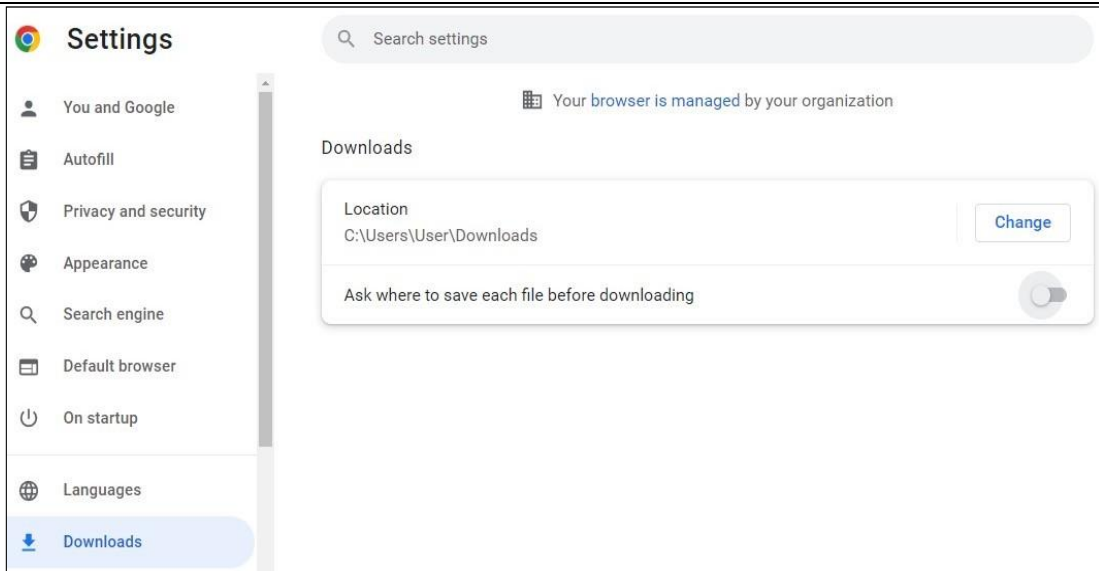
- These are small data files stored on your browser. Websites use cookies in order to remember your accounts and passwords, browsing history and to track user behaviour on their site. Because of the information they contain, cookies are prime targets for cybercriminals, especially the ones that contain emails, account names and passwords.
- When you disable and clear cookies you cut down on the personal data cybercriminals can obtain. One thing you will want to keep in mind is that there are two types of cookies: First party and third party cookies. First party cookies are placed by the site you visit, for instance you get a first party cookie by cnn.com while visiting cnn.com. Third party cookies are placed by other

sites, for example you get a cookie from amazon.com while visiting cnn.com.

- First party cookies are frequently used to remember your login information so you don't have to enter it every time you visit a site. But we can't stress this enough, don't allow your browser to save passwords!
- Third party cookies are almost always placed on your computer by advertisers or marketers interested in tracking your movement online, so nothing bad will happen if you block them. Browser extensions and add-ons add extra functionality to your browser such as ad blocking or search bars. However, these add-ons pose a security risk, since they can open up windows into your PC which can be exploited to inject malware.

Chrome hacks and tips for better security

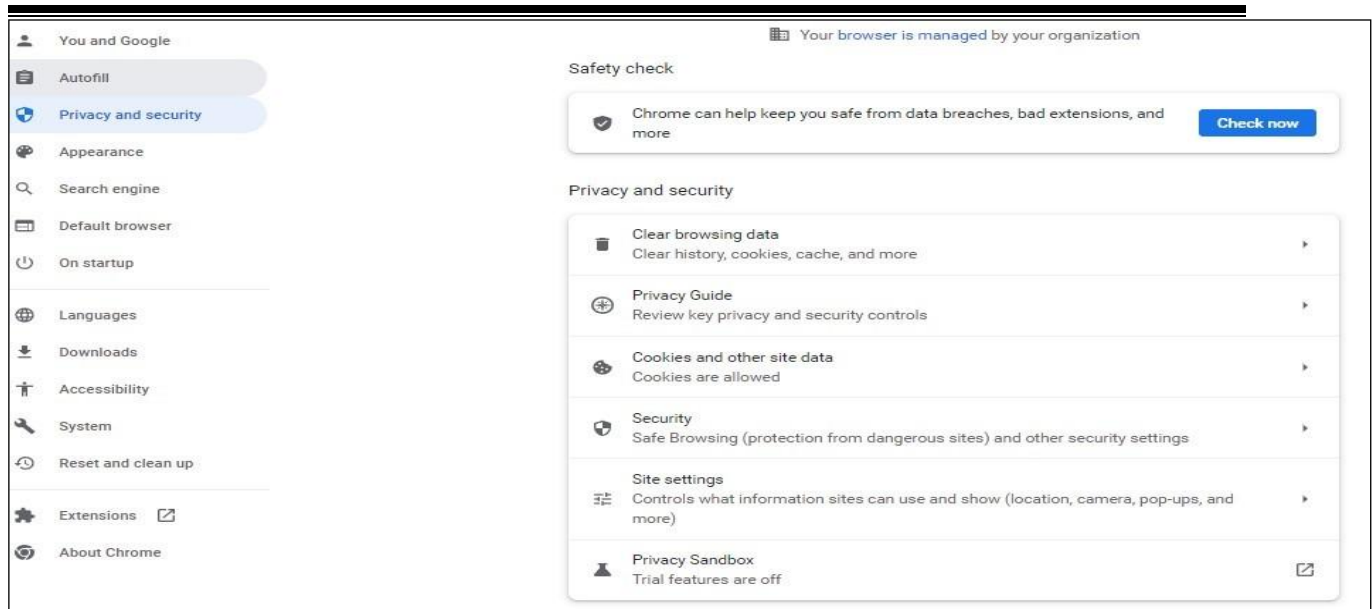
- If you use Google Chrome and want to improve your browser security settings, then go to browsersettings.
- At the Downloads section, press **“Ask where to save each file before downloading”**. This way, you won't have a web location try to automatically save dangerous content to your computer. At the same time, this gives you the option to place suspicious content in a safe location where you can analyze it afterwards.



Next, go to the Autofill tab.

- By enabling “Offer to save passwords” browser will ask to save your password before login and vice versa.
- By enabling “Auto Sign-in” browser will automatically sign in to websites using stored credentials.
- On clicking “Check passwords” will keep your passwords safe from data breaches and other security issues.
- We can “View and manage saved passwords of our Google Account”.

In privacy and security tab, we can observe for cookies, browsing data, security and privacy.



4. Set up password protection.

Steps to set up Password protection

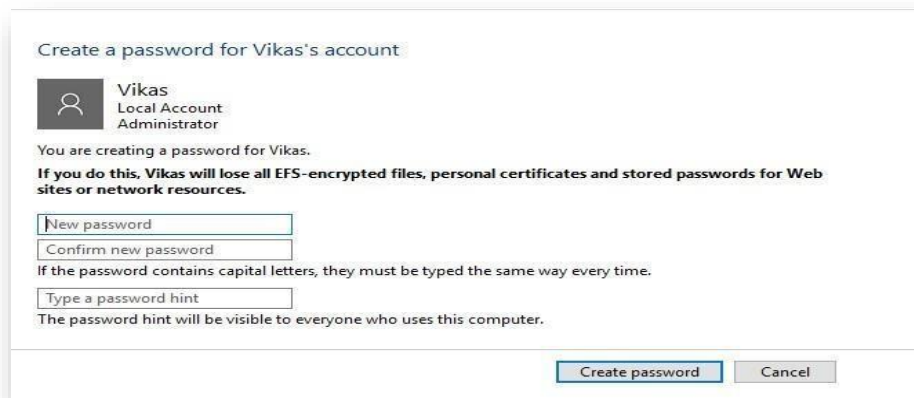
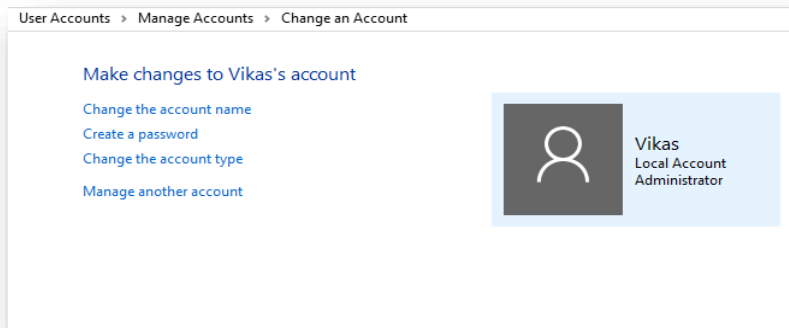
Step 1: Go to Start and open Control Panel.

Step 2: Select User Accounts > click Change Account Type.



Step 3: Double click on User account > click on create password.

Step 4: Enter new password and confirm the password > click create password.

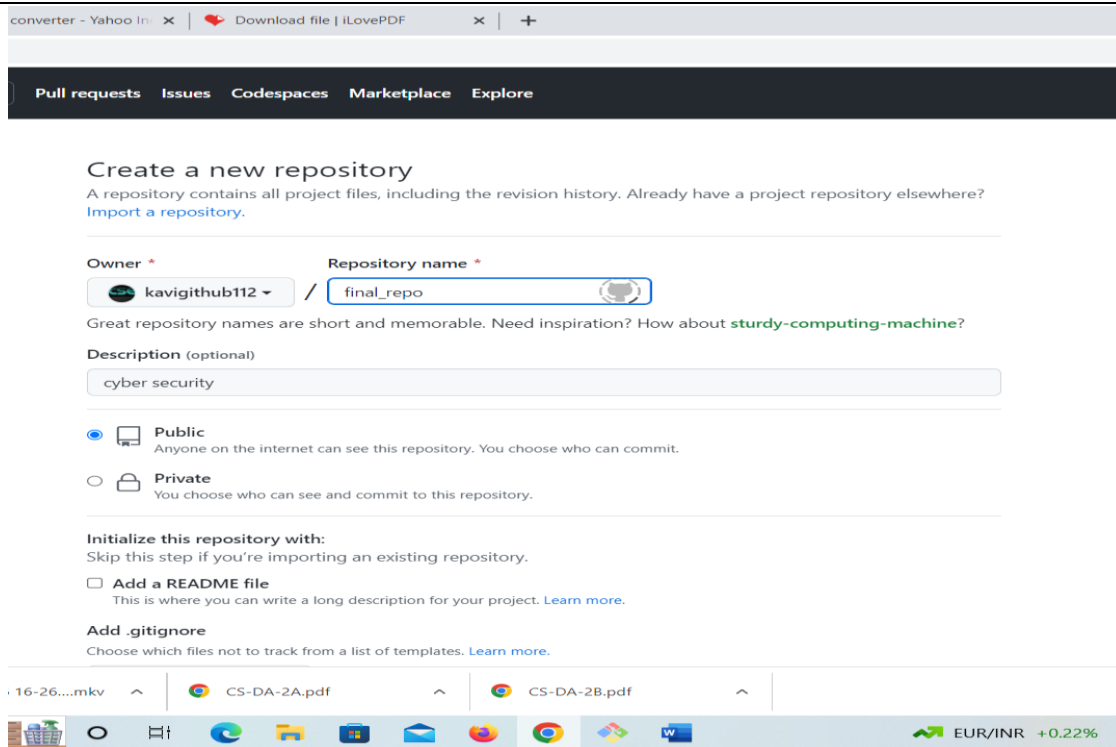


2) Install and setup Git. Perform the following operations.

- a) creating a repository**
- b) making and recording changes**
- c) staging and committing changes**
- d) viewing the history of all the changes and undoing changes**
- e) cloning a repository**

a) creating a repository

- Click on + icon, select New repository
- Enter Repository name
- Check add README file option
- Click Create repository option



converter - Yahoo In: x | Download file | iLovePDF x | +

Pull requests Issues Codespaces Marketplace Explore

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Owner * kavigithub112 / Repository name * final_repo

Great repository names are short and memorable. Need inspiration? How about [sturdy-computing-machine?](#)

Description (optional)
cyber security

☒ **Public**
Anyone on the internet can see this repository. You choose who can commit.

☐ **Private**
You choose who can see and commit to this repository.

Initialize this repository with:
Skip this step if you're importing an existing repository.

☒ **Add a README file**
This is where you can write a long description for your project. [Learn more.](#)

Add .gitignore
Choose which files not to track from a list of templates. [Learn more.](#)

16-26....mkv ^ | CS-DA-2A.pdf ^ | CS-DA-2B.pdf ^

16-26....mkv ^ | CS-DA-2A.pdf ^ | CS-DA-2B.pdf ^

EUR/INR +0.22%

→ Create a local directory using the following command in command line:

\$ mkdir test

\$ cd test

→ The next step is to initialize the directory:

\$ git init

The above command will create a new subdirectory named .git that holds all necessary repository files. The .git subdirectory can be understood as a Git repository skeleton.

MINGW64/c:/Users/kavit/desktop/test

```
kavit@DESKTOP-9HAQ5GN MINGW64 ~  
$ cd desktop  
  
kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop  
$ mkdir test  
  
kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop  
$ cs test  
bash: cs: command not found  
  
kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop  
$ cd test  
  
kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test  
$ git init  
Initialized empty Git repository in C:/Users/kavit/Desktop/test/.git/
```

→ Now Go to the folder where "test" is created and create a text document named "demo." Open "demo" and put any content, like "Hello Cyber security specialist." Save and close the file.

→ If we want to start version-controlling for existing files, we should track these files with git add command, followed by a commit. We can list all the untracked files by git status command.

→ Enter the Git bash interface and type in the following command to check the status:

\$ git status

To share these files on the version control system, we have to track it with git add command followed by a commit. To track the files, operate git add command as follows:

\$ git add demo.txt

To commit a file (it's like save file), perform the git commit command as follows:

\$ git commit -m "first commit"

```
kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      demo.txt

nothing added to commit but untracked files present (use "git add" to track)

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git add demo.txt

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git commit -m "first commit"
[master (root-commit) bebecda] first commit
1 file changed, 1 insertion(+)
create mode 100644 demo.txt
```

Pushing- From Local system to remote location (Git Hub)

Step 1: Before pushing local file to remote location or GitHub, we have to first add and commit file.

Step 2: Link the Git to a Github account using following command.

\$ git config --global user.username

Ex: \$ git config --global user.kavigithub112

Where username is name of user account on GitHub.

Step 3: Now Copy repository link of final_repo which was created on GitHub. Go back to Git bash and link the remote and local repository using the following command:

\$ git remote add origin <link> or

\$ git remote add origin https://github.com/kavigithub112/final_repo.git

Step 4: Push the local file onto the remote repository using the following command:

\$ git push origin master

Step 5: Move back to Github and click on "final_repo" and check if the local file "demo.txt" is pushed to this repository. We can find demo.txt file on branches option.

```
kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git config --global user.kavigithub112

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git remote add origin https://github.com/kavigithub112/final_repo.git

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git push origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 218 bytes | 109.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:   https://github.com/kavigithub112/final_repo/pull/new/master
remote:
To https://github.com/kavigithub112/final_repo.git
 * [new branch]      master -> master
```

Make some changes to demo text file and save again to repository.

- ➔ Open demo text file and make some changes and save it.
- ➔ Open git bash and type following commands:

\$ git status

\$ git add demo.txt

\$ git commit -m "second commit"

\$ git push origin master

```
kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   demo.txt

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git add demo.txt

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git commit -m "second commit"
[master fe14b4e] second commit
1 file changed, 2 insertions(+), 1 deletion(-)

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 263 bytes | 263.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/kavigithub112/final_repo.git
bebecda..fe14b4e  master -> master

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$
```

kavigithub112 / final_repo Public

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

👤 master had recent pushes less than a minute ago [Compare & pull request](#)

👤 master 2 branches 0 tags [Go to file](#) [Add file](#) [Code](#)

This branch is 2 commits ahead, 1 commit behind main. [Contribute](#)

 kavigithub112 second commit fe14b4e 23 minutes ago 🕒 2 commits

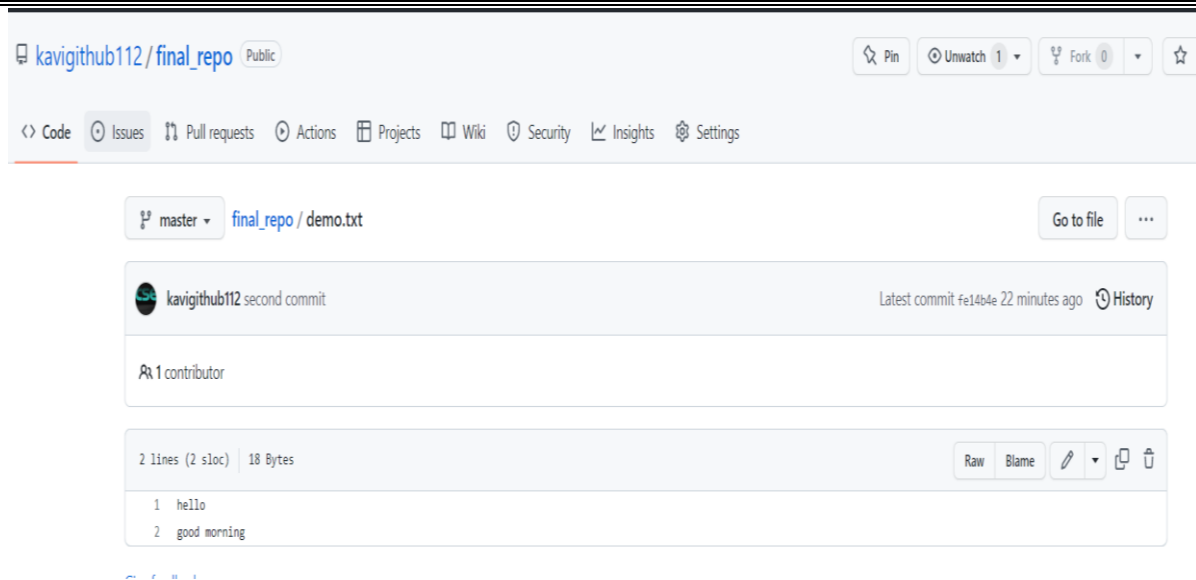
 demo.txt second commit 23 minutes ago

Help people interested in this repository understand your project by adding a README. [Add a README](#)

About
hi
0 stars
1 watching
0 forks

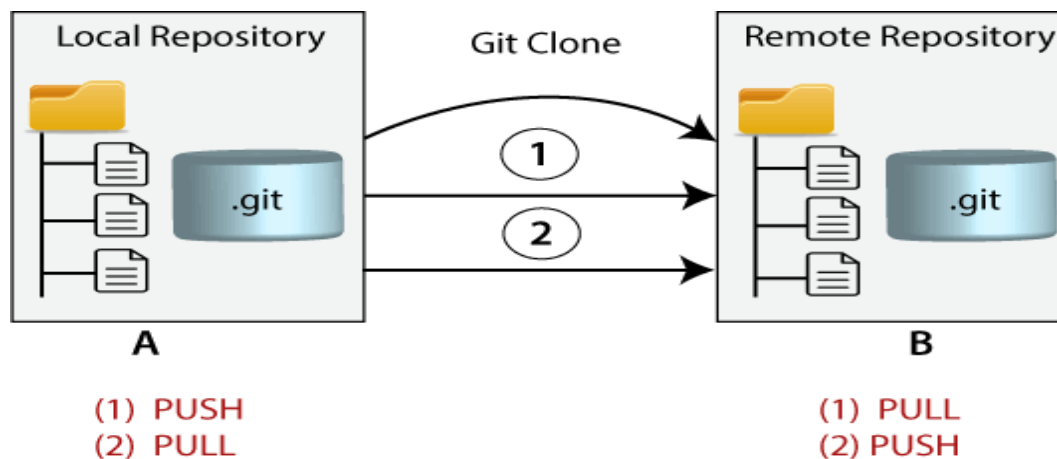
Releases
No releases published
[Create a new release](#)

Packages
No packages published
[Publish your first package](#)



Cloning a repository

In Git, cloning is the act of making a copy of any target repository. The target repository can be remote or local. You can clone your repository from the **remote repository to create a local copy on your system**. Also, you can sync between the two locations.



Pulling- From remote location to local system

Suppose, you want to clone a repository from GitHub, or have an existing repository owned by any other user you would like to contribute.

Steps to clone a repository are as follows:

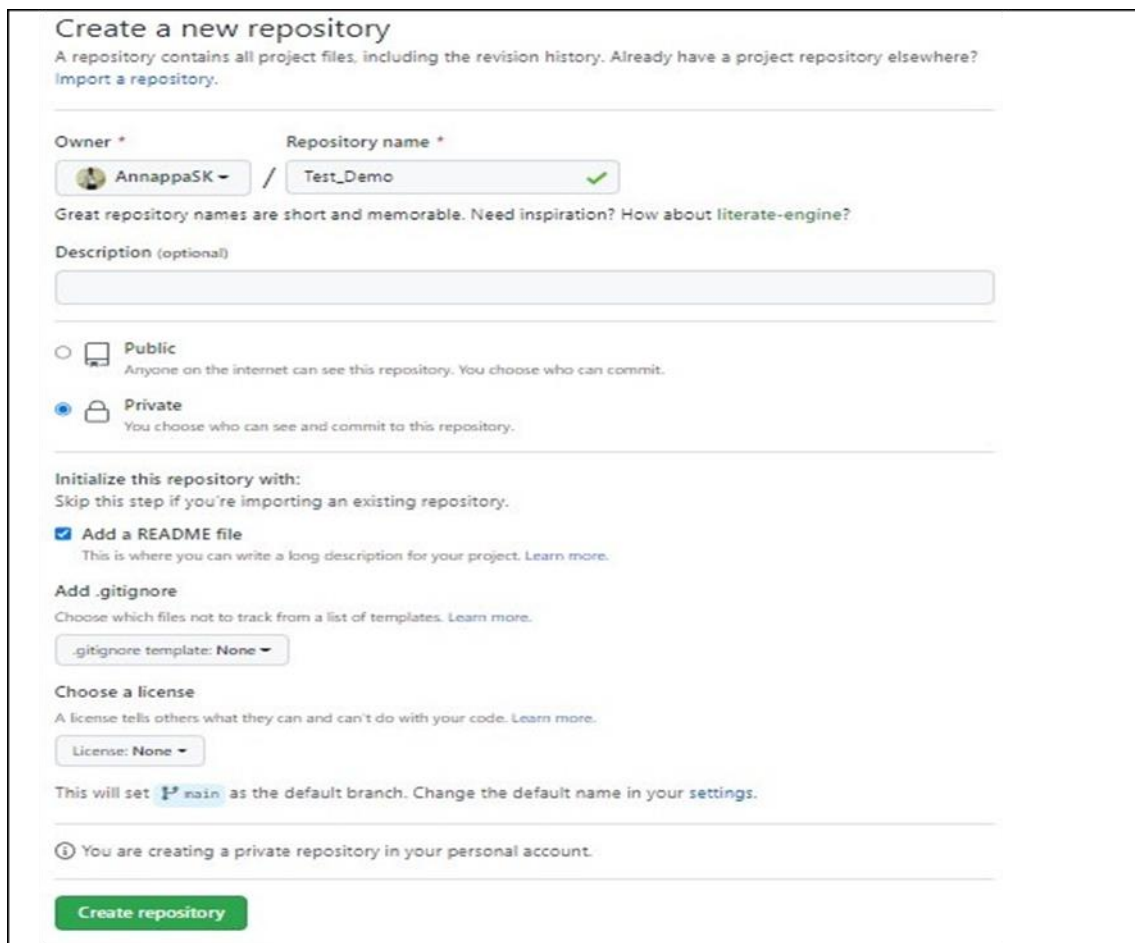
Step 1: Open GitHub website and login with user account and navigate to the main page of

the repository.

Step 2: After logging into GitHub account, click on **New** button to create new repository.

Step 3: Now give any name for your repository such as Test_Demo. Choose repository as public or private.

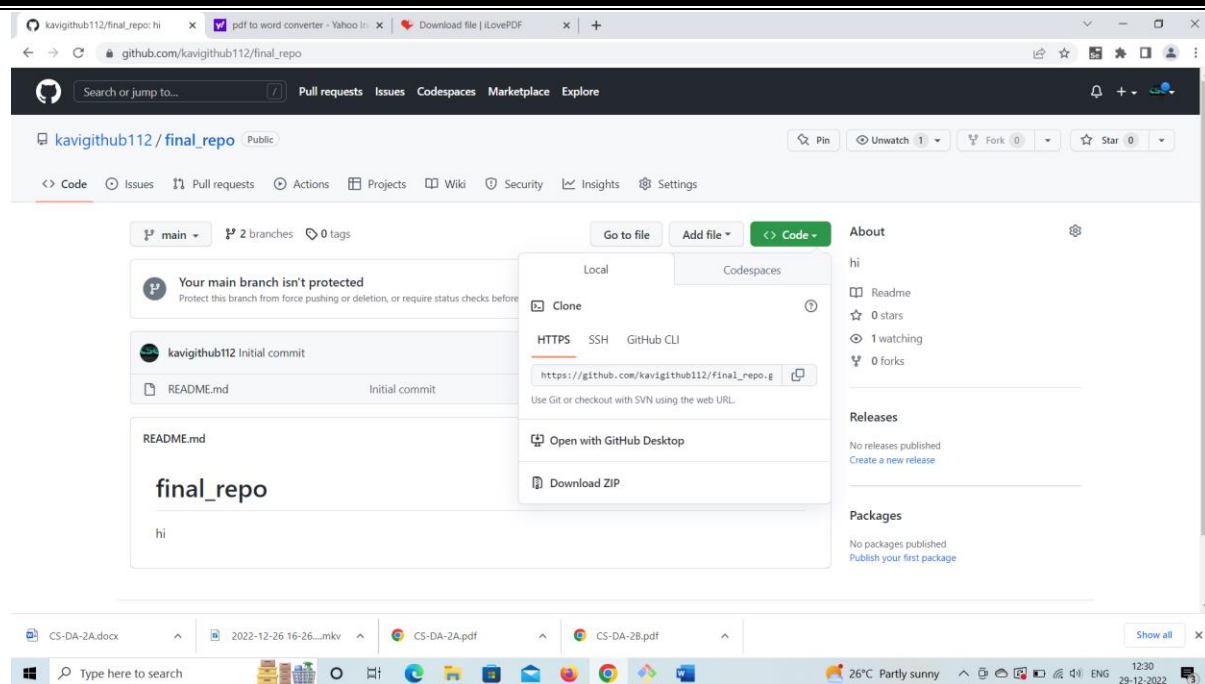
Then check **Add a README file** and click on create repository.



The screenshot shows the GitHub 'Create a new repository' page. At the top, it says 'Create a new repository' and 'A repository contains all project files, including the revision history. Already have a project repository elsewhere? Import a repository.' Below this, there are two input fields: 'Owner *' with a dropdown menu showing 'AnnappaSK' and 'Repository name *' with a text input 'Test_Demo' and a green checkmark. A note says 'Great repository names are short and memorable. Need inspiration? How about [literate-engine](#)?'. There is a 'Description (optional)' text area. Below that, there are two radio buttons: 'Public' (unselected) and 'Private' (selected). A note says 'You choose who can see and commit to this repository.' Underneath, it says 'Initialize this repository with:' and 'Skip this step if you're importing an existing repository.' There is a checked checkbox for 'Add a README file' with a note 'This is where you can write a long description for your project. Learn more.' Below that, it says 'Add .gitignore' and 'Choose which files not to track from a list of templates. Learn more.' There is a dropdown menu for '.gitignore template: None'. Then, it says 'Choose a license' and 'A license tells others what they can and can't do with your code. Learn more.' There is a dropdown menu for 'License: None'. A note says 'This will set `main` as the default branch. Change the default name in your settings.' At the bottom, there is a note 'You are creating a private repository in your personal account.' and a green 'Create repository' button.

Step 4: Now you can add any files to main tab using add file option.

Step 5: Next click on code to copy link of Test_Demo repository-> select HTTPS and copy link



Step 6: Open Git Bash and use git clone command as follows.

\$ git clone https://github.com/AnnappaSK/Test_Demo.git and press enter as shown below figure.

Step 7: Now go to the test folder where you have downloaded repository called final_demo from remote location.

```
kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ git clone https://github.com/kavigithub112/final_repo.git
Cloning into 'final_repo'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 9 (delta 0), reused 6 (delta 0), pack-reused 0
Receiving objects: 100% (9/9), done.

kavit@DESKTOP-9HAQ5GN MINGW64 ~/desktop/test (master)
$ |
```



3) Inspect and download digital certificates using a web browser and visiting popular websites- Identify and write down the crypto algorithms in TLS.

What is a Digital Certificate?

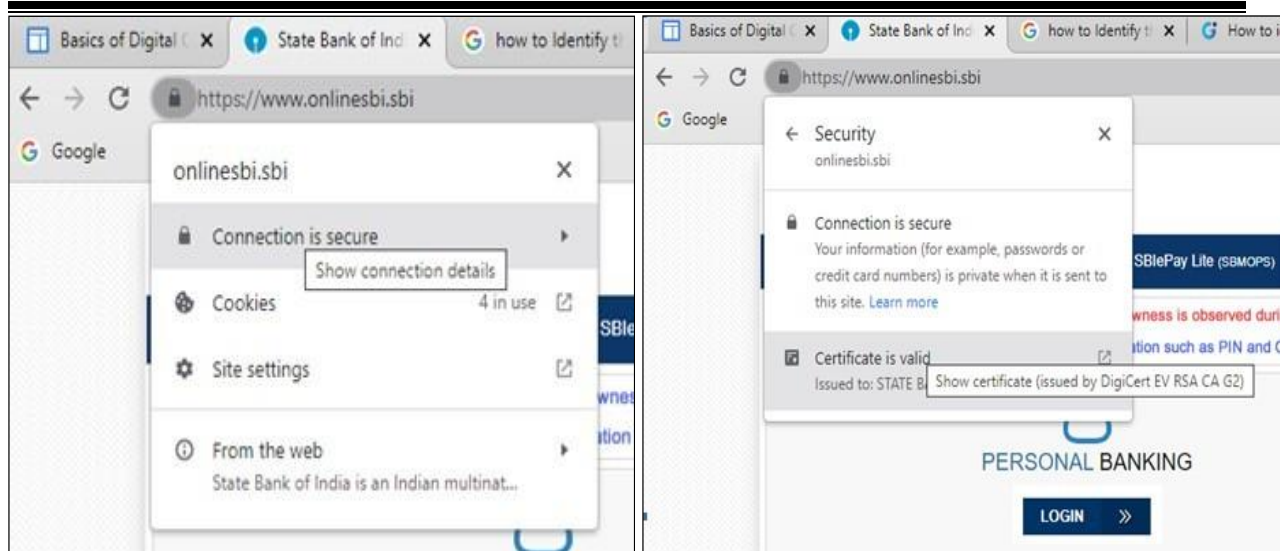
- Digital certificates are electronic credentials that are used to assert the online identities of individuals, computers, and other entities on a network. Digital certificates function similarly to identification cards such as passports, Adhaar cards and drivers licenses.
- Most commonly they contain a public key and the identity of the owner. They are issued by certification authorities (CAs) that must validate the identity of the certificate-holder both before the certificate is issued and when the certificate is used.
- Common uses include business scenarios requiring authentication, encryption, and digital signing.

Certificate Authority

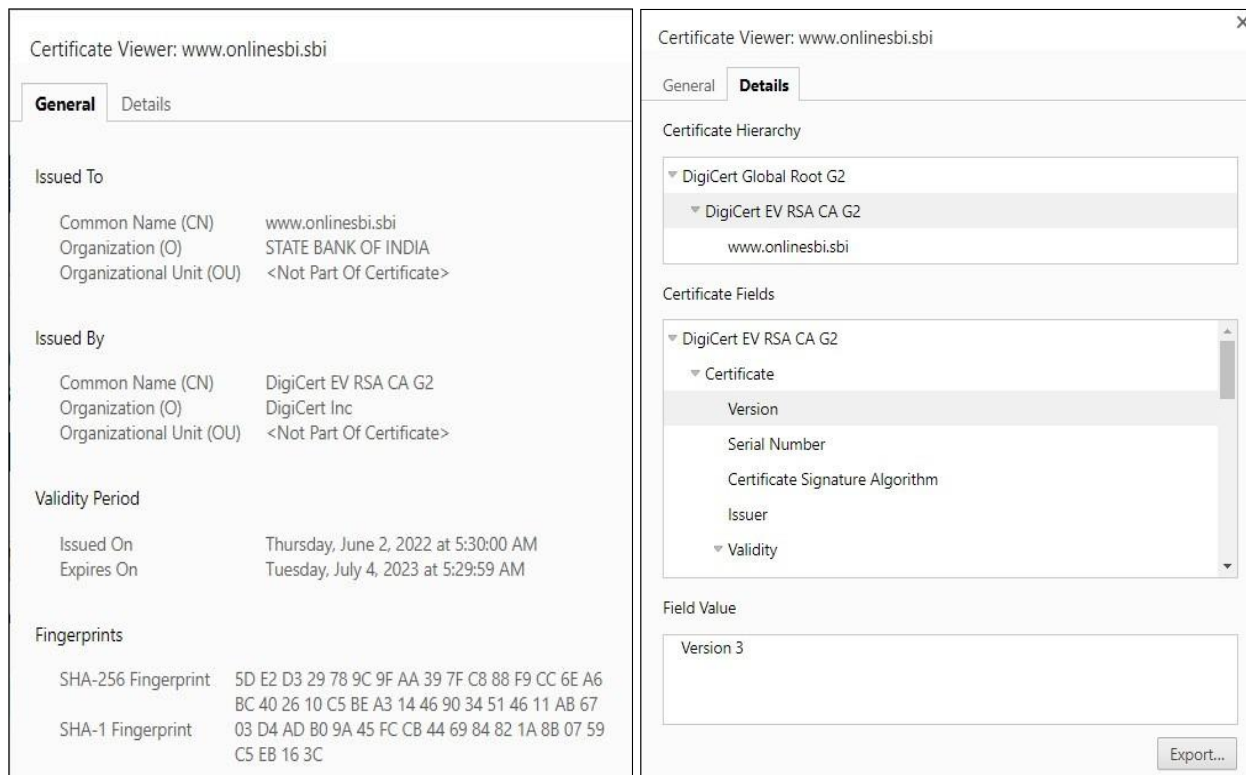
- A Certificate Authority (CA) issues digital certificates that contain a public key and the identity of the owner. The matching private key is not made available publicly, but kept secret by the end user who generated the key pair. The certificate is also a confirmation or validation by the CA that the public key contained in the certificate belongs to the person, organization, server or other entity noted in the certificate. CAs use a variety of standards and tests to do so. In essence, the Certificate Authority is responsible for saying "yes, this person is who they say they are, and we, the CA, verify that".
- If the user trusts the CA and can verify the CA's signature, then he can also verify that a certain publickey does indeed belong to whoever is identified in the certificate. Browsers maintain list of well known CAs root certificates. Aside from commercial CAs, some providers issue digital certificates to the public at no cost. Large institutions or government entities may have their own CAs.

Real Examples:

- Let us check a real certificate, its details and its chain. There are certificate viewer tools that read those archaic encoding formats and show the certificates nicely! You can actually check any https url in any browser to check a X.509 digital certificate. Here we are going to check internet banking site of **State Bank of India** in Chrome.
- Go to <https://www.onlinesbi.sbi/> and click on the padlock icon to view certificate as shown below.

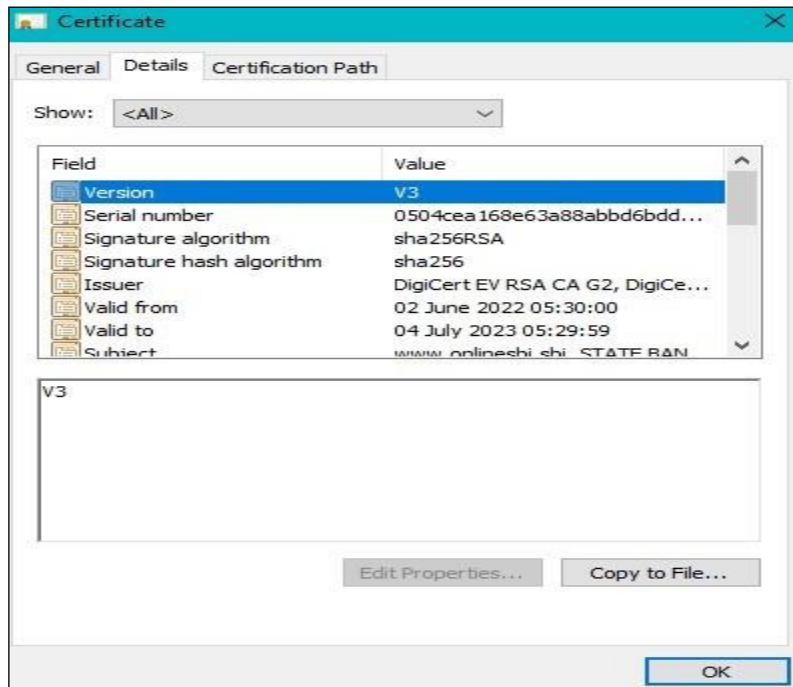


- Once you click on the certificate is valid link, Windows certificate viewer tool will open and show the certificate owned by State Bank of India. This certificate, as you can see in "Issued by" field is issued by DigiCert EV RSA CA G2.



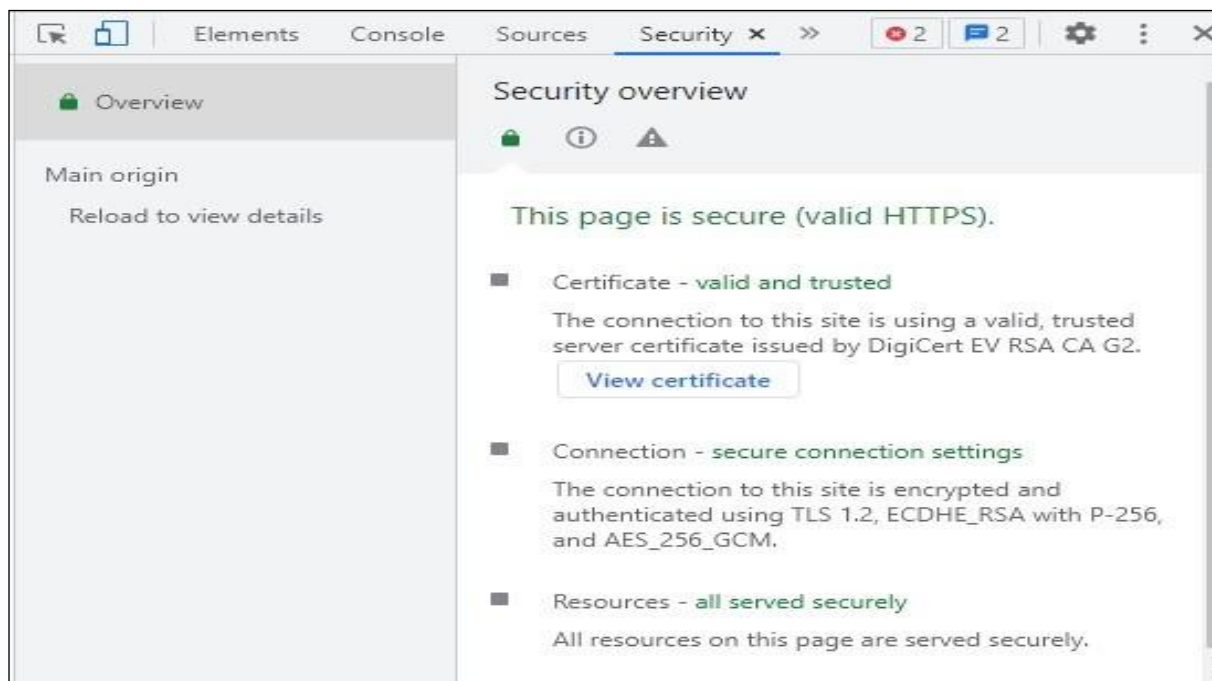
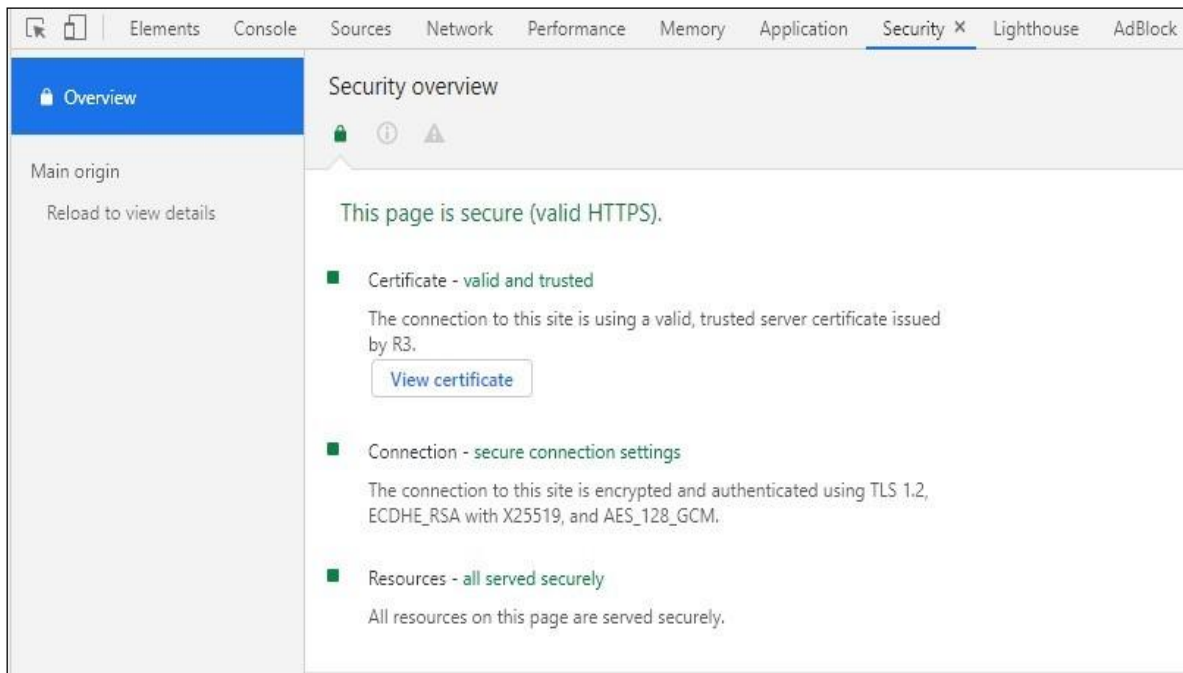
- Now you can download/export certificate to know more details like subject, SBI, and its detail Distinguished Name (DN).
- Open downloaded file and Click on details tab to explore version, serial number,

issuer, valid from, valid to, subject, public key and many more.



Identify and write down the crypto algorithms in TLS:

- Launch **Chrome**.
- Enter the URL you wish to check in the browser.
- Click on the ellipsis located on the top-right in the browser.
- Select More tools > Developer tools > Security
- Look for the line "Connection...". This will describe the version of TLS or SSL used.



4) Design a Simple Cryptosystem (Including Key Generation, Encryption, Decryption, Digital Signature or Hash Function) Using any Tool

JCRYPT TOOL

- Jcrypt is a *framework for developing cryptological and cryptographic programs*.
- JCrypt is a free software
- JCrypt Tool enables students, teachers, developers, and anyone else interested in cryptography to apply and analyze cryptographic algorithms.

Installation

1. The JCrypTool installation is very simple
2. Select any web browser ex : google browser.
3. In search type jcrypt tool free download from softpedia.
4. Download and extract the zip file.
5. Launch the main program and get started.
6. Admin rights are not required.

Cryptography is the study of secure communications techniques that allow only the sender and intended receiver of a message to view its contents.

Types Of Cryptography

1. Symmetric Cryptography
2. Asymmetric Cryptography
3. Digital Signature
4. Hash Function

Symmetric Cryptography: also known as secret key cryptography

- ☐ Encrypting and decrypting a message using single key
- ☐ The single key is known as secret key
- ☐ Ex: AES- Advanced Encryption Standard,
- ☐ RC4- Rivest Cipher 4
- ☐ DES- Data Encryption Standard ETC

Steps: For Encryption (Plaintext Is Converted Into Ciphertext)

1. Go To File
2. Select New Empty Text Editor File
3. Save The File
4. Open A Saved File Type A Message You Want To Send
5. Go To Algorithm -> Select Symmetric Algorithm ->Select Aes
6. In Dialog Box Select Encrypt Option
7. Click On Key Generation
8. Generate A Secret Key
9. Finish

Steps: For Decryption (Ciphertext Is Converted Into Plaintext)

1. Go To File
2. Select Encrypted File
3. Go To Algorithm -> Select Symmetric Algorithm ->Select Aes
4. In Dialog Box Select Decrypt Option
5. Enter Secet Key
6. Finish

Asymmetric Algorithm Also Known As Public Key Cryptography

- ☐ Encrypting and decrypting a message using double key.
- ☐ The two keys are Private and public key.
- ☐ Example: RSA - Rivest, Shamir, Adleman, DSS - Digital Signature Standard etc

Steps: For Encryption (Plaintext Is Converted Into Ciphertext)

1. Go To File
 2. Select New Empty Text Editor File
 3. Save The File
 4. Open A Saved File Type A Message You Want To Send
 5. Go To Algorithm -> Select Asymmetric Algorithm ->Select Rsa
 6. In Dialog Box Select Encrypt Option
 7. Click On Key Generation
-

8. Generate A Private Key
9. Finish

Steps: For Decryption (Ciphertext Is Converted Into Plaintext)

1. Go To File
2. Select Encrypted File
3. Go To Algorithm -> Select Asymmetric Algorithm -> Select Rsa
4. In Dialog Box Select Decrypt Option
5. Enter Private Key
6. Finish

DIGITAL SIGNATURE

- ☐ A digital signature is a cryptographic output used to verify the authenticity of data.
- ☐ Digital signature algorithm consists of two operations: SIGN and VERIFY OPERATION.

Steps: Signing Of Data

1. Go To File – Select A New File Text Editor
2. Type A Message
3. Go To Algorithm
4. Select Signature - > Select Dsa
5. Select Sign -> Choose Path To Save A Cryptographic Value
6. Finish

Steps: Verifying The Data

1. Go To File – Select A New File Text Editor
2. Type A Message
3. Go To Algorithm
4. Select Signature - > Select Dsa
5. Select Verify -> Open The Saved Path
6. Finish

Result: Valid Means Authenticated

Invalid Means Unauthenticated Wrong Sender

Hash Function

- ☐ A cryptographic hash function is a mathematical function used in cryptography.
- ☐ It is one way function.
- ☐ It's also a process that takes plaintext data of any size and converts it into a unique cipher text of a specific length.

Steps For Hash Function

1. Go To File – Select A New File Text Editor
2. Type A Message
3. Go To Algorithm
4. Go To Hash->Select Md5

Finish

5) Attacks and vulnerabilities: Injection attacks : SQL, HTTP header, OS command**OS Command Injection:**

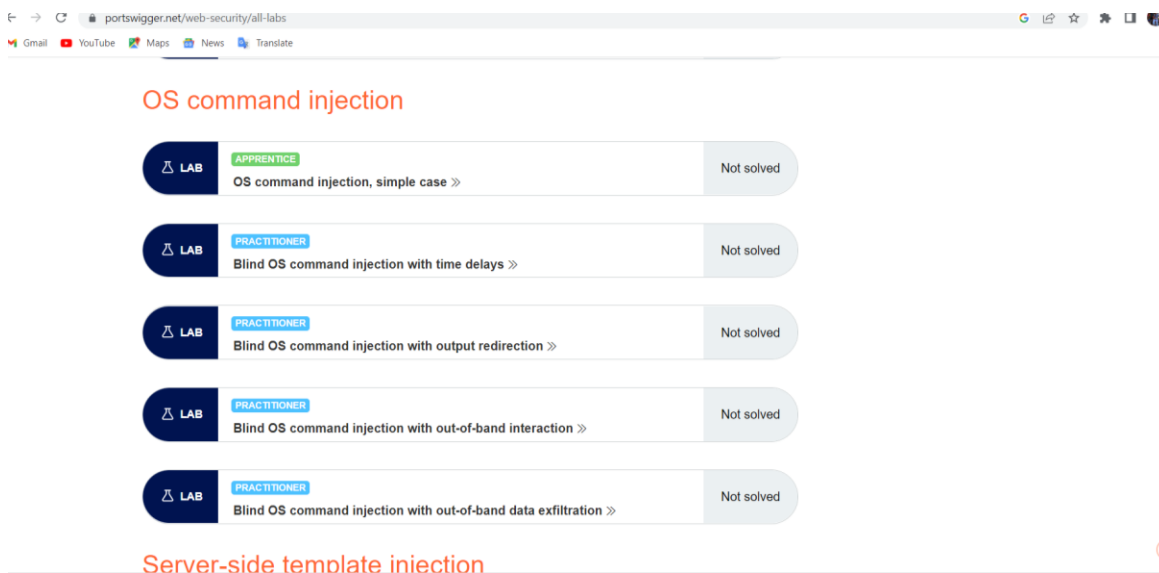
Step 1: Download the burp suite app from the chrome

Step 2: Open the burp suite app > click next > start burp > go to proxy > click on open browser

Step 3: Then the burp suite browser will open, minimize that

Step 4: Go to normal chrome, search “/all labs”

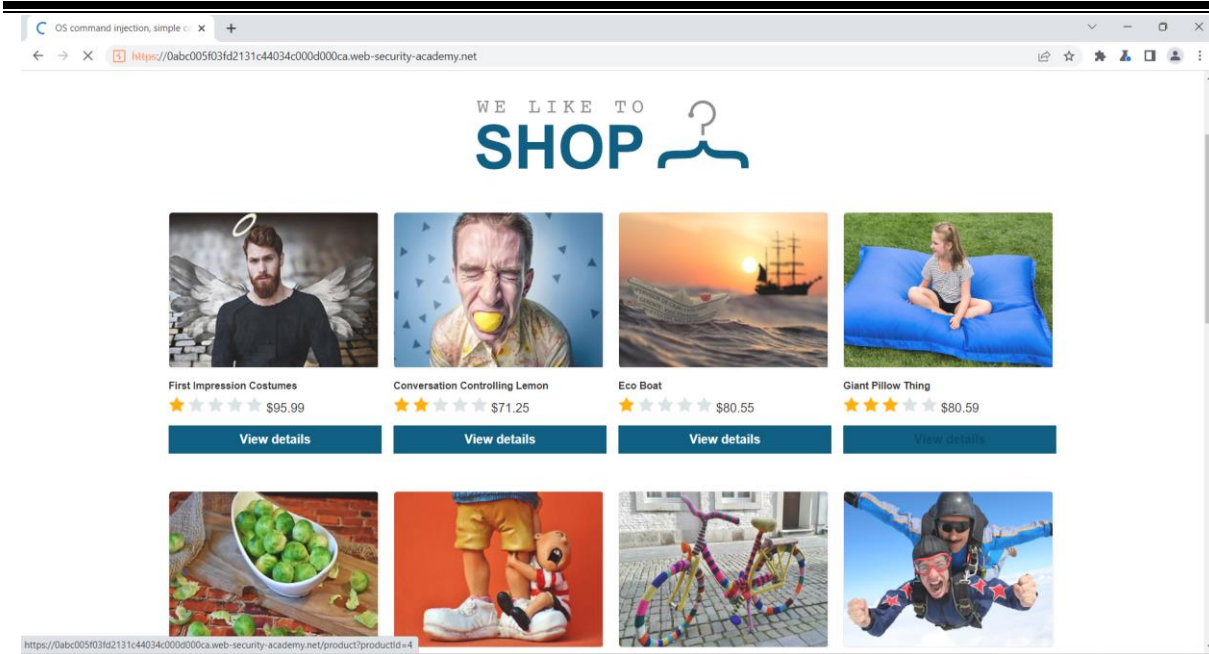
Step 5: Select OS Command Injection and it will ask for port swigger login



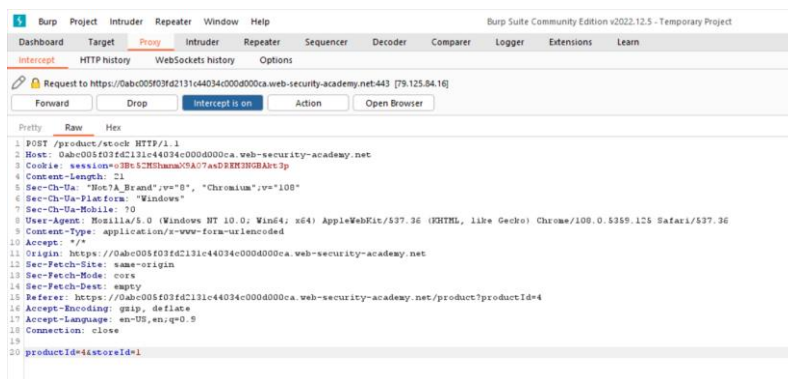
Step 6: Create an account on port swigger and login to that

Step 7: Then click on Access lab and copy the path

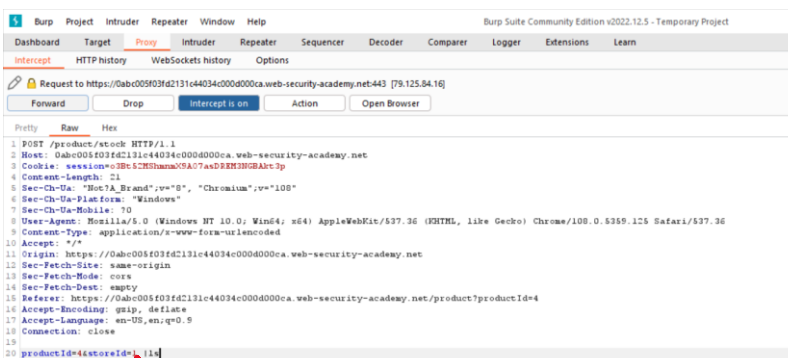
Step 8: Come to burp suite browser, paste that path > select one > click on view details



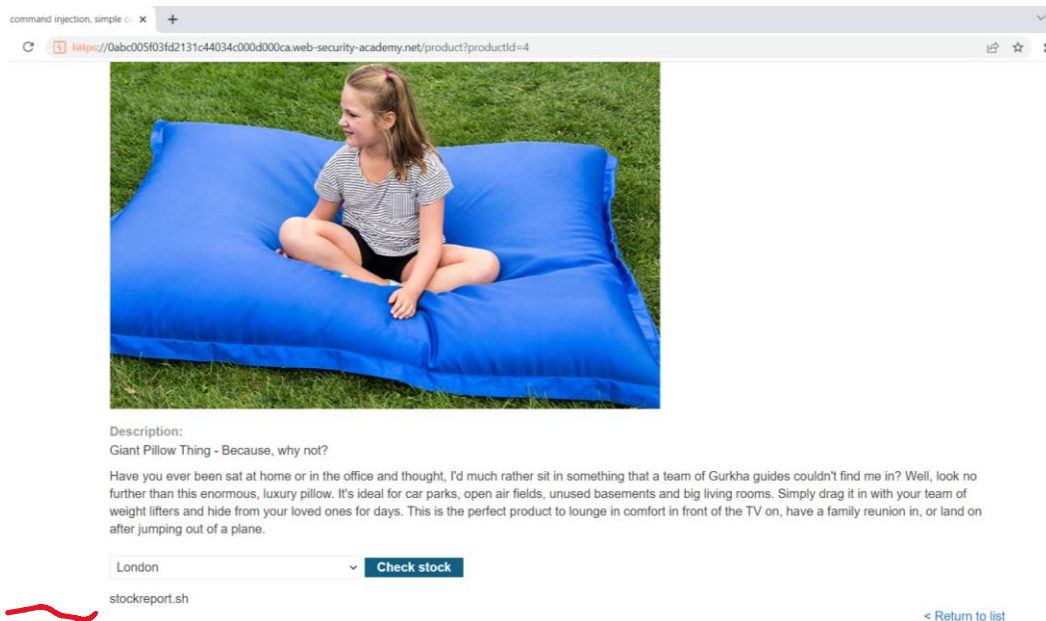
Step 9: Click on check stock and then come to the burp suite > turn on the intercept > it will show some codes which is shown in the below



Step 10: In the 20th line go to the end and give one space and type “|s” and then click forward



Step 11: Go to burp suite browser again click on check stock , it will show the result like below



HTTP Header Injection:

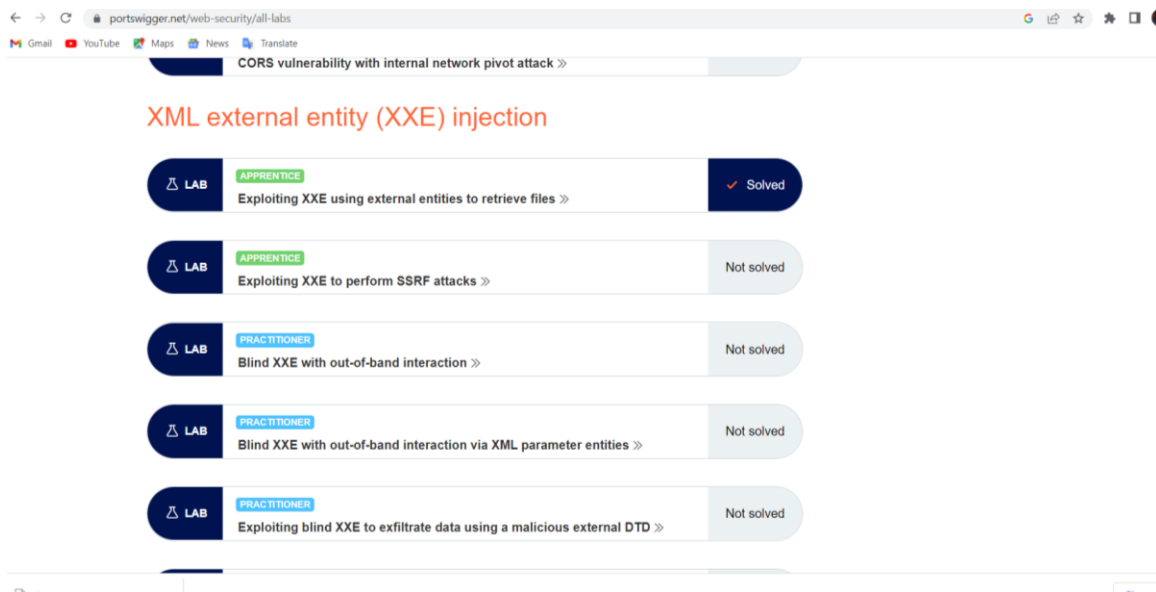
Step 1: Download the burp suite app from the chrome

Step 2: Open the burp suite app > click next > start burp > go to proxy > click on open browser

Step 3: Then the burp suite browser will be open , minimize that

Step 4: Go to normal chrome, search “/alllabs”

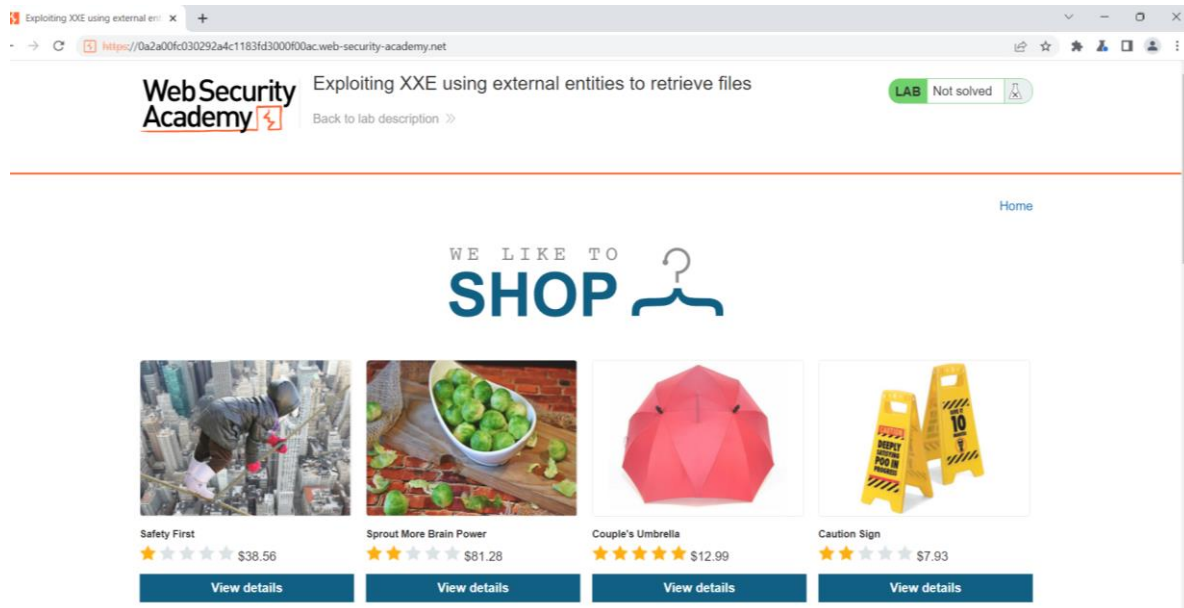
Step 5: Select XML external entity (XXE) Injection and it will ask for port swigger login



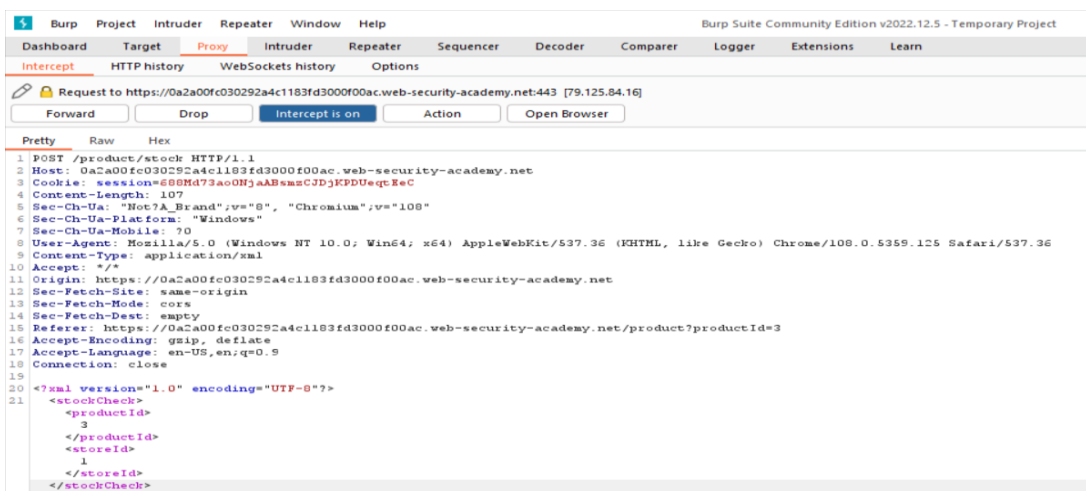
Step 6: Create an account on port swigger and login to that

Step 7: Then click on Access lab and copy the path

Step 8: Come to burp suite browser,> paste that path > select one > click on view details



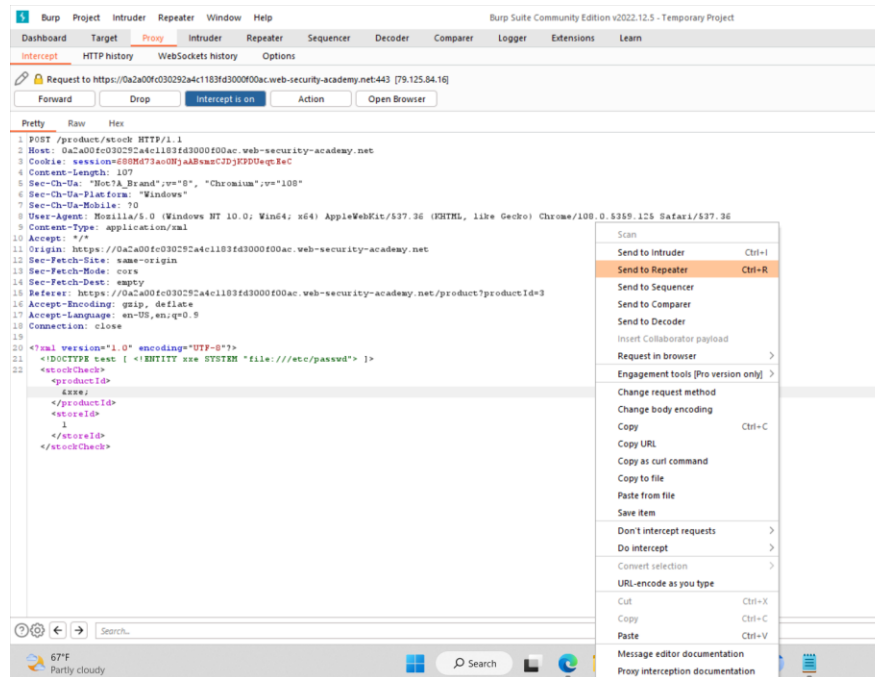
Step 9: Click on check stock and then come to the burp suite > turn on the intercept > it will show some codes which is shown in the below picture



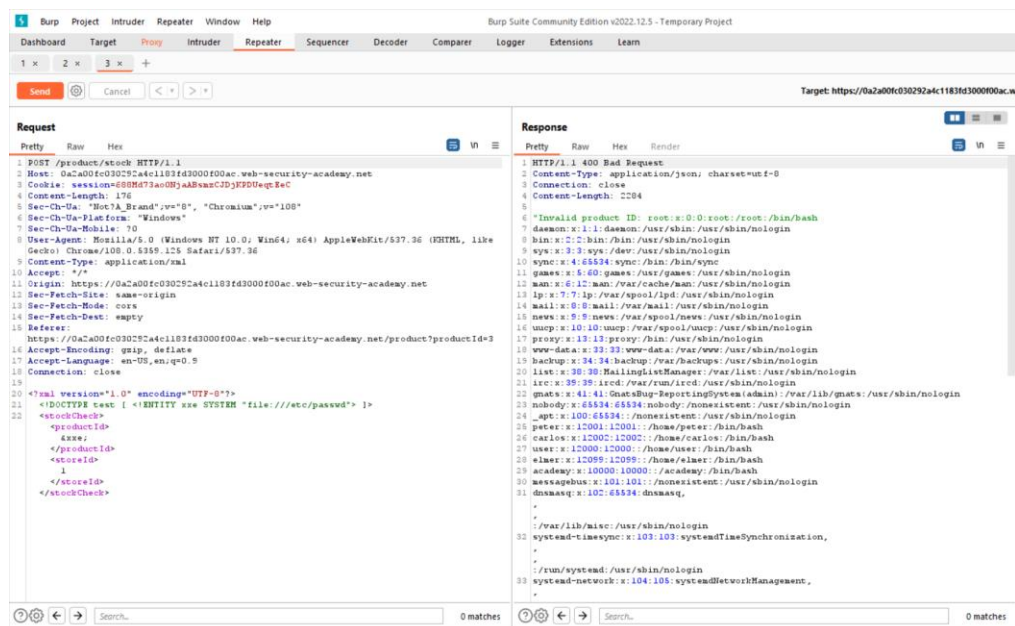
Step 10: In the 21st line add these below code

➤ `<!DOCTYPE test [ <ENTITY xxe SYSTEM "file:///etc/passwd"> ]>`

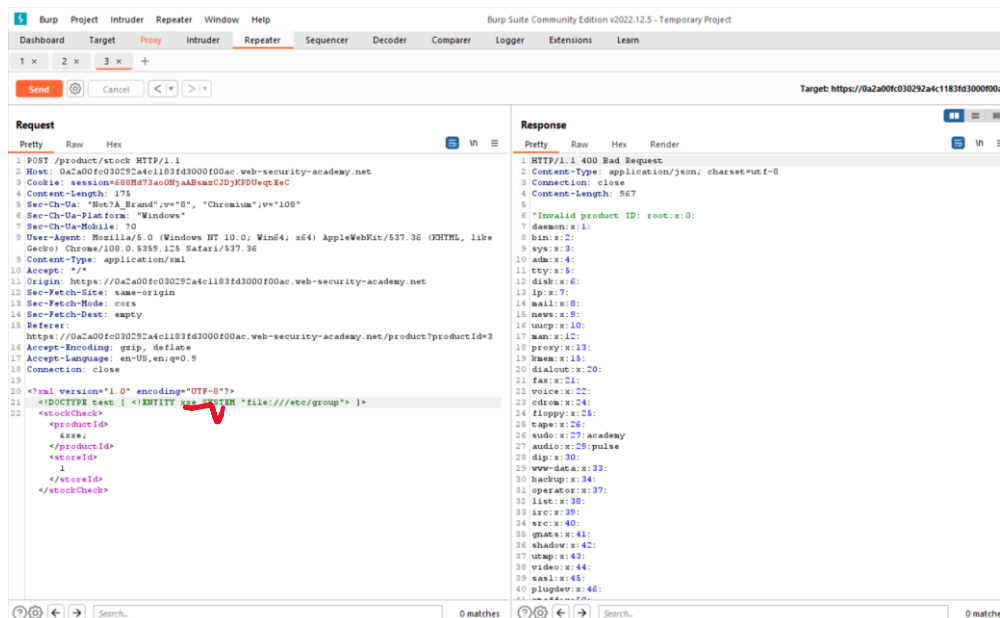
- And in the product id , remove that number and add the below command
- **&xxe;**
- Right click > click on send to Repeater



Step 10: Go to Repeater > click on send > it will show the response



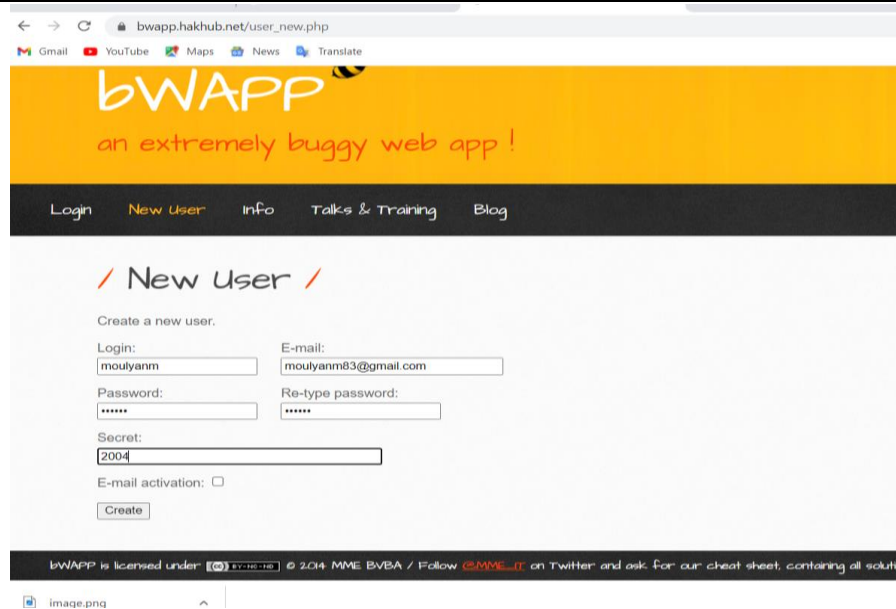
Step 11: In the 21st line you can remove the psswd and type “group” or admin and then click send, it will show the response



SQL Injection :

Step 1: Go to chrome > search “BWAPP LOGIN”

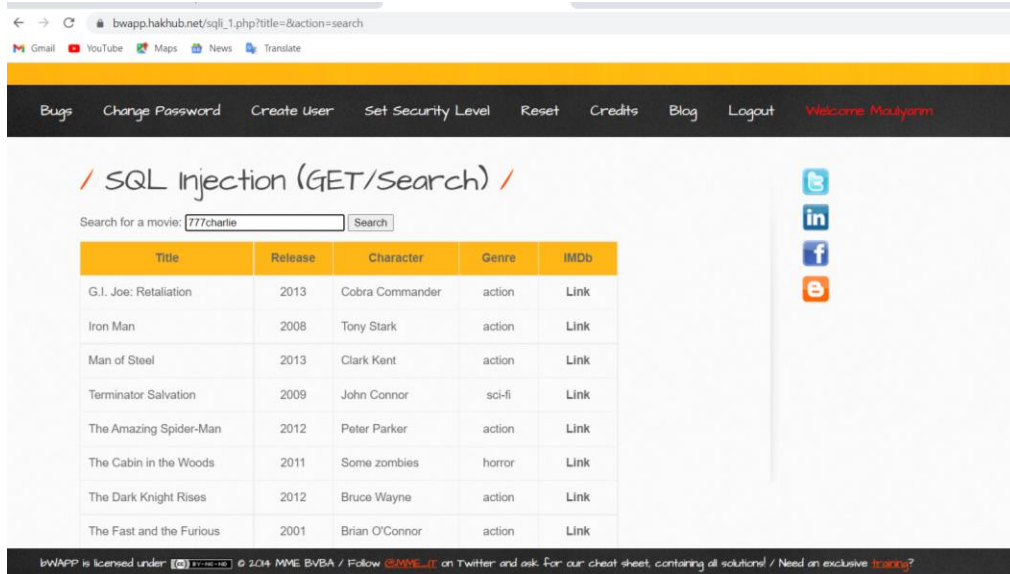
Step 2: Click on new user > create an account > and then login to that



Step 3: In the top right corner > Select SQL Injection > set security level to high > click on Hack

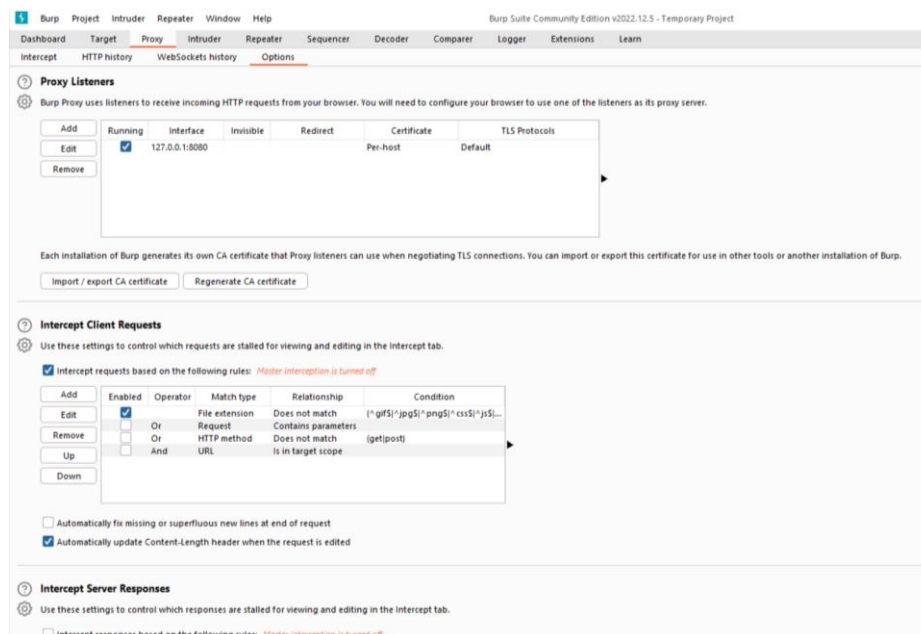


Step 4: Search for a movie > search one movie name

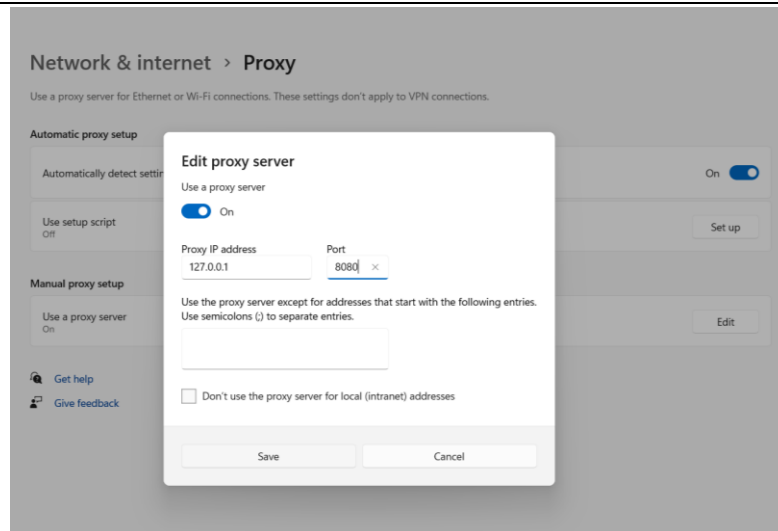


Step 5: Open the burp suite > go to proxy > click on options

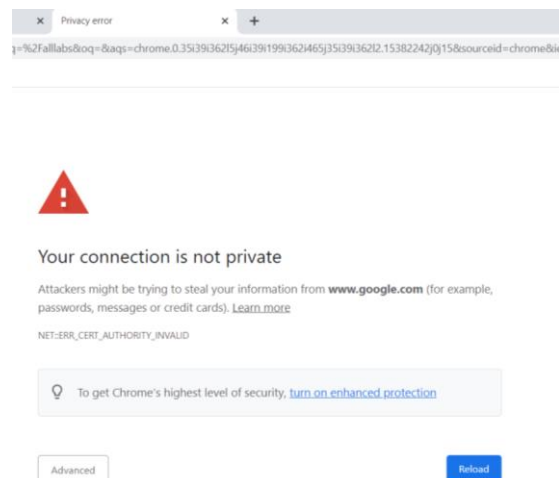
Step 6: Note down the number [127.0.0.1:8080]



Step 7: Go to setting > search proxy settings > turn on manual proxy setup > enter the proxy IP address and port number which is note downed from the burp suite > click on save

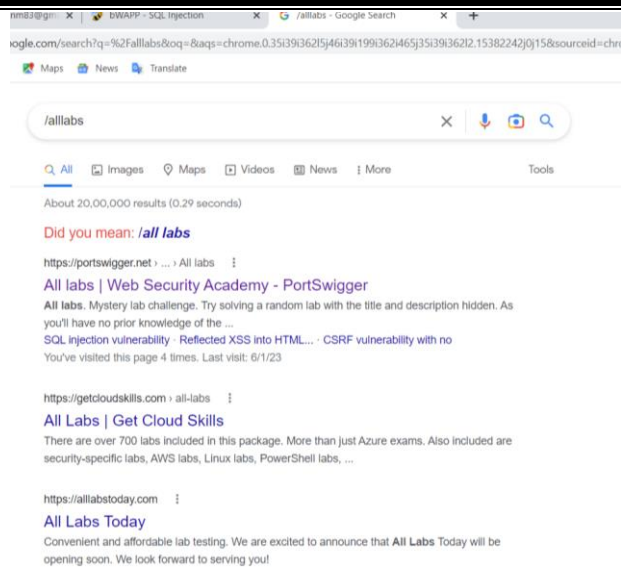


Step 8: Now go to chrome, search something, it will not load



Step 9: Go to settings > proxy settings > turn off the manual proxy setup

Step 10: Now go to chrome > reload that > now it will work

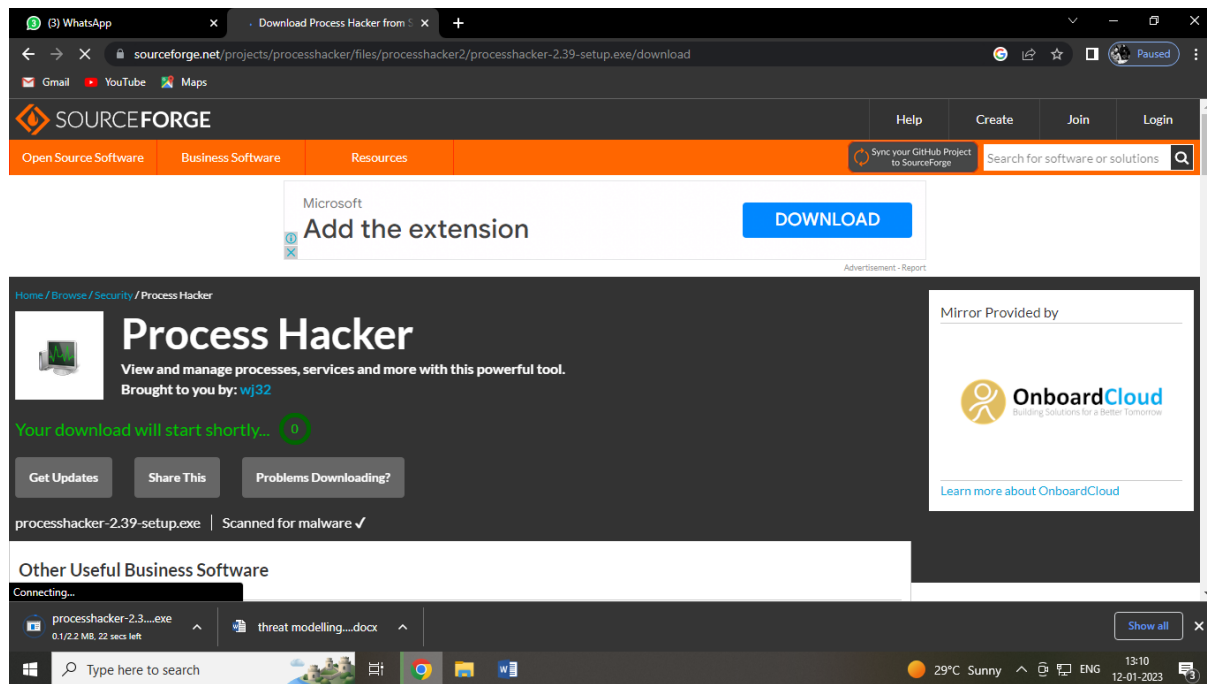


6) Process observation and analysis with Process Hacker

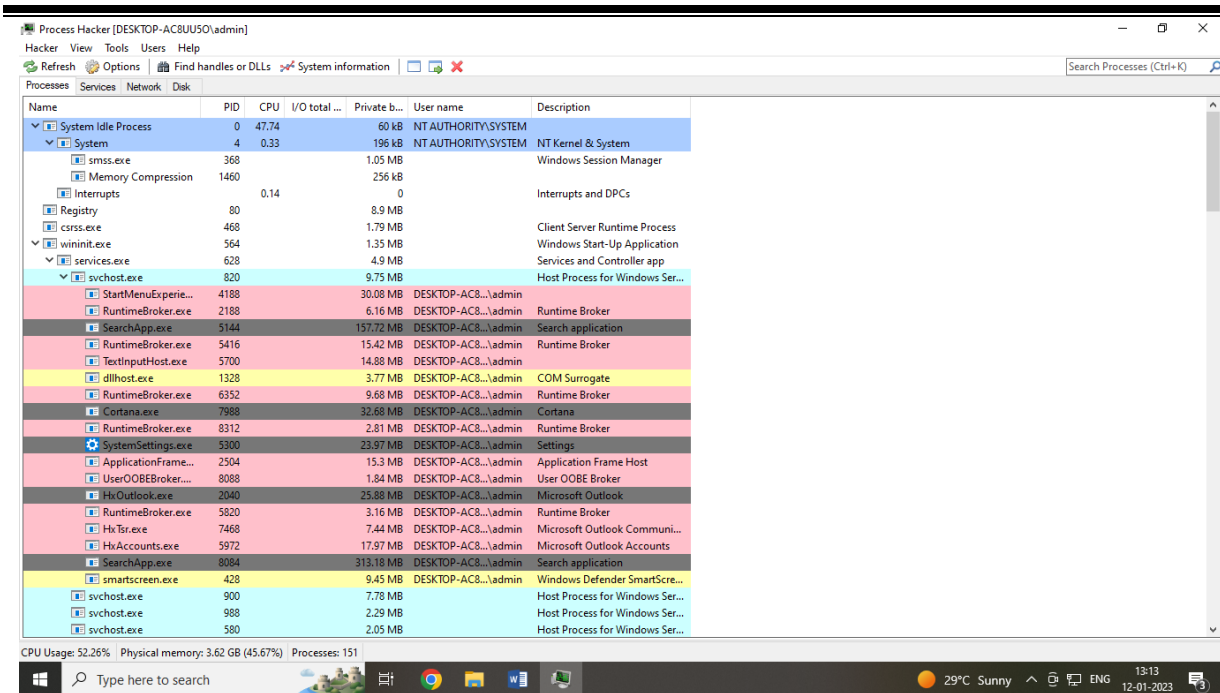
Process Hacker

Process Hacker is an open-source tool that will **allow you to see what processes are running on a device, identify programs that are eating up CPU resources and identify network connections that are associated with a process.** These types of features make Process Hacker an ideal tool for monitoring malware on a device.

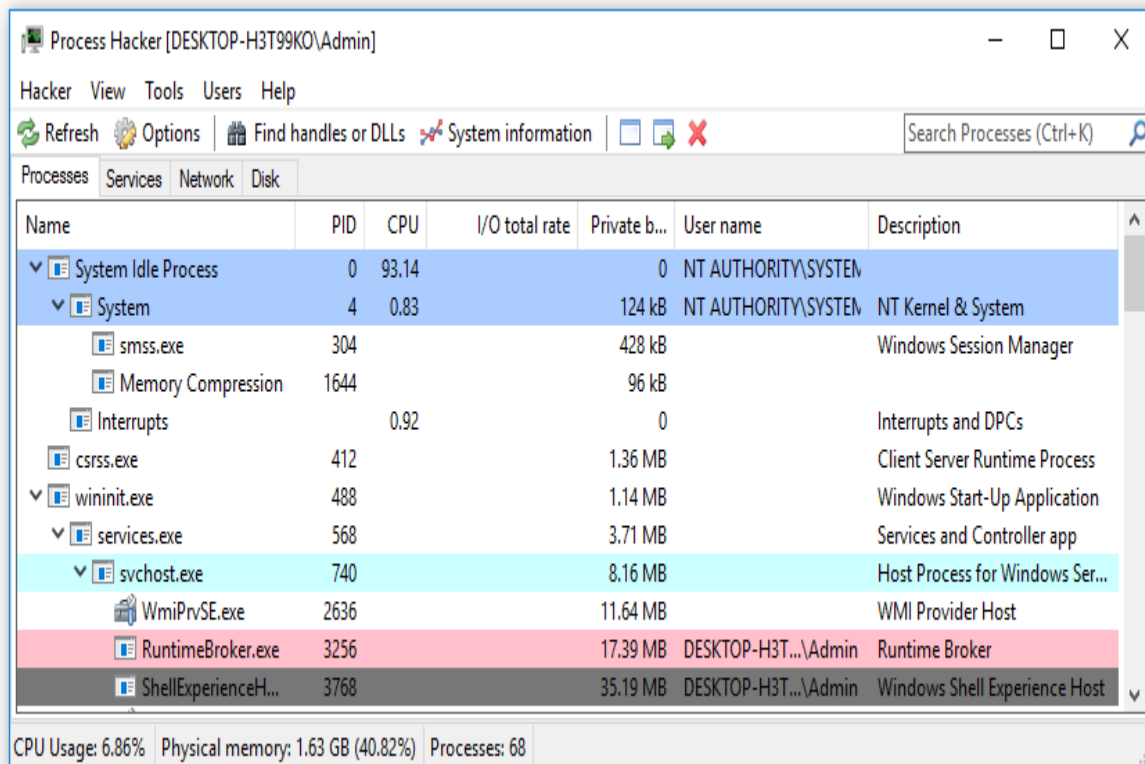
Step 1 – Go To Chrome And Download Process Hacker Tool



Step 2 – Open The Application



Below is the default display shown for Process Hacker when it is launched on a device:



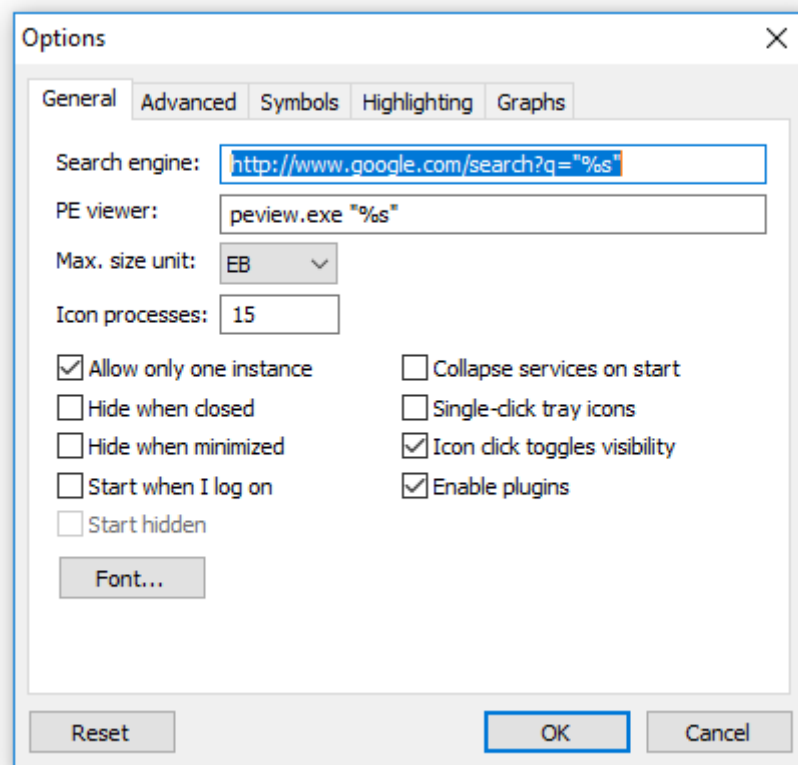
The first tab named ‘Processes’ gives an overview of what processes are running on the device which contains the following information:

- **Name** of the running process
- **The PID** is the process ID, this is a unique number assigned to the process
- **The CPU** tab displays the amount of CPU being consumed by the process
- **The I/O total output** tab
- **The Private bytes** tab
- **The User name** tab displays which account was used to launch the process
- **The Description** tab displays information relating to what the process is

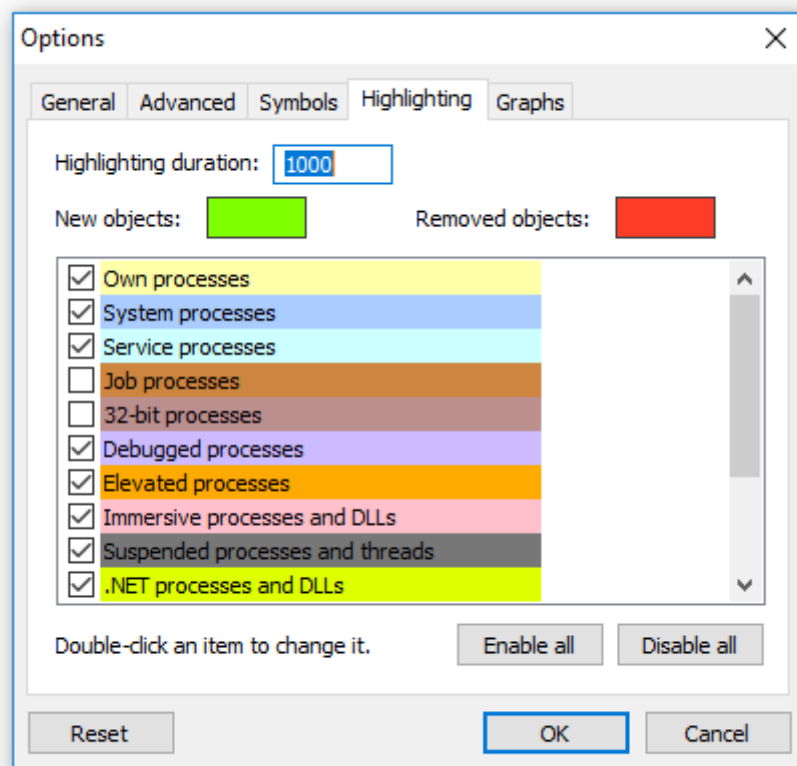
The ‘Processes’ tab also colour codes the listed processes. By navigating to ‘Hacker’ and then ‘Options’ menu you can identify what each colour represents in Process Hacker.



This then opens the ‘Options’ menu.

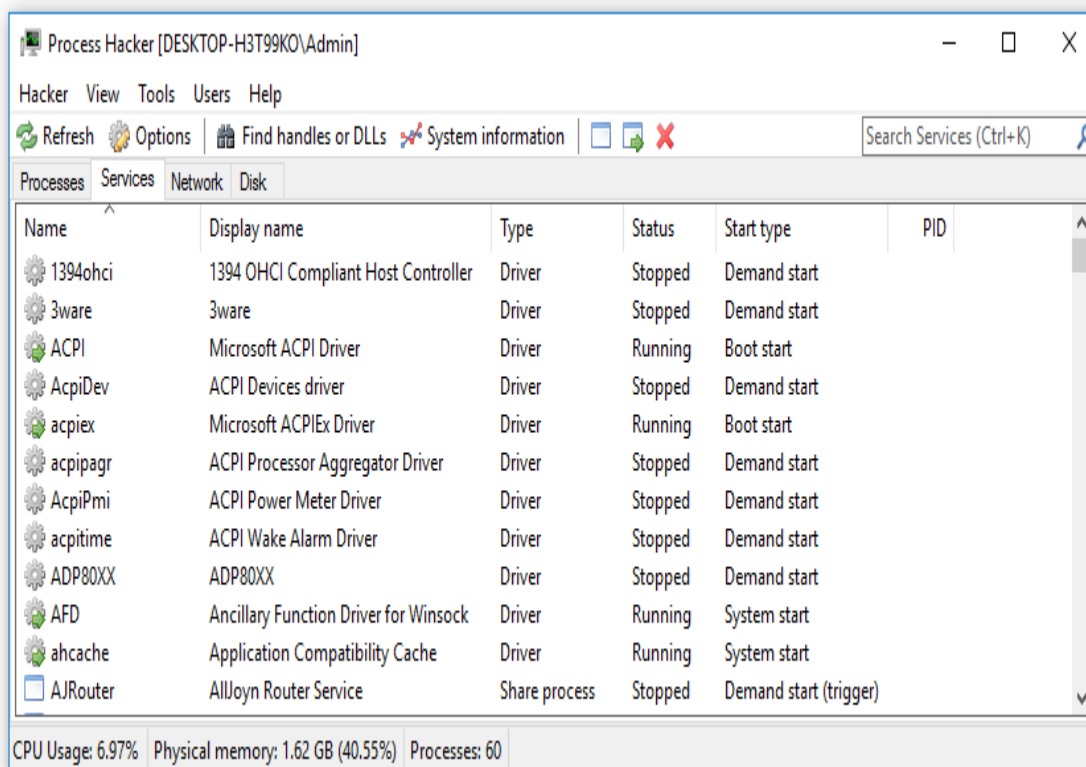


Select the 'Highlighting' tab to view what each color represents:



I won't cover what each colour represents but this is useful to quickly identify what processes are expected system processes compared to say a packed process.

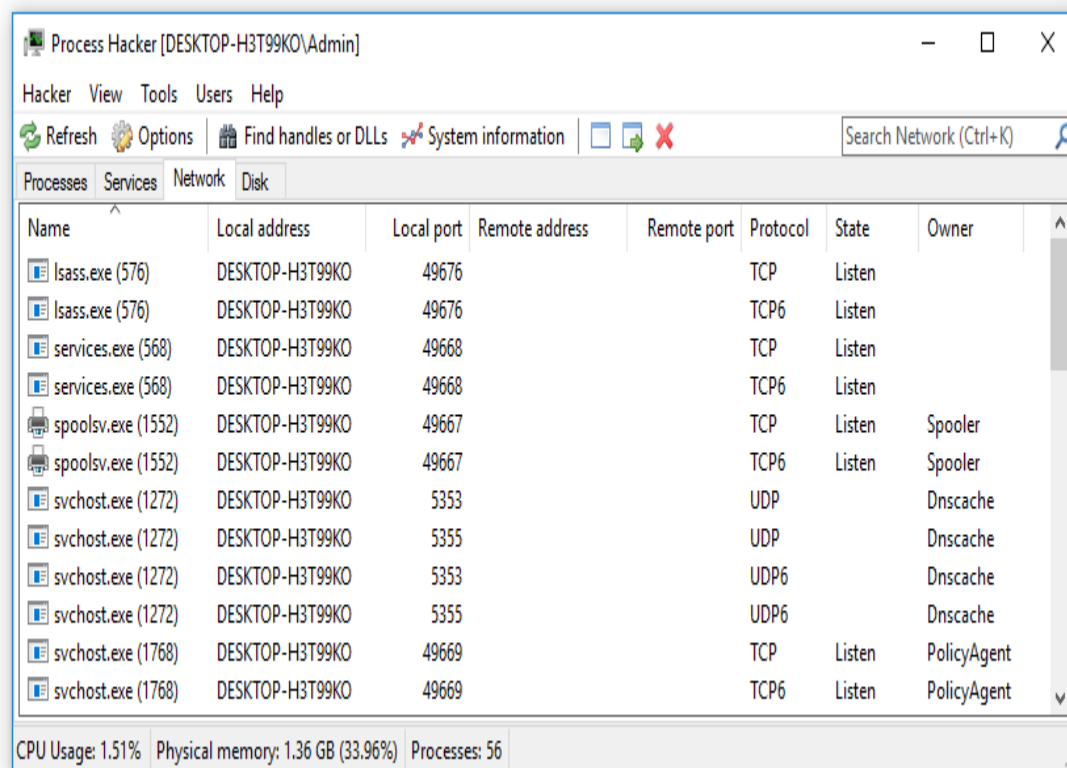
The image below displays the **services** identified by Process Hacker, services run in the background and don't interact with the desktop.



The ‘Services’ tab displays the following information:

- **Name** of identified service
- **Display name** of service
- **Type** of service identified i.e. Driver
- **Status** of service i.e. Running
- **Start type** i.e. Boot start
- **Process** identifier of service if available

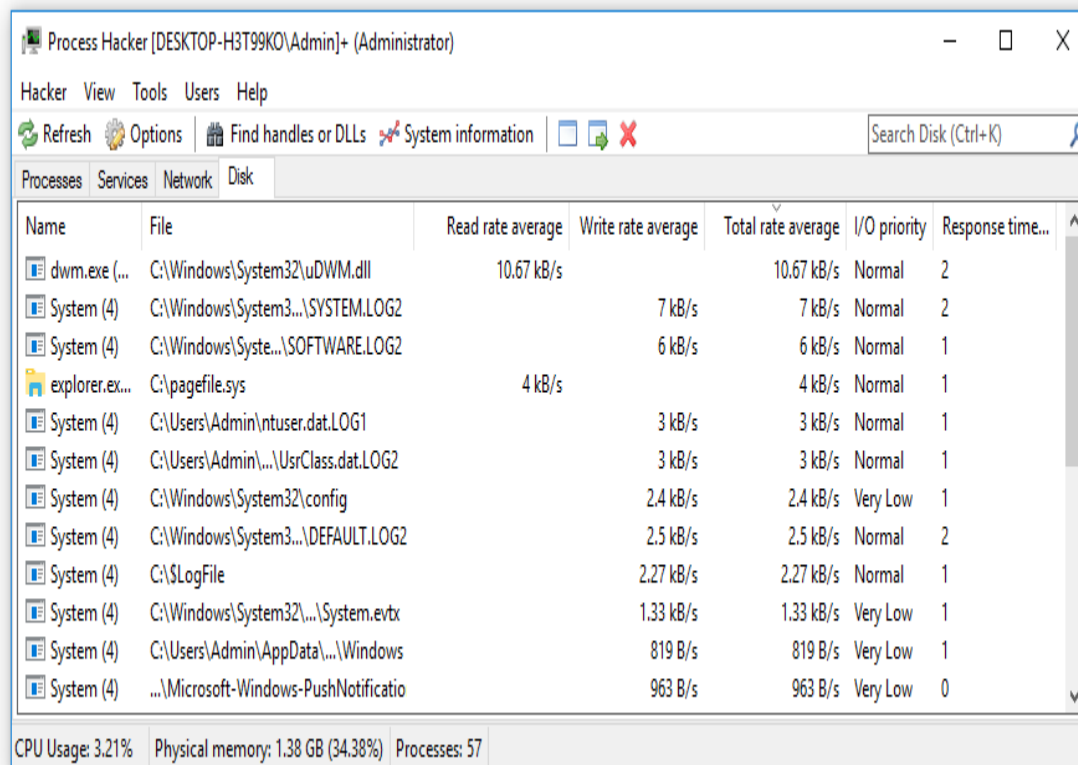
The ‘Network’ tab is useful for malware analysis as malware will often try to call home to the bad guy’s command and control (c2) infrastructure.



The 'Network' tab displays the following information:

- **Process name** and PID
- **Local address**
- **Local port** used by the process
- **Remote address** the process is connecting to
- **Remote port** of network connection
- **Protocol** used by the process
- **State** of identified network connection
- **Owner**

The 'Disk' tab displays information relating to files on the device hard drive which are being used:



The screenshot shows the 'Disk' tab in Process Hacker. The table lists processes and their disk activity. The status bar at the bottom indicates CPU Usage: 3.21%, Physical memory: 1.38 GB (34.38%), and Processes: 57.

Name	File	Read rate average	Write rate average	Total rate average	I/O priority	Response time...
dwm.exe (...)	C:\Windows\System32\udwm.dll	10.67 kB/s		10.67 kB/s	Normal	2
System (4)	C:\Windows\System32\SYSTEM.LOG2		7 kB/s	7 kB/s	Normal	2
System (4)	C:\Windows\Syste...\SOFTWARE.LOG2		6 kB/s	6 kB/s	Normal	1
explorer.ex...	C:\pagefile.sys	4 kB/s		4 kB/s	Normal	1
System (4)	C:\Users\Admin\ntuser.dat.LOG1		3 kB/s	3 kB/s	Normal	1
System (4)	C:\Users\Admin\...\UsrClass.dat.LOG2		3 kB/s	3 kB/s	Normal	1
System (4)	C:\Windows\System32\config		2.4 kB/s	2.4 kB/s	Very Low	1
System (4)	C:\Windows\System32\DEFAULT.LOG2		2.5 kB/s	2.5 kB/s	Normal	2
System (4)	C:\\$LogFile		2.27 kB/s	2.27 kB/s	Normal	1
System (4)	C:\Windows\System32\...\System.evtx		1.33 kB/s	1.33 kB/s	Very Low	1
System (4)	C:\Users\Admin\AppData\...\Windows		819 B/s	819 B/s	Very Low	1
System (4)	...\Microsoft-Windows-PushNotificatio		963 B/s	963 B/s	Very Low	0

The 'Disk' tab displays the following information:

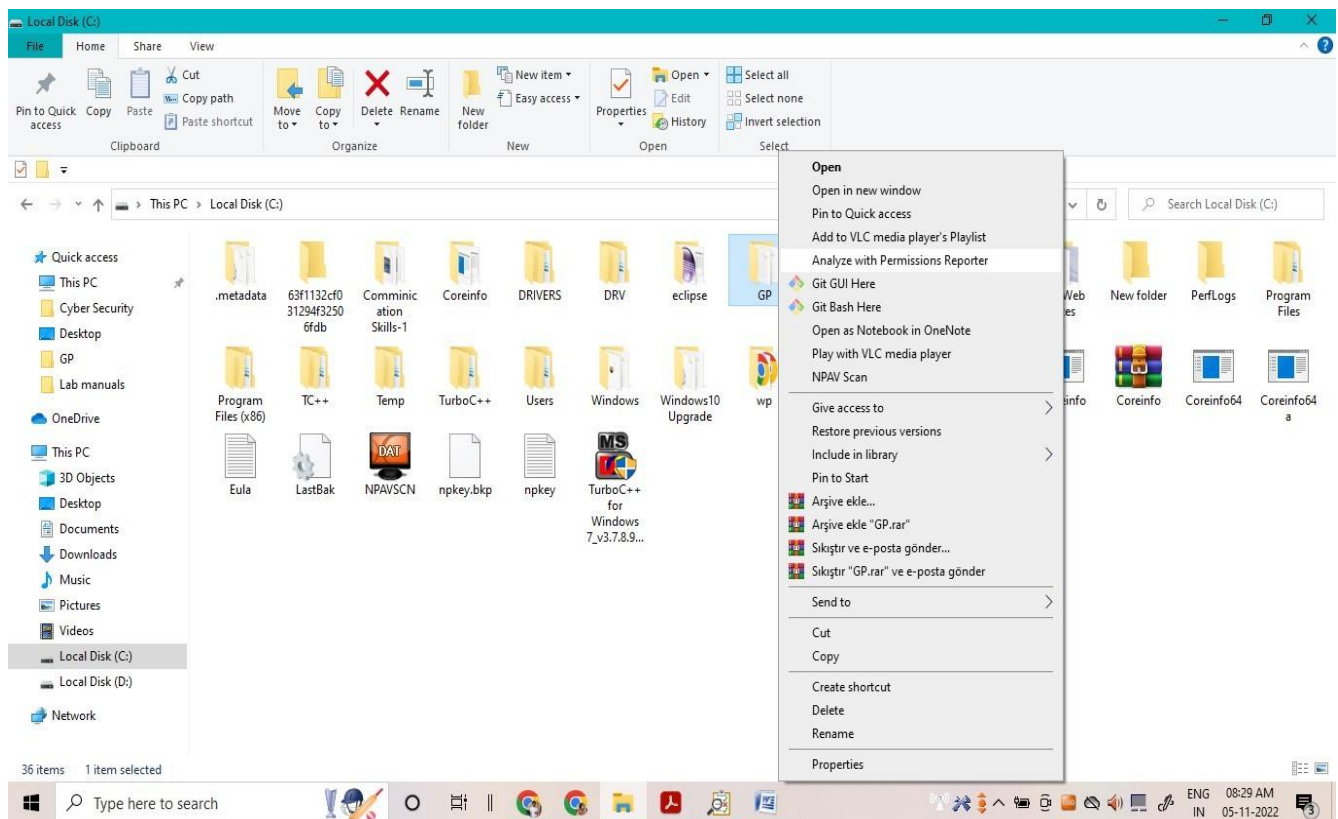
- **Process name** and PID
- **File** location on disk
- **Read rate average** in real-time of the hard drive
- **Write rate average** in real-time of the hard drive
- **Total rate average** of read and write output
- **I/O priority**
- **Response time**

7) NTFS file system practical using NTFS Permissions Reporter

- NTFS Permissions reporting is good ways of auditing the level of access that user have on files and folders so that maintaining NTFS folder security is enhanced.
- Managing folders is a difficult task since it requires constant monitoring of the NTFS permissions to avoid unauthorized access. However, if you have a good tool, you can conveniently present easy-to-read reports listing the permissions granted to a user or a group of users.

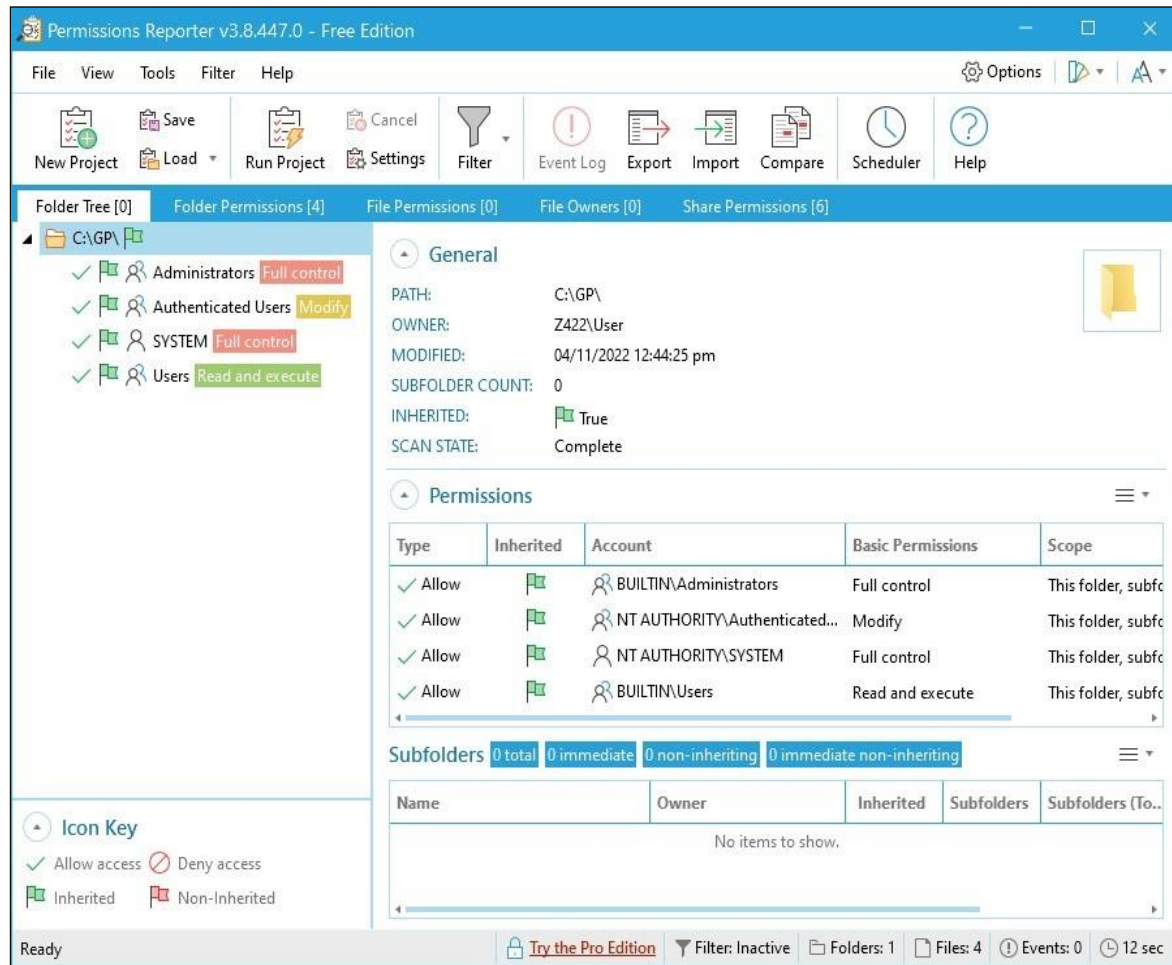
TOOL: NTFS PERMISSIONS REPORTER

- The NTFS Permissions Reporter by is an excellent tool that allows you to export file and folder permissions for further reviewing.
- Once installed, you can right click on any folder in your Windows Explorer and select the “Analyze with Permissions Reporter” option. Thereafter, you’ll be directed to the tool’s main page for you to see the various permissions associated with the folder.

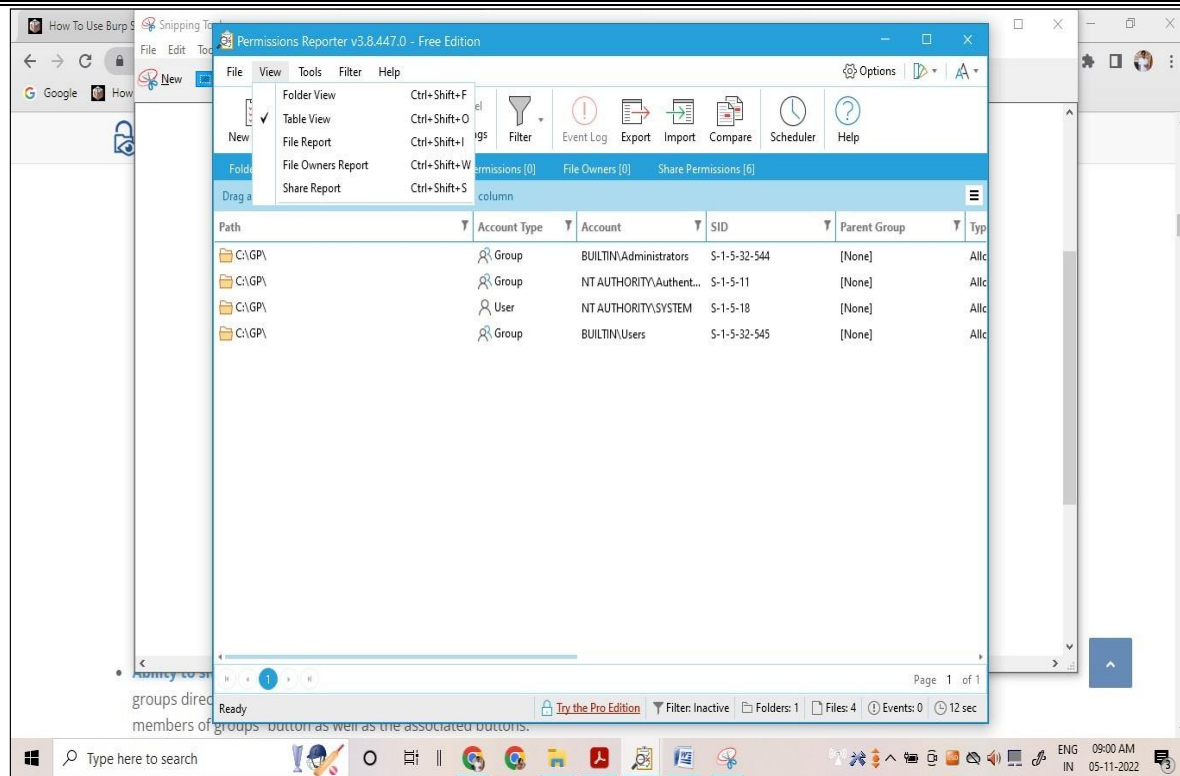


- **Colorized report results**—After selecting the folder you want to view its NTFS

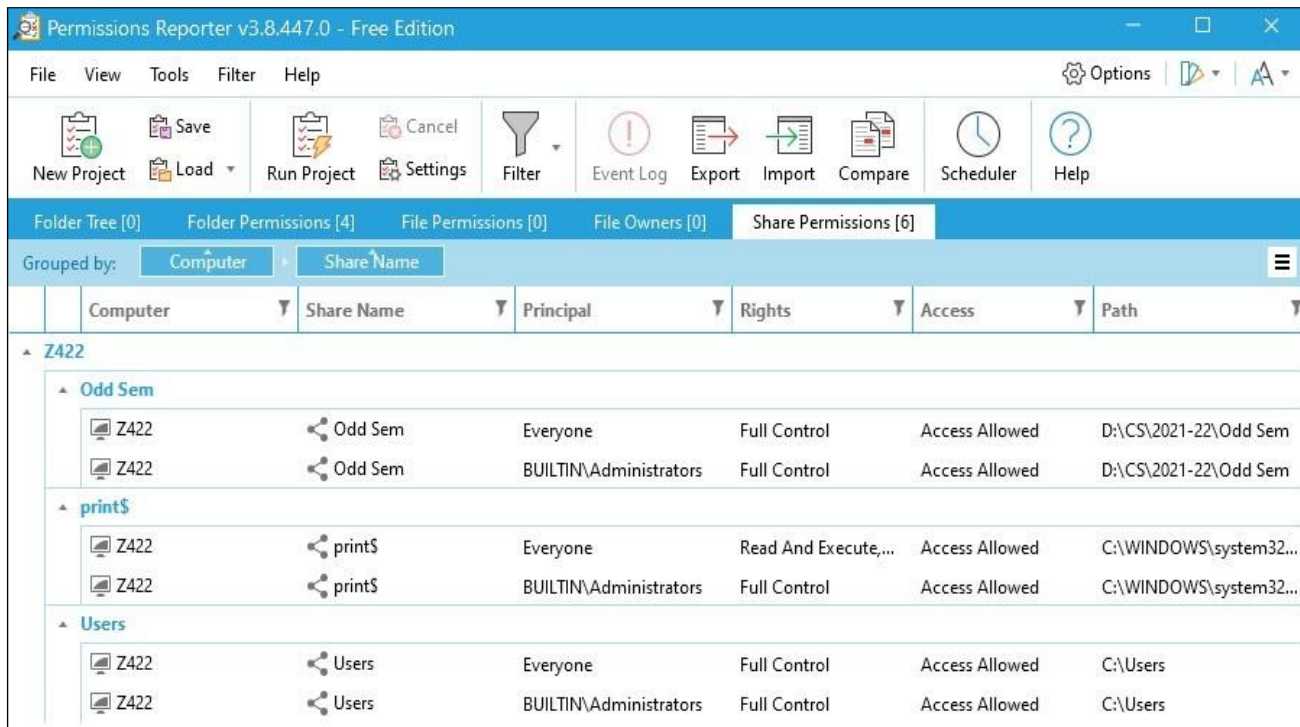
permissions by clicking “Run Project from File menu”, you’ll be presented with a report of the permissions in various colors, allowing you to make a proper analysis. For example, Full Control permission is colored in **red** while Read and Execute permission is colored in **green**.



- **Varied reporting formats**—depending on your preferences, you can choose either the Folder View or the Table View report format on the View option.



- **Share Permissions:** The tool has option that allows you to view share permissions.



- The NTFS Permissions Reporter exports the folder permissions reports as HTML files. Just click the “Export Report” button to export the results.

File Permissions					
Path	Inherited	Owner	Modified	Description	
No items to show.					
File Owners					
Extension	Size	Count			
No items to show.					
Shares Report					
Computer	Share Name	Principal	Rights	Access	Path
Z422	Odd Sem	Everyone	Full Control	Access Allowed	D:\CS\2021-22\Odd Sem
Z422	Odd Sem	BUILTIN\Administrators	Full Control	Access Allowed	D:\CS\2021-22\Odd Sem
Z422	print\$	Everyone	Read And Execute, Synchronize	Access Allowed	C:\WINDOWS\system32\spool\drivers
Z422	print\$	BUILTIN\Administrators	Full Control	Access Allowed	C:\WINDOWS\system32\spool\drivers
Z422	Users	Everyone	Full Control	Access Allowed	C:\Users
Z422	Users	BUILTIN\Administrators	Full Control	Access Allowed	C:\Users
Group Members					
Event Log					
Date	Severity	Message	Path		
No items to show.					

8) PowerShell scripting and automation techniques

POWERSHELL AUTOMATION

Automation with PowerShell is **executing your commands via a script**; those commands you type every day in PowerShell to do something. You can also use your PowerShell scripts and run them with Task Scheduler at desired times of the day.

POWERSHELL TO AUTOMATE SCA AND SAST TOOLS

PREREQUISITES: OWASP DEPENDENCY-TOOL (SCA TOOL), VISUAL CODE GREPPER TOOL (SAST TOOL), JAVA, NX PARSER , JAVA FILE

Step 1 – Prepare A Script To Perform The Scans

#Sample Automation script

```
$OWASPD_path = "C:\dependency-check\bin"
```

```
$VGC_Path = "C:\Program Files (x86)\VisualCodeGrepper"
```

```
Write-Host " "
```

```
Write-Host " "
```

```
Write-Host "=====
```

```
Write-Host "PowerShell to Automate the SCA and SAST Scans"
```

```
Write-Host "=====
```

```
Write-Host " "
```

```
Write-Host ""
```

```
Write-Host "SCA Scan : OWASP Dependency Checker"
```

```
Write-Host "-----"
```

```
$dependency_path = Read-Host "Please enter the path for Dependency checker with pattern"
```

```
Write-Host "Your Dependency path:"
```

```
Write-Host $dependency_path
```

```
Set-Location -Path $OWASPD_path
```

```
Write-Host "The current working directory"
```

```
pwd
```

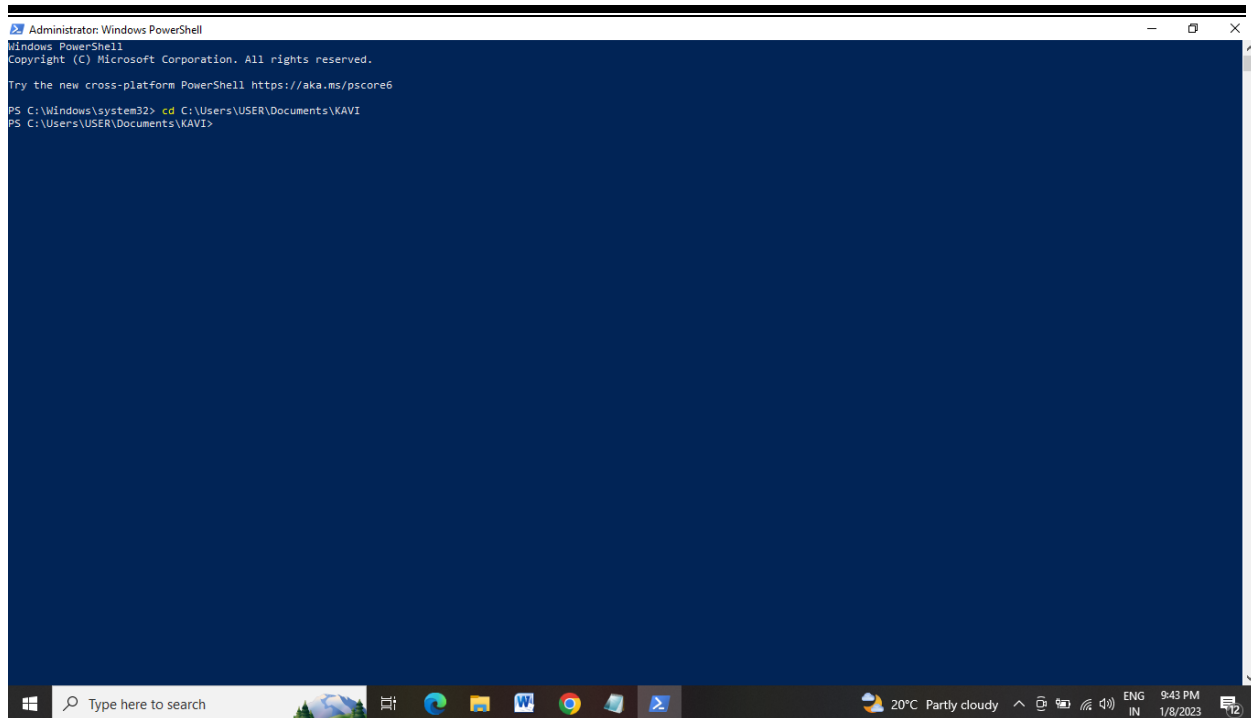
```
dependency-check.bat --scan $dependency_path
```

```
#C:\Users\Administrator\Downloads\Demo\nxparser-master\nxparser-master\**\*.jar
```

```
Write-Host " "
```

```
Write-Host "SAST Scan: Visual Code Grepper "  
Write-Host "-----"  
$VGC_code_path = Read-Host "Please enter the directory path for the Visual Code Grepper"  
$VGC_l = Read-Host "Please enter the programming language [Enter any one from the list:  
CPP, PLSQL, JAVA, CS, VB, PHP, COBOL"  
Write-Host "Your Scanning folder"  
Write-Host $VGC_code_path  
Write-Host "Current Working directory"  
Set-Location -Path $VGC_Path  
pwd  
./Visualcodegrepper.exe -c -l $VGC_l -t $VGC_code_path --results  
C:\Users\Administrator\Downloads\Demo\dvja-master\result.csv  
Write-Host " "  
Write-Host "-----"  
Write-Host "End of Sample Automation script"  
NOTE-CHANGES TO BE MADE  
    • DEPENDENCY-CHECK PATH  
    • VISUAL CODE GREPPER PATH  
    • RESULT TO BE SAVED PATH  
After changing the script save it as extention.ps1
```

STEP 2 –OPEN POWERSHELL and RUN IT AS ADMINISTRATOR



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

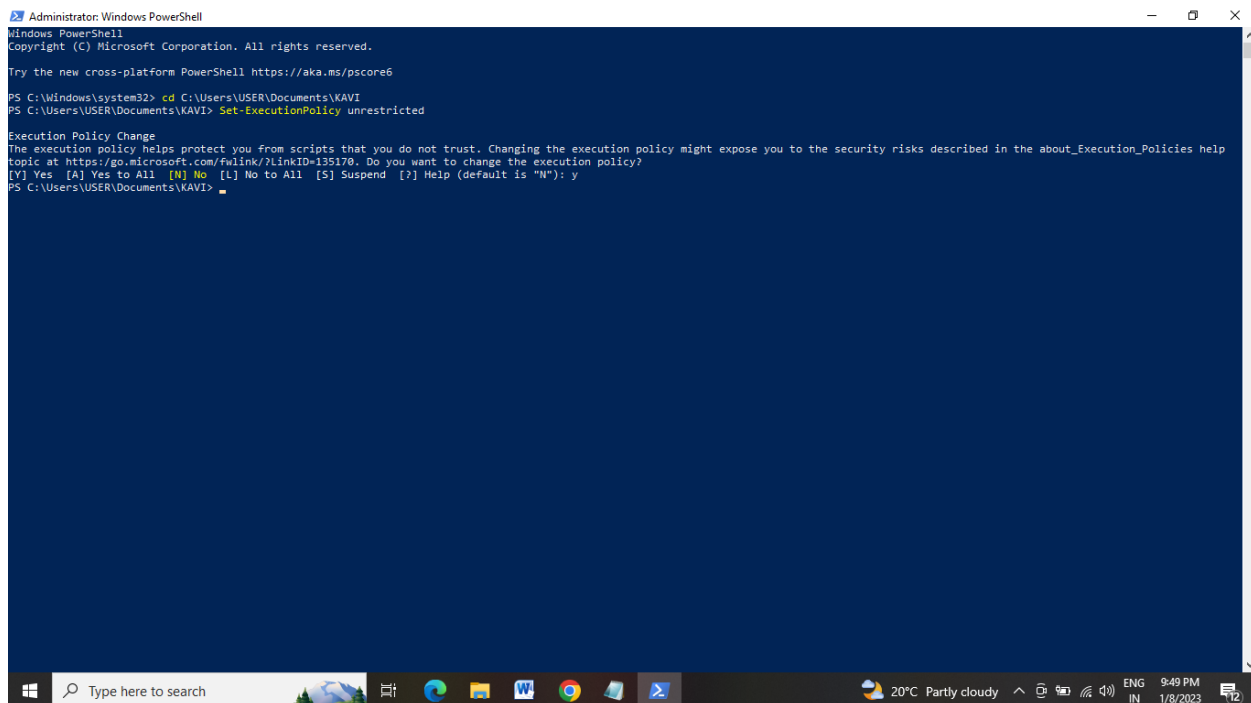
PS C:\Windows\system32> cd C:\Users\USER\Documents\KAVI
PS C:\Users\USER\Documents\KAVI>
```

Perform following commands

Cd C:\Users\USER\Documents\KAVI (path where you have saved your script)

we must enable the execution policy like below command. **Please disable it post your lab, as its vulnerable to keep this kind of policies.**

Set-ExecutionPolicy unrestricted (command to enable execution policy)



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

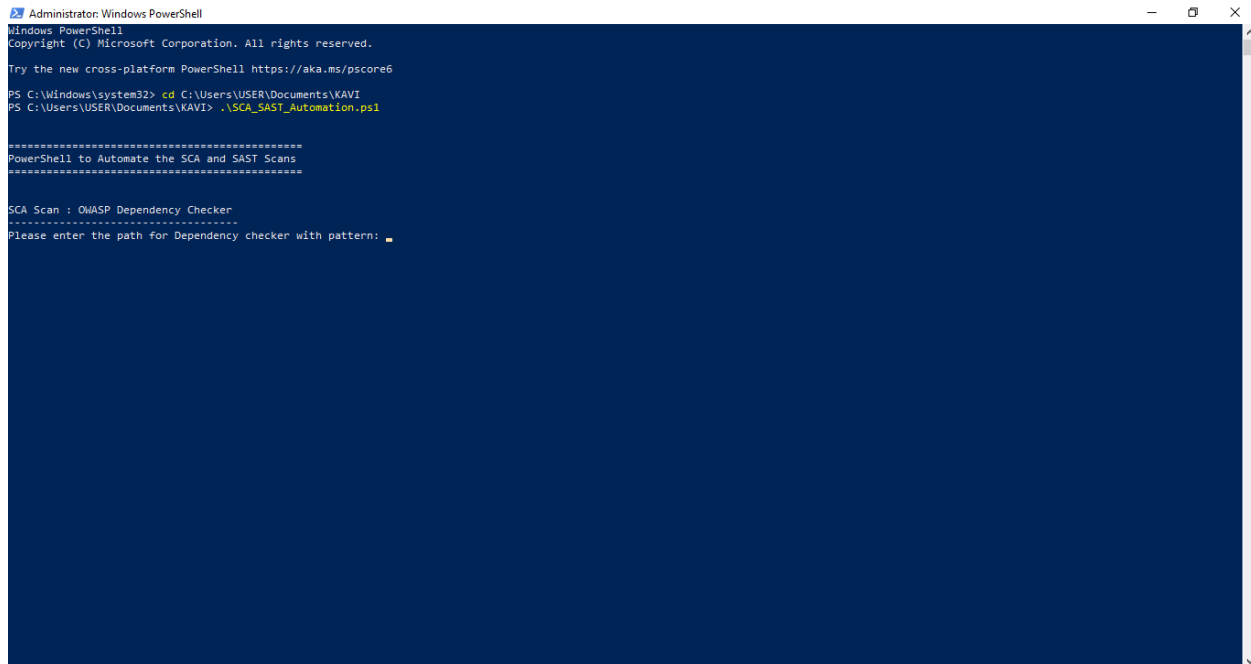
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> cd C:\Users\USER\Documents\KAVI
PS C:\Users\USER\Documents\KAVI> Set-ExecutionPolicy unrestricted

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose you to the security risks described in the about_Execution_Policies help
topic at https://go.microsoft.com/fwlink/?linkID=135170. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): y
PS C:\Users\USER\Documents\KAVI>
```

STEP 3 -Now,execute the following command to start the scan

./SCA_SAST_Automation.ps1 (script title)



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/powershell

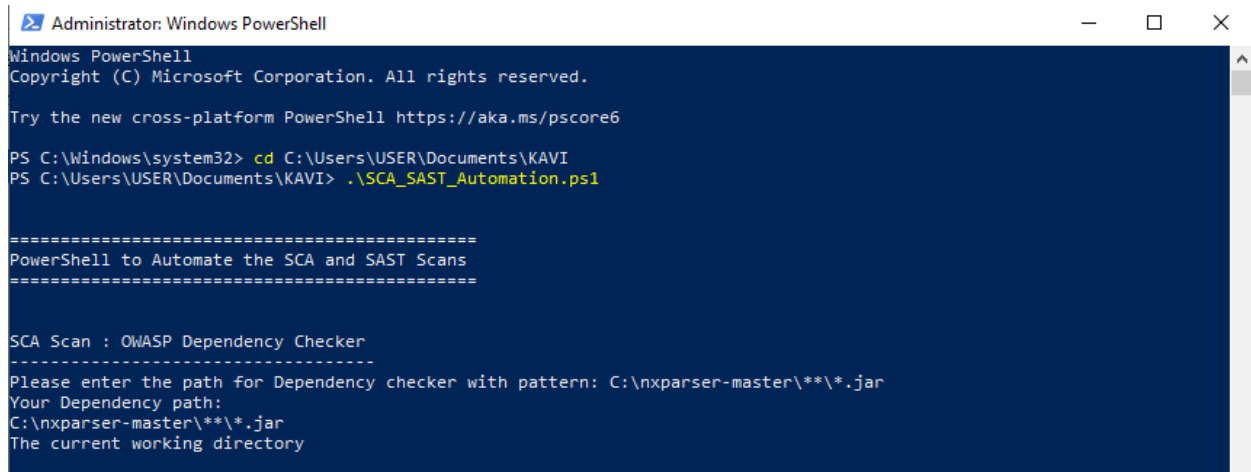
PS C:\Windows\system32> cd C:\Users\USER\Documents\KAVI
PS C:\Users\USER\Documents\KAVI> .\SCA_SAST_Automation.ps1

=====
PowerShell to Automate the SCA and SAST Scans
=====

SCA Scan : OWASP Dependency Checker
-----
Please enter the path for Dependency checker with pattern: 
```

STEP 4 – Now dependency-scan will take place,we must give the path where the nx parser master is located.

\nxparser-master*.jar** (to run dependency scan)



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

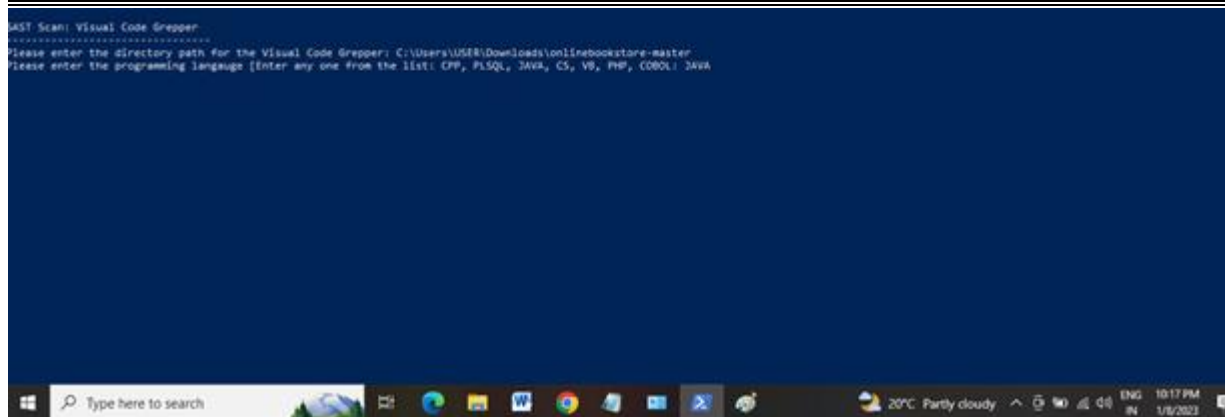
Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Windows\system32> cd C:\Users\USER\Documents\KAVI
PS C:\Users\USER\Documents\KAVI> .\SCA_SAST_Automation.ps1

=====
PowerShell to Automate the SCA and SAST Scans
=====

SCA Scan : OWASP Dependency Checker
-----
Please enter the path for Dependency checker with pattern: C:\nxparser-master\**\*.jar
Your Dependency path:
C:\nxparser-master\**\*.jar
The current working directory
```

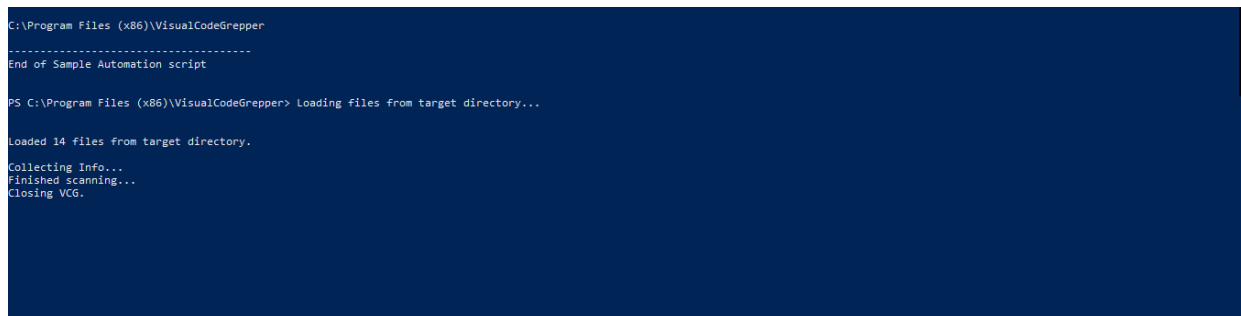
After SCA scanning,



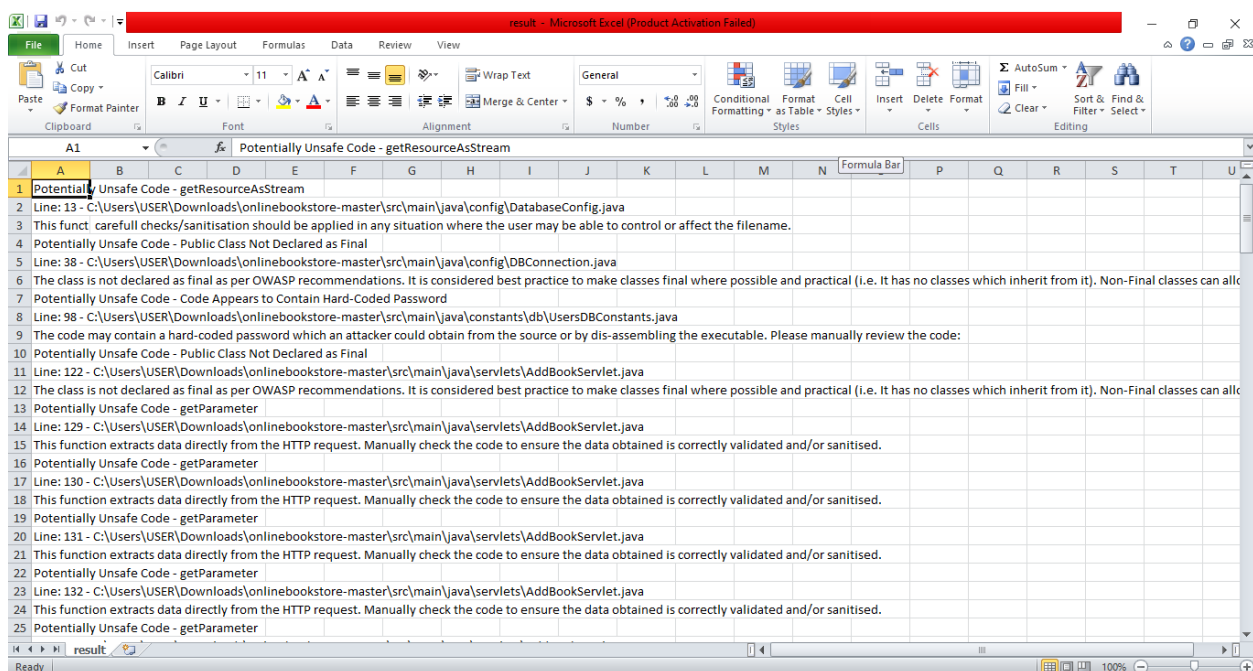
Now For Sast Scan, Copy The Path Where Your Source Code Is Located And Paste It In Powershell.

C:\Users\USER\Downloads\onlinebookstore-master

And Specify The Code Language-JAVA (In This Context) You Can Also Select Php,Cs, Etc.



The result of sast scan will be located at the place where we have specified in the script.



SAST report .

Dependency-Check Report

File | C:/dependency-check/bin/dependency-check-report.html

DEPENDENCY-CHECK

Dependency-Check is an open source tool performing a best effort analysis of 3rd party dependencies; false positives and false negatives may exist in the analysis performed by the tool. Use of the tool and the reporting provided constitutes acceptance for use in an AS IS condition, and there are NO warranties, implied or otherwise, with regard to the analysis or its use. Any use of the tool and the reporting provided is at the user's risk. In no event shall the copyright holder or CNVASP be held liable for any damages whatsoever arising out of or in connection with the use of this tool, the analysis performed, or the resulting report.

[How to read the report](#) | [Suppressing false positives](#) | [Getting Help: github issues](#)

[Sponsor](#)

Project:

Scan Information ([show all](#)):

- dependency-check version: 7.4.4
- Report Generated On: Sun, 8 Jan 2023 20:34:53 +0530
- Dependencies Scanned: 1 (1 unique)
- Vulnerable Dependencies: 1
- Vulnerabilities Found: 7
- Vulnerabilities Suppressed: 0
- ...

Summary

Display: [Showing Vulnerable Dependencies \(click to show all\)](#)

Dependency	Vulnerability IDs	Package	Highest Severity	CVE Count	Confidence	Evidence Count
gradle-wrapper.jar	cpe:2.3:a:gradle:gradle:4.10.2:*:*:*:*:*		CRITICAL	7	High	8

Dependencies

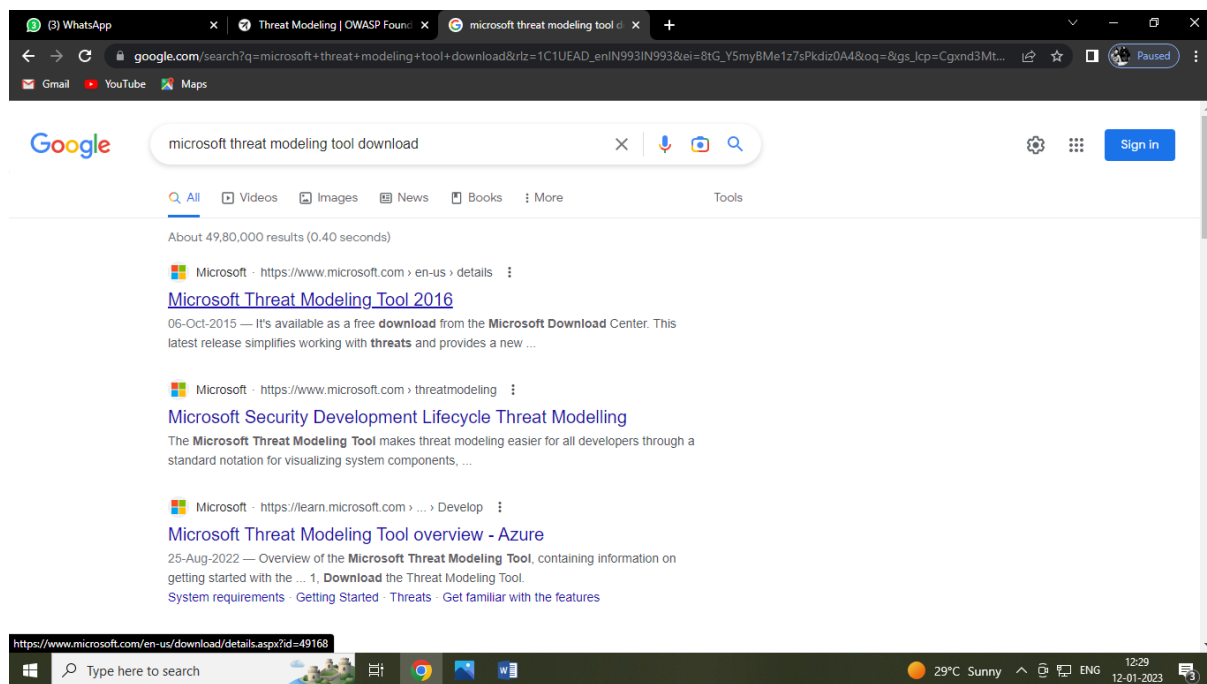
Dependency-check report (it will be saved in dependency-check\bin).

9) Using the Microsoft Threat Modeling methodology, execute a threat model for a given application architecture using Microsoft threat modeling tool.**Threat Modelling:**

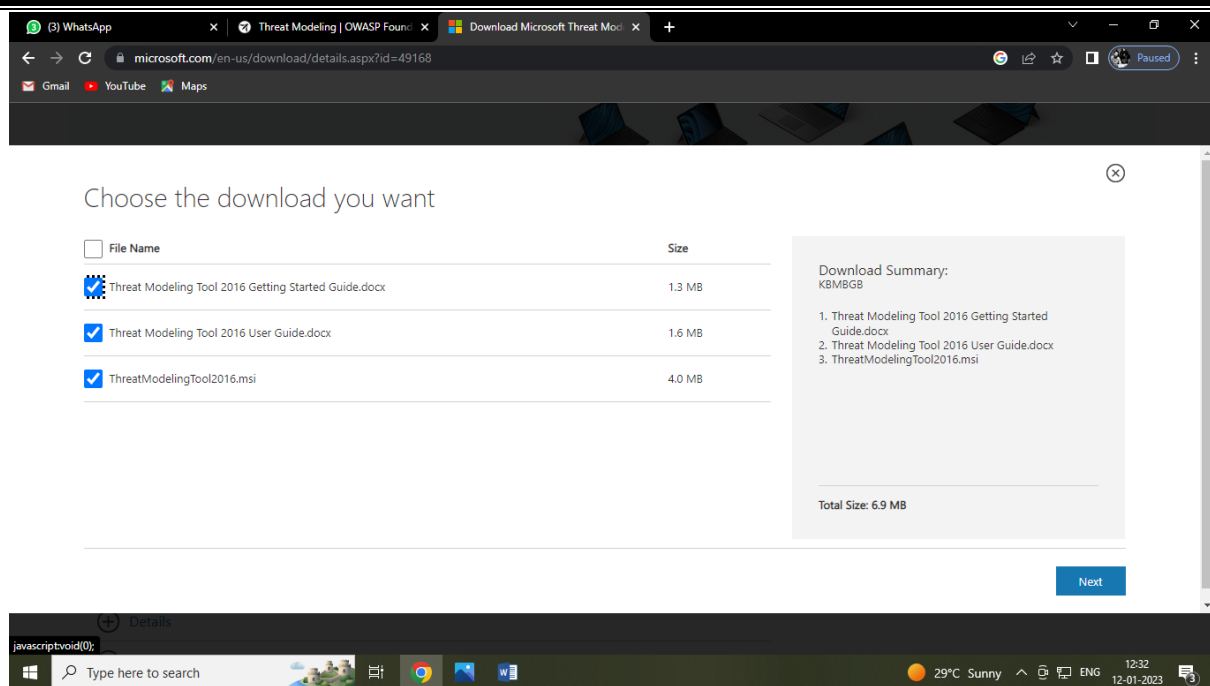
Threat modeling works to identify, communicate, and understand threats and mitigations within the context of protecting something of value. A threat model is a structured representation of all the information that affects the security of an application. In essence, it is a view of the application and its environment through the lens of security. Threat modeling can be applied to a wide range of things, including software, applications, systems, networks, distributed systems, Internet of Things (IoT) devices, and business processes.

STEPS TO PERFORM THREAT MODELLING

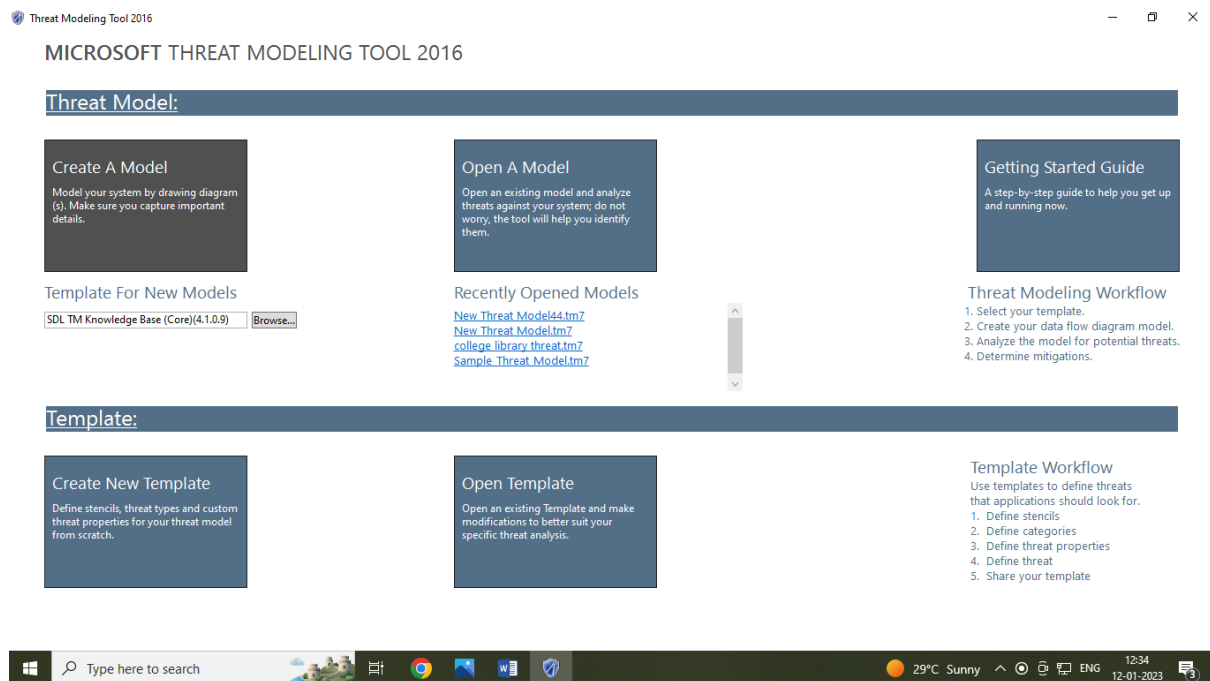
STEP 1 – Go to chrome and download microsoft threat modelling tool 2016.



STEP 2 – Download all the 3 files.

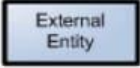


STEP 3 – After installing, open the application and select creat a model option.



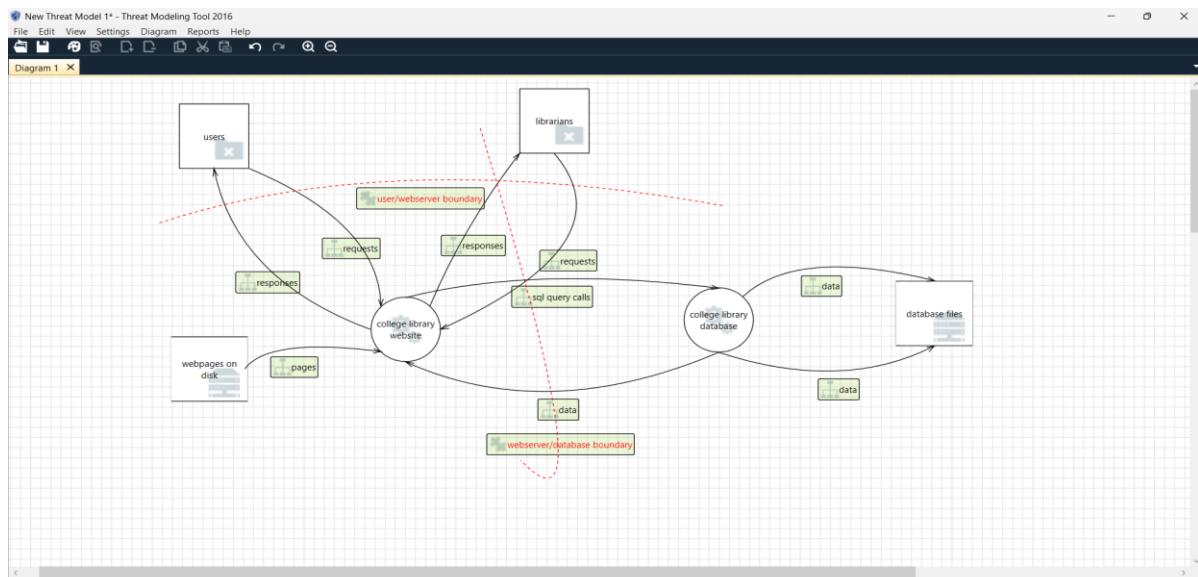
STEP 4 – Create the diagram using the below mentioned stencils.

There are a number of symbols that are used in [DFDs](#) for threat modeling.

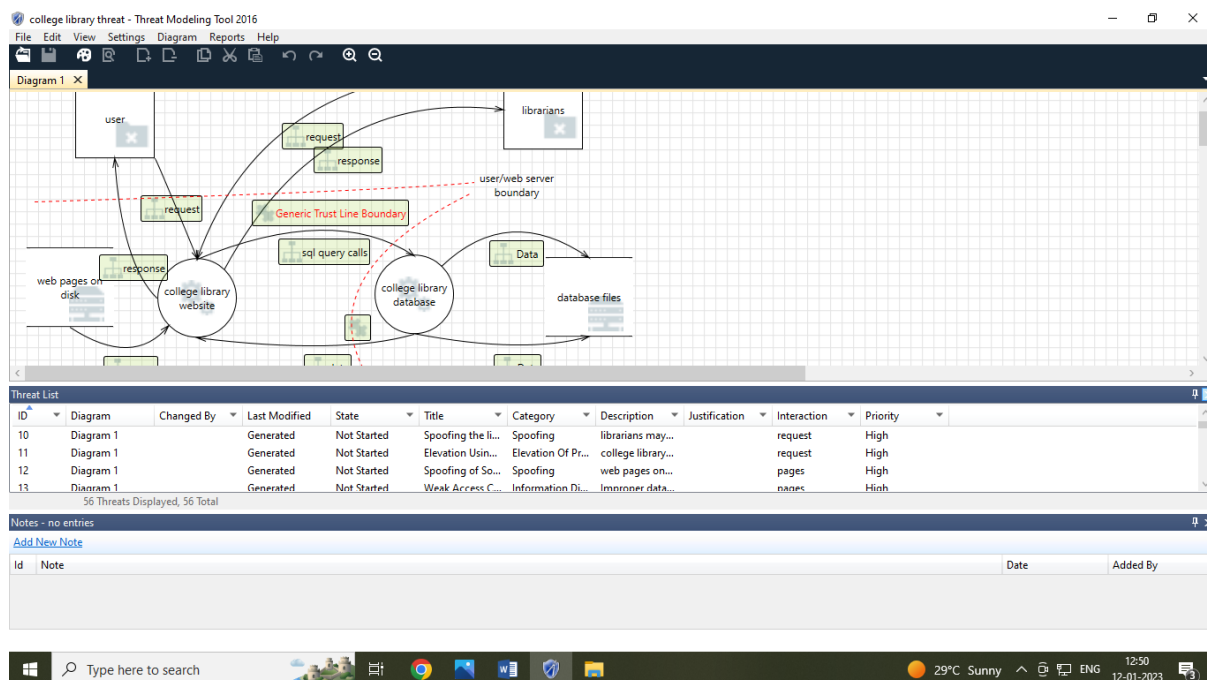
Symbol	Name	Description
	External Entity	The external entity shape is used to represent any entity outside the application that interacts with the application via an entry point.

	Process	The process shape represents a task that handles data within the application. The task may process the data or perform an action based on the data.
	Multiple Process	The multiple process shape is used to present a collection of subprocesses. The multiple process can be broken down into its subprocesses in another DFD.
	Data Store	The data store shape is used to represent locations where data is stored. Data stores do not modify the data, they only store data.
	Data Flow	The data flow shape represents data movement within the application. The direction of the data movement is represented by the arrow.
	Privilege Boundary	The privilege boundary (or trust boundary) shape is used to represent the change of trust levels as the data flows through the application. Boundaries show any location where the level of trust changes.

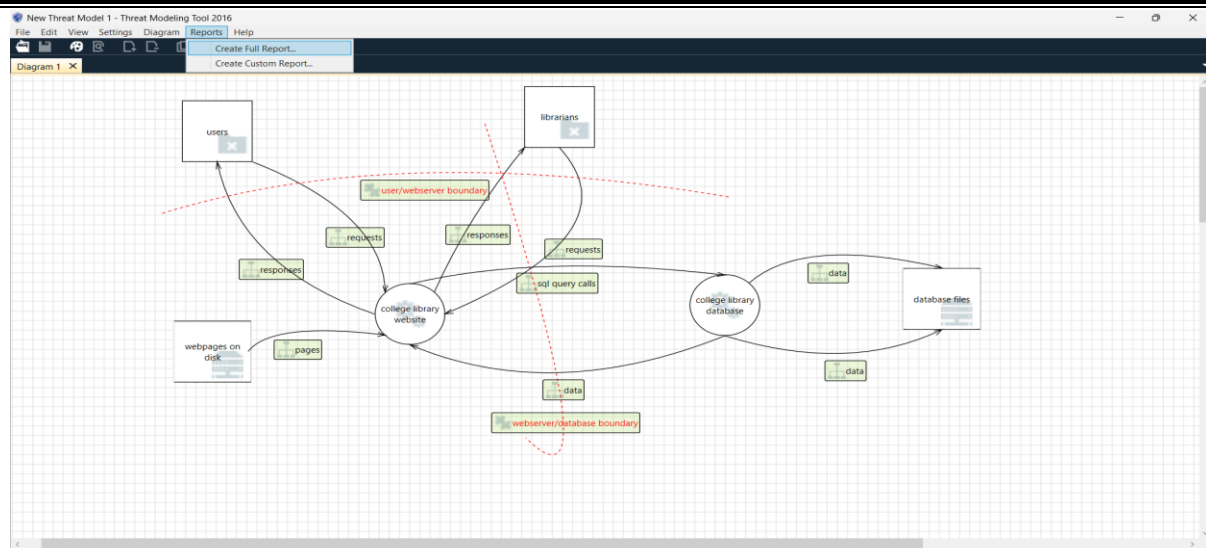
STEP 5 – Create the model as you require



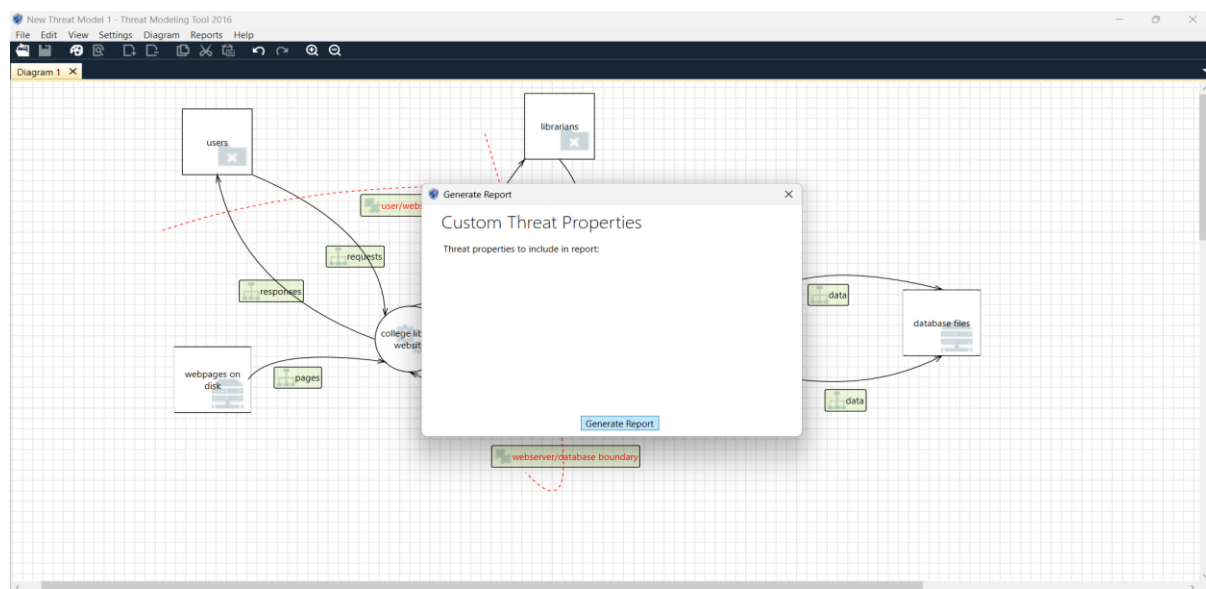
STEP 6 – Go to view>analysis view



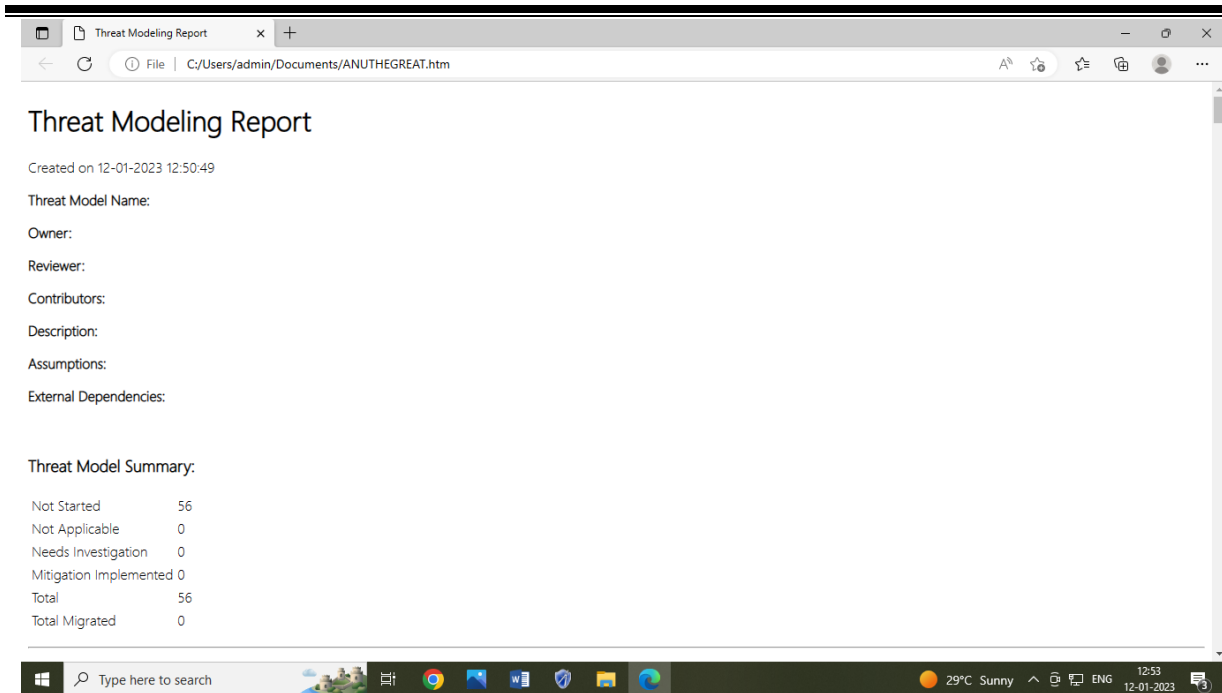
STEP 6 – Next generate a report, click on report > create full report.



Step 7 – Click on generate report.



The report will be generated.



Threat Modeling Report

Created on 12-01-2023 12:50:49

Threat Model Name:

Owner:

Reviewer:

Contributors:

Description:

Assumptions:

External Dependencies:

Threat Model Summary:

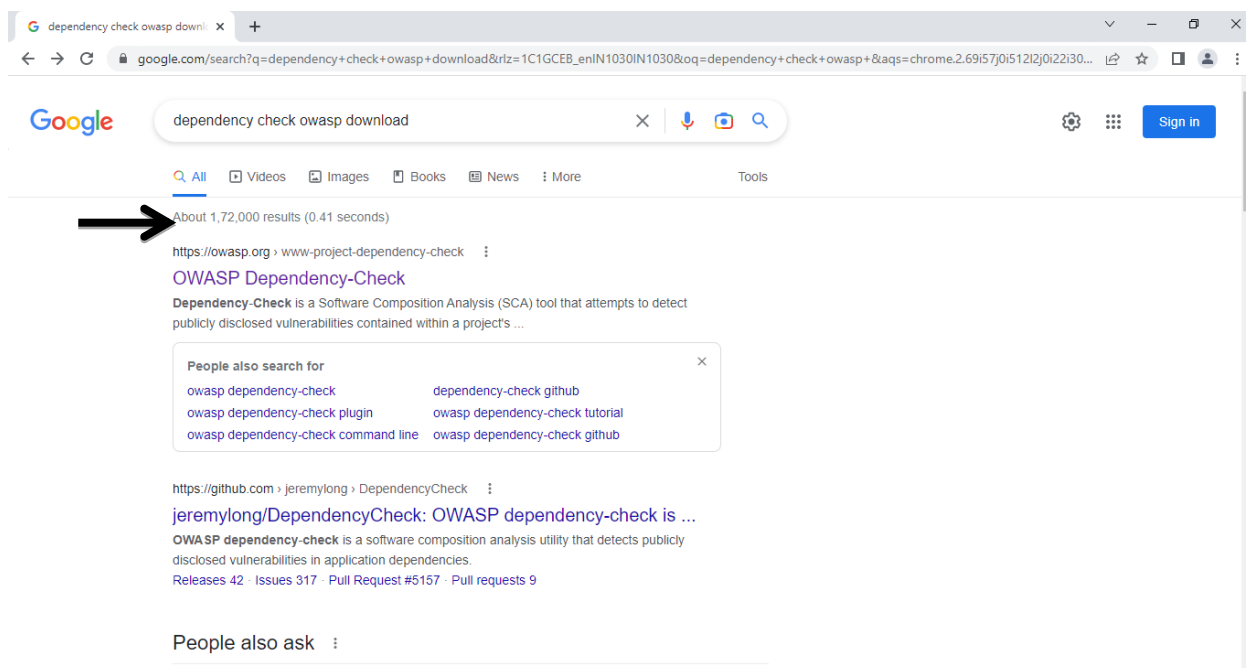
Not Started	56
Not Applicable	0
Needs Investigation	0
Mitigation Implemented	0
Total	56
Total Migrated	0

10) Demonstrate a tool like OWASP Dependency Check.**OWASP Dependency-Check**

Dependency-Check is a Software Composition Analysis (SCA) tool that attempts to detect publicly disclosed vulnerabilities contained within a project's dependencies. It does this by determining if there is a Common Platform Enumeration (CPE) identifier for a given dependency. If found it will generate a report linking to the associated CVE entries.

PREREQUISITES – Java version 7+.

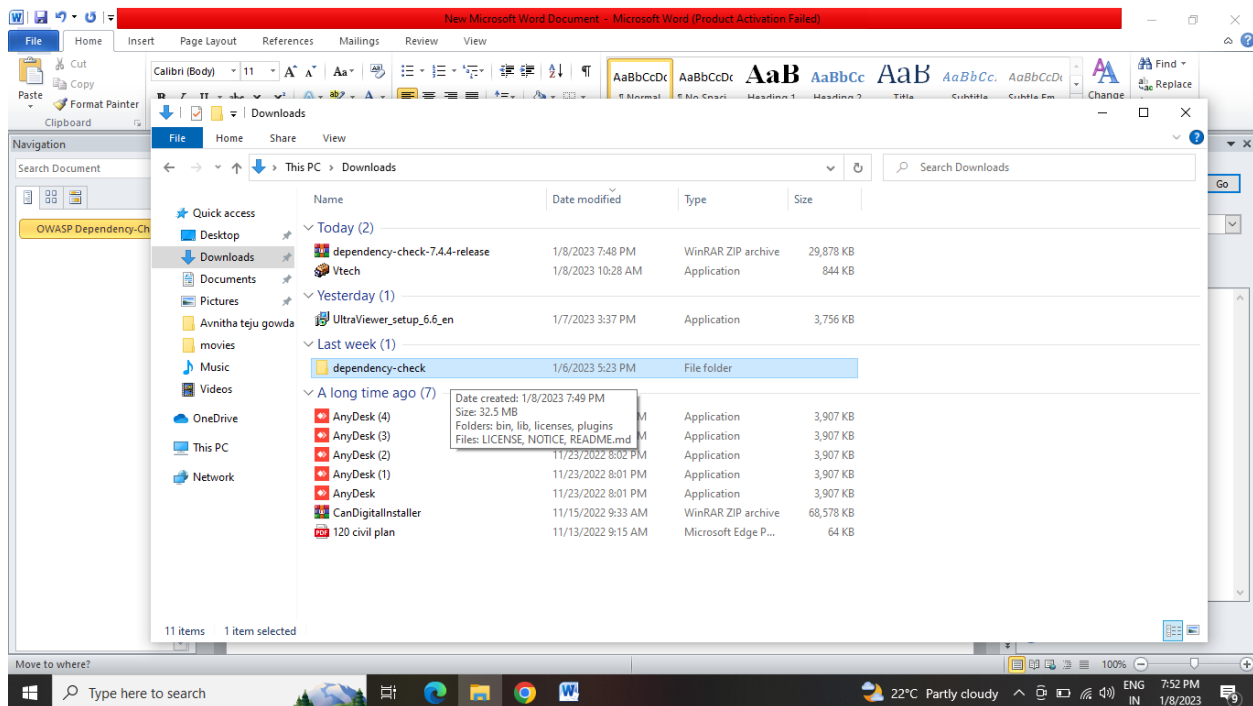
Step 1 – Go to chrome and search for OWASP DEPENDENCY CHECK TOOL.



Step 2 – In the downloads section, select COMMAND LINE option.



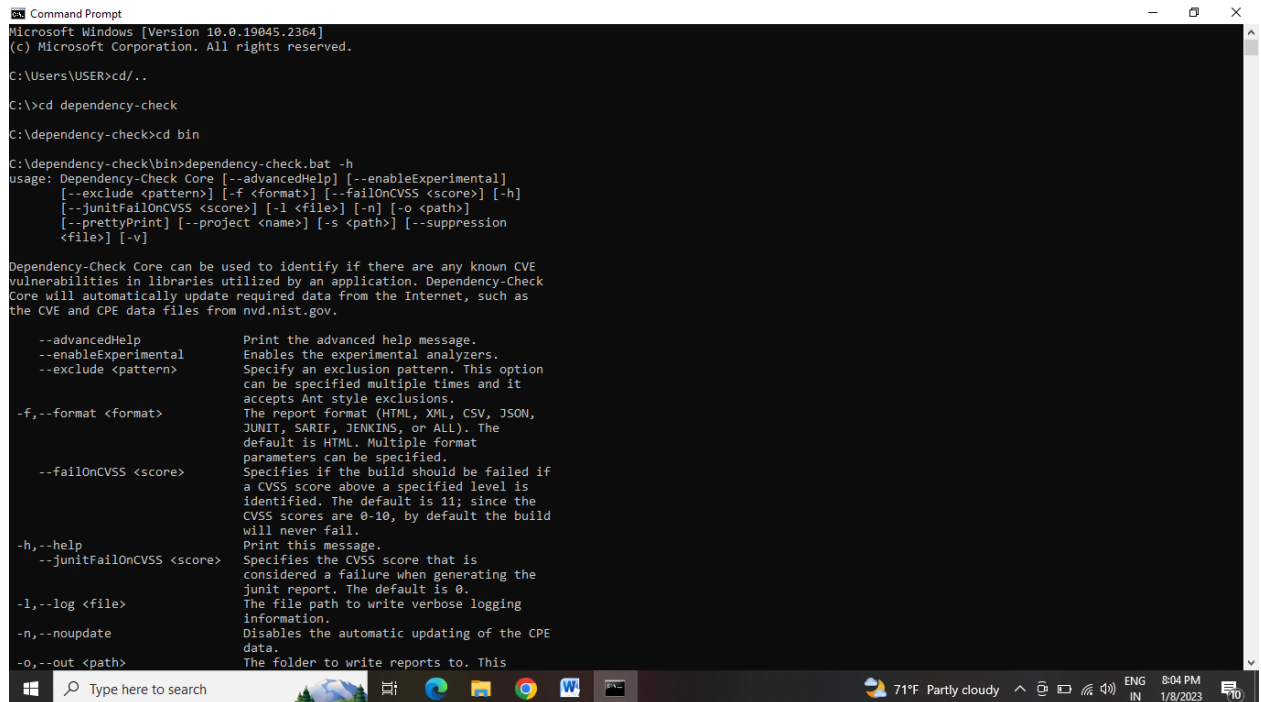
Step 3 – Dependency-check tool will be downloaded in the ZAR format, extract that.



Step 4 – After extracting, place the folder in Local Disk C.

Step 5 – Open command prompt, and execute following commands

- Cd/.. (to go back to local disk c).
- Cd dependency-check (change directory to dependency check).
- Cd bin (inside dependency check, change to bin).



```
Microsoft Windows [Version 10.0.19045.2364]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd/..

C:\>cd dependency-check

C:\dependency-check>cd bin

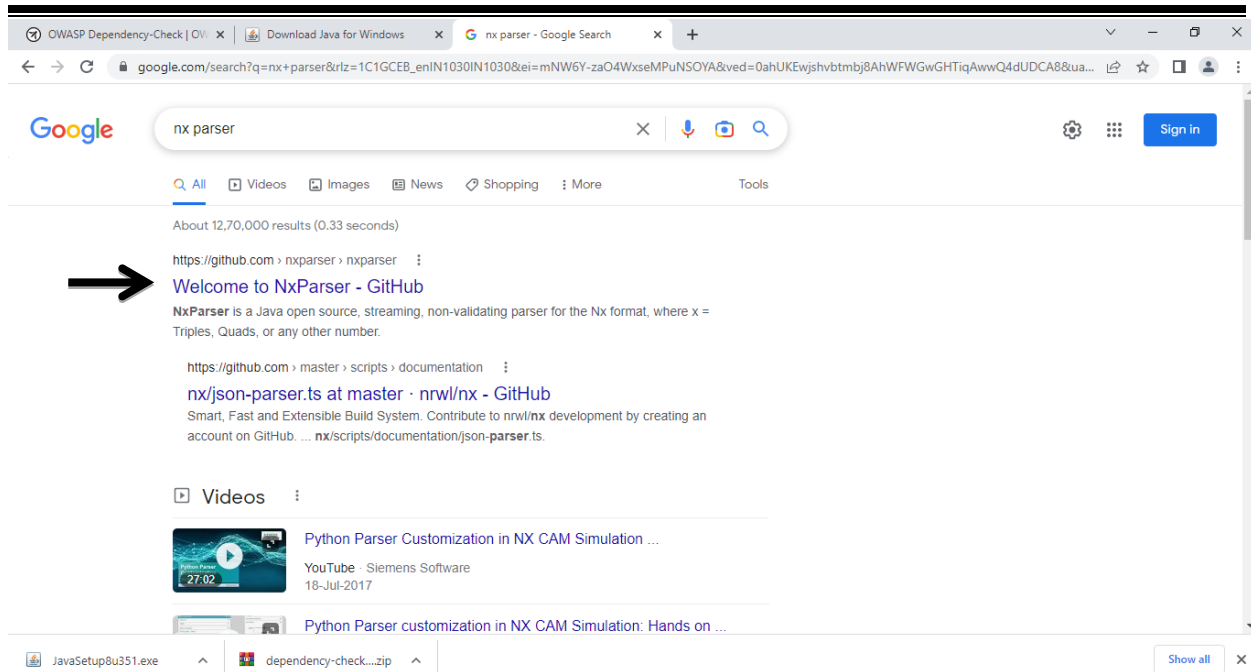
C:\dependency-check\bin>dependency-check.bat -h
usage: Dependency-Check Core [-h] [--advancedHelp] [--enableExperimental]
                             [--exclude <pattern>] [-f <format>] [--failOnCVSS <score>] [-h]
                             [--junitFailOnCVSS <score>] [-l <file>] [-n] [-o <path>]
                             [--prettyPrint] [--project <name>] [-s <path>] [--suppression
                             <file>] [-v]

Dependency-Check Core can be used to identify if there are any known CVE
vulnerabilities in libraries utilized by an application. Dependency-Check
Core will automatically update required data from the Internet, such as
the CVE and CPE data files from nvd.nist.gov.

--advancedHelp      Print the advanced help message.
--enableExperimental Enables the experimental analyzers.
--exclude <pattern> Specify an exclusion pattern. This option
                    can be specified multiple times and it
                    accepts Ant style exclusions.
-f,--format <format> The report format (HTML, XML, CSV, JSON,
                    JUNIT, SARIF, JENKINS, or ALL). The
                    default is HTML. Multiple format
                    parameters can be specified.
--failOnCVSS <score> Specifies if the build should be failed if
                    a CVSS score above a specified level is
                    identified. The default is 11; since the
                    CVSS scores are 0-10, by default the build
                    will never fail.
-h,--help           Print this message.
--junitFailOnCVSS <score> Specifies the CVSS score that is
                    considered a failure when generating the
                    junit report. The default is 0.
-l,--log <file>     The file path to write verbose logging
                    information.
-n,--noupdate       Disables the automatic updating of the CPE
                    data.
-o,--out <path>     The folder to write reports to. This
```

To check the installation of tool – dependency-check.bat -h

Step6 – DOWNLOAD NX-parser to scan the dependency.



After downloading nx parser, extract that and paste it to local disk C.

Step7 – run the following command to perform dependency scan

- Dependency-check.bat --scan \nx parser-master***.jar

```

Command Prompt - dependency-check.bat --scan \nxparser-master\**\*.jar

C:\dependency-check\bin>dependency-check.bat --scan \nxparser-master\**\*.jar
[INFO] Checking for updates
[INFO] NVD CVE requires several updates; this could take a couple of minutes.
[INFO] Download Started for NVD CVE - 2002
[INFO] Download Complete for NVD CVE - 2002 (2569 ms)
[INFO] Processing Started for NVD CVE - 2002
[INFO] Download Started for NVD CVE - 2003
[INFO] Download Complete for NVD CVE - 2003 (2098 ms)
[INFO] Processing Started for NVD CVE - 2003
[INFO] Download Started for NVD CVE - 2004
[INFO] Processing Complete for NVD CVE - 2003 (5616 ms)
[INFO] Download Complete for NVD CVE - 2004 (2471 ms)
[INFO] Processing Started for NVD CVE - 2004
[INFO] Download Started for NVD CVE - 2005
[INFO] Processing Complete for NVD CVE - 2002 (18915 ms)
[INFO] Processing Complete for NVD CVE - 2004 (10129 ms)
[INFO] Download Complete for NVD CVE - 2005 (9107 ms)
[INFO] Processing Started for NVD CVE - 2005
[INFO] Download Started for NVD CVE - 2006
[INFO] Download Complete for NVD CVE - 2006 (8360 ms)
[INFO] Processing Started for NVD CVE - 2006
[INFO] Processing Complete for NVD CVE - 2005 (15355 ms)
[INFO] Download Started for NVD CVE - 2007
[INFO] Download Complete for NVD CVE - 2007 (6529 ms)
[INFO] Processing Started for NVD CVE - 2007
[INFO] Download Started for NVD CVE - 2008
[INFO] Download Complete for NVD CVE - 2008 (6311 ms)
[INFO] Download Started for NVD CVE - 2009
[INFO] Processing Complete for NVD CVE - 2006 (29756 ms)
[INFO] Processing Started for NVD CVE - 2008
[INFO] Download Complete for NVD CVE - 2009 (6561 ms)
[INFO] Download Started for NVD CVE - 2010
[INFO] Processing Complete for NVD CVE - 2007 (27445 ms)
[INFO] Processing Started for NVD CVE - 2009
[INFO] Download Complete for NVD CVE - 2010 (5982 ms)
[INFO] Download Started for NVD CVE - 2011
[INFO] Download Complete for NVD CVE - 2011 (6154 ms)
[INFO] Download Started for NVD CVE - 2012
[INFO] Download Complete for NVD CVE - 2012 (6435 ms)
[INFO] Download Started for NVD CVE - 2013
[INFO] Processing Complete for NVD CVE - 2008 (38706 ms)
[INFO] Processing Started for NVD CVE - 2010
[INFO] Download Complete for NVD CVE - 2013 (7388 ms)

```

Step 8 – After performing scan , the report will be generated and stored in dependency-check folder automatically.

```

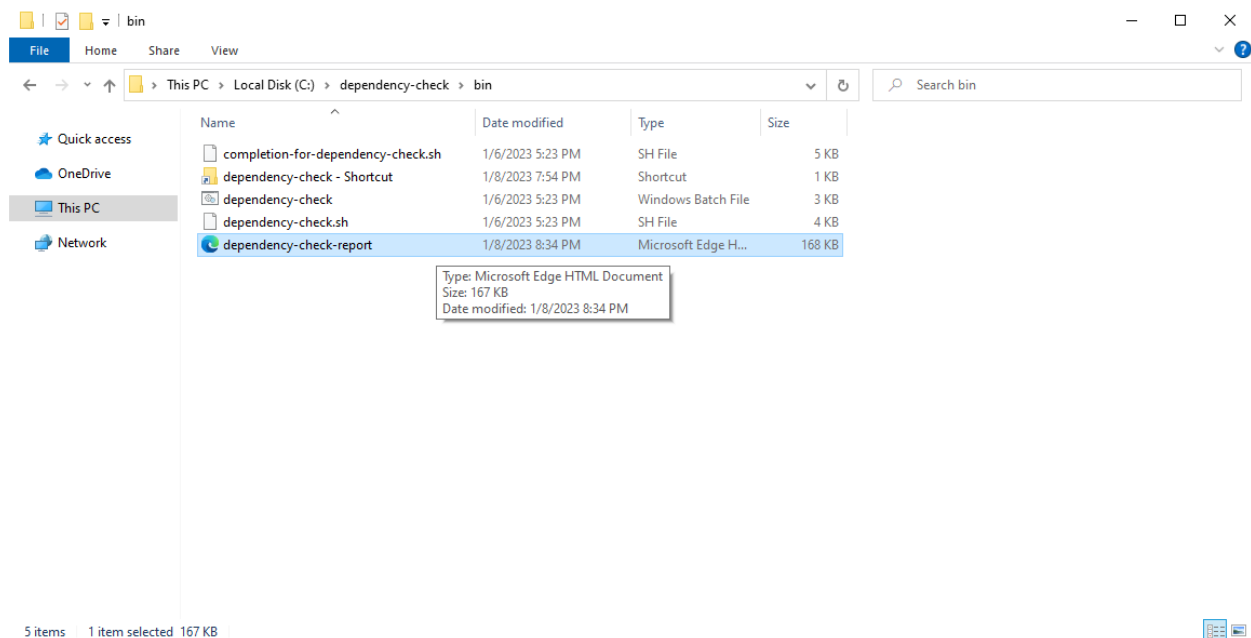
C:\dependency-check\bin>
C:\dependency-check\bin>
C:\dependency-check\bin>
C:\dependency-check\bin>
C:\dependency-check\bin>dependency-check.bat --scan \nparser-master\*\*.jar
[INFO] Checking for updates
[INFO] Skipping NVD check since last check was within 4 hours.
[INFO] Skipping RetireJS update since last update was within 24 hours.
[INFO] Check for updates complete (31 ms)
[INFO]

Dependency-Check is an open source tool performing a best effort analysis of 3rd party dependencies; false positives and false negatives may exist in the analysis performed by the tool. Use of the tool and the reporting provided constitutes acceptance for use in an AS IS condition, and there are NO warranties, implied or otherwise, with regard to the analysis or its use. Any use of the tool and the reporting provided is at the user's risk. In no event shall the copyright holder or OWASP be held liable for any damages whatsoever arising out of or in connection with the use of this tool, the analysis performed, or the resulting report.

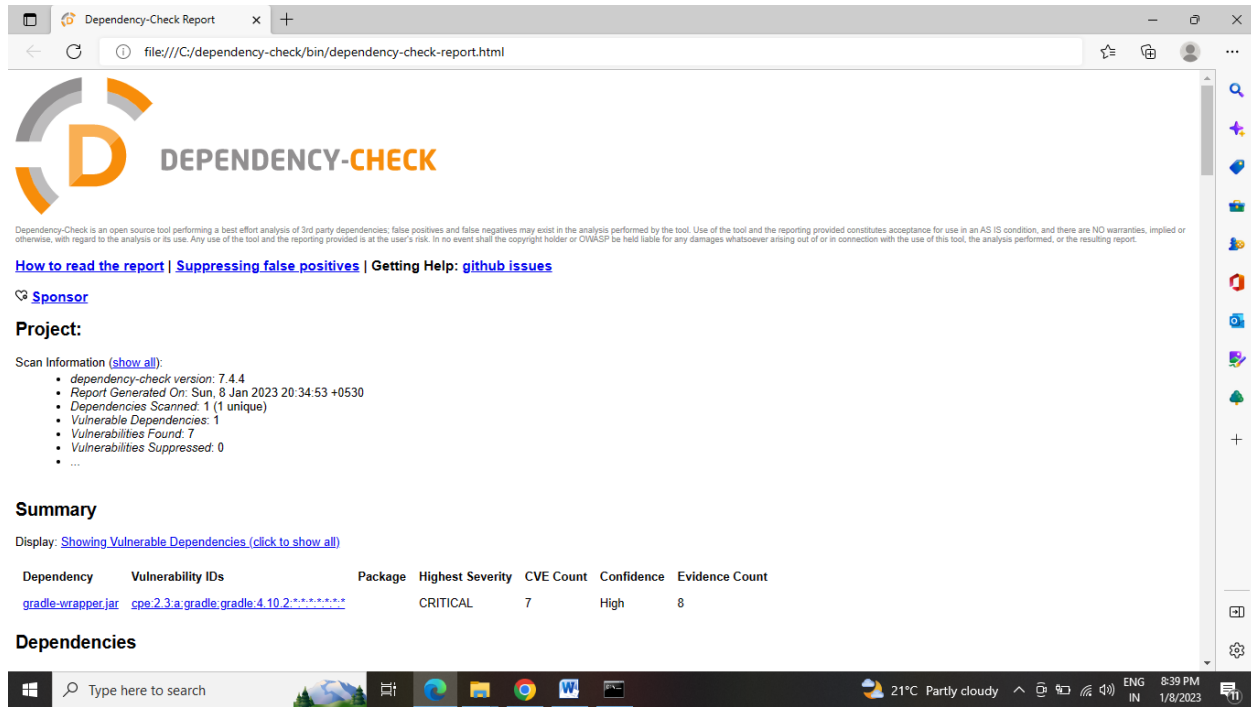
About ODC: https://jeremylong.github.io/DependencyCheck/general/internals.html
False Positives: https://jeremylong.github.io/DependencyCheck/general/suppression.html
? Sponsor: https://github.com/sponsors/jeremylong

[INFO] Analysis Started
[INFO] Finished Archive Analyzer (0 seconds)
[INFO] Finished File Name Analyzer (0 seconds)
[INFO] Finished Jar Analyzer (0 seconds)
[INFO] Finished Central Analyzer (2 seconds)
[INFO] Finished Dependency Merging Analyzer (0 seconds)
[INFO] Finished Version Filter Analyzer (0 seconds)
[INFO] Finished Hint Analyzer (0 seconds)
[INFO] Created CPE Index (5 seconds)
[INFO] Finished CPE Analyzer (5 seconds)
[INFO] Finished False Positive Analyzer (0 seconds)
[INFO] Finished NVD CVE Analyzer (0 seconds)
[WARN] Unable to determine Package-URL identifiers for 1 dependencies
[INFO] Finished Sonatype OSS Index Analyzer (0 seconds)
[INFO] Finished Vulnerability Suppression Analyzer (0 seconds)
[INFO] Finished Dependency Bundling Analyzer (0 seconds)
[INFO] Finished Unused Suppression Rule Analyzer (0 seconds)
[INFO] Analysis Complete (8 seconds)
[INFO] Writing report to: C:\dependency-check\bin\.\dependency-check-report.html
C:\dependency-check\bin>
  
```

You can view the report here.



<file:///C:/dependency-check/bin/dependency-check-report.html>



Dependency-Check Report

file:///C:/dependency-check/bin/dependency-check-report.html

DEPENDENCY-CHECK

Dependency-Check is an open source tool performing a best effort analysis of 3rd party dependencies; false positives and false negatives may exist in the analysis performed by the tool. Use of the tool and the reporting provided constitutes acceptance for use in an AS IS condition, and there are NO warranties, implied or otherwise, with regard to the analysis or its use. Any use of the tool and the reporting provided is at the user's risk. In no event shall the copyright holder or CNVASP be held liable for any damages whatsoever arising out of or in connection with the use of this tool, the analysis performed, or the resulting report.

[How to read the report](#) | [Suppressing false positives](#) | [Getting Help: github issues](#)

[Sponsor](#)

Project:

Scan Information ([show all](#)):

- dependency-check version: 7.4.4
- Report Generated On: Sun, 8 Jan 2023 20:34:53 +0530
- Dependencies Scanned: 1 (1 unique)
- Vulnerable Dependencies: 1
- Vulnerabilities Found: 7
- Vulnerabilities Suppressed: 0
- ...

Summary

Display: [Showing Vulnerable Dependencies \(click to show all\)](#)

Dependency	Vulnerability IDs	Package	Highest Severity	CVE Count	Confidence	Evidence Count
gradle-wrapper.jar	cpe:2.3:a:gradle:gradle:4.10.2:*:*:*:*		CRITICAL	7	High	8

Dependencies

21°C Partly cloudy 8:39 PM 1/8/2023

gradle-wrapper.jar

File Path: C:\nxparser-master\gradle\wrapper\gradle-wrapper.jar
 MD5: 4dc3e0f882e58aa20422b7f5b85336
 SHA1: 5f084ee091f052df54bac48c3411bc7dc570840
 SHA256: ad63ba21fb91e490e0f6d0ca7d4049241f0f68a454b0b3075c041c4554e611c

Evidence

Identifiers

- cpe:2.3:a:gradle:gradle:4.10.2:*:*:*:* (Confidence: High) [\[suppress\]](#)

Published Vulnerabilities

[CVE-2019-15052](#) [\[suppress\]](#)

The HTTP client in Gradle before 5.6 sends authentication credentials originally destined for the configured host. If that host returns a 30x redirect, Gradle also sends those credentials to all subsequent hosts that the request redirects to. This is similar to CVE-2018-1000007.

CWE-522 Insufficiently Protected Credentials

CVSSv2:

- Base Score: MEDIUM (5.0)
- Vector: /AV:N/AC:L/Au:N/C:P/I:N/A:N

CVSSv3:

- Base Score: CRITICAL (9.8)
- Vector: CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H

References:

- MISC - <https://github.com/gradle/gradle/issues/19278>
- MISC - <https://github.com/gradle/gradle/pull/10176>
- MISC - <https://github.com/gradle/gradle/security/advisories/GHSA-4cwq-17qc-6r95>

Vulnerable Software & Versions:

- cpe:2.3:a:gradle:gradle:*:*:*:*:* versions up to (excluding) 5.6

[CVE-2021-29428](#) [\[suppress\]](#)

In Gradle before version 7.0, on Unix-like systems, the system temporary directory can be created with open permissions that allow multiple users to create and delete files within it. Gradle builds could be vulnerable to a local privilege escalation from an attacker quickly deleting and recreating files in the system temporary directory. This vulnerability impacted builds using precompiled script plugins written in Kotlin DSL and tests for Gradle plugins written using ProjectBuilder or TestKit. If you are on Windows or modern versions of macOS, you are not vulnerable. If you are on a Unix-like operating system with the "sticky" bit set on your system temporary directory, you are not vulnerable. The problem has been patched and released with Gradle 7.0. As a workaround, on Unix-like operating systems, ensure that the "sticky" bit is set. This only allows the original user (or root) to delete a file. If you are unable to change the permissions of the system temporary directory, you can move the Java temporary directory by setting the System Property "java.io.tmpdir". The new path needs to limit permissions to the build user only. For additional details refer to the referenced GitHub Security Advisory.

CWE-378 Creation of Temporary File With Insecure Permissions, CWE-379 Creation of Temporary File in Directory with Incorrect Permissions

CVSSv2:

- Base Score: MEDIUM (4.4)
- Vector: /AV:L/AC:M/Au:N/C:P/I:P/A:P

CVSSv3:

- Base Score: HIGH (7.8)
- Vector: CVSS:3.1/AV:L/AC:M/PR:L/UI:N/S:U/C:H/I:H/A:H

References:

- CONFIRM - <https://github.com/gradle/gradle/security/advisories/GHSA-89gm-pxvm-p336>
- MISC - <https://docs.gradle.org/7.0/release-notes.html#security-advisories>
- MISC - <https://github.com/gradle/gradle/pull/15240>
- MISC - <https://github.com/gradle/gradle/pull/15654>

Vulnerable Software & Versions: [\(show all\)](#)

- cpe:2.3:a:gradle:gradle:*:*:*:*:* versions up to (excluding) 7.0
-

[CVE-2020-11979](#) [\[suppress\]](#)

As mitigation for CVE-2020-1945 Apache Ant 1.10.8 changed the permissions of temporary files it created so that only the current user was allowed to access them. Unfortunately the fixcrlf task deleted the temporary file and created a new one without said protection, effectively nullifying the effort. This would still allow an attacker to inject modified source files into the build process.

NVD-CWE-Other

CVSSv2:

- Base Score: MEDIUM (5.0)

Dependency-Check Report

file:///C:/dependency-check/bin/dependency-check-report.html

CVSSv2:

- Base Score: HIGH (7.5)
- Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:H/A:N

References:

- FEDORA - [FEDORA-2020-2640aa4e19](#)
- FEDORA - [FEDORA-2020-3ce0f55bc5](#)
- FEDORA - [FEDORA-2020-92b1d001b3](#)
- GENTOO - [GLSA-202011-18](#)
- MISC - [https://github.com/gradle/gradle/security/advisories/GHSA-j45w-grgf-25vm](#)
- MISC - [https://lists.apache.org/thread.html#r3c8e9f724b5b1e171529b4714b35cb7920edfb6e917fa21eb6c64ea%40%3Cdev.apache.org%3F](#)
- MISC - [https://www.oracle.com/security-alerts/cpuapr2021.html](#)
- MISC - [https://www.oracle.com/security-alerts/cpuapr2022.html](#)
- MISC - [https://www.oracle.com/security-alerts/cpujan2021.html](#)
- MISC - [https://www.oracle.com/security-alerts/cpujan2022.html](#)
- MISC - [https://www.oracle.com/security-alerts/cpuoct2021.html](#)
- MLIST - [\[cveadur-dev\] 20201006 \[jira\] \[Assigned\] \(RAT-274\) Update to at least Ant 1.10.8/1.9.15 in order to fix CVE-2020-11979](#)
- MLIST - [\[cveadur-dev\] 20201006 \[jira\] \[Commented\] \(RAT-274\) Update to at least Ant 1.10.8/1.9.15 in order to fix CVE-2020-11979](#)
- MLIST - [\[cveadur-dev\] 20201006 \[jira\] \[Resolved\] \(RAT-274\) Update to at least Ant 1.10.8/1.9.15 in order to fix CVE-2020-11979 / raise compiler level to JDK8](#)
- MLIST - [\[cveadur-dev\] 20201006 \[jira\] \[Updated\] \(RAT-274\) Update to at least Ant 1.10.8/1.9.15 in order to fix CVE-2020-11979](#)
- MLIST - [\[cveadur-dev\] 20201006 \[jira\] \[Updated\] \(RAT-274\) Update to at least Ant 1.10.8/1.9.15 in order to fix CVE-2020-11979 / raise compiler level to JDK8](#)
- MLIST - [\[cveadur-dev\] 20210419 \[jira\] \[Commented\] \(RAT-274\) Update to at least Ant 1.10.8/1.9.15 in order to fix CVE-2020-11979 / raise compiler level to JDK8](#)
- MLIST - [\[cveadur-dev\] 20210621 \[jira\] \[Commented\] \(RAT-274\) Update to at least Ant 1.10.8/1.9.15 in order to fix CVE-2020-11979 / raise compiler level to JDK8](#)
- N/A - N/A

Vulnerable Software & Versions: ([show all](#))

- [cpe:2.3:a:gradle:gradle:*:*:*:*:* versions up to \(excluding\) 6.8.0](#)
- ...

[CVE-2021-32751](#) [suppress](#)

Gradle is a build tool with a focus on build automation. In versions prior to 7.2, start scripts generated by the 'application' plugin and the 'gradlew' script are both vulnerable to arbitrary code execution when an attacker is able to change environment variables for the user running the script. This may impact those who use 'gradlew' on Unix-like systems or use the scripts generated by Gradle in their application on Unix-like systems. For this vulnerability to be exploitable, an attacker needs to be able to set the value of particular environment variables and have those environment variables be seen by the vulnerable scripts. This issue has been patched in Gradle 7.2 by removing the use of 'eval' and requiring the use of the 'bash' shell. There are a few workarounds available. For CI/CD systems using the Gradle build tool, one may ensure that untrusted users are unable to change environment variables for the user that executes 'gradlew'. If one is unable to upgrade to Gradle 7.2, one may generate a new 'gradlew' script with Gradle 7.2 and use it for older versions of Gradle. Applications using start scripts generated by Gradle, one may ensure that untrusted users are unable to change environment variables for the user that executes the start script. A vulnerable start script could be manually patched to remove the use of 'eval' or the use of environment variables that affect the application's command-line. If the application is simple enough, one may be able to avoid the use of the start scripts by running the application directly with Java command.

CWE-78 Improper Neutralization of Special Elements used in an OS Command ('OS Command Injection')

CVSSv2:

[cpe:2.3:a:gradle:gradle:*:*:*:*:* versions from \(including\) 1.4: versions up to \(including\) 5.3.1](#)

[CVE-2019-16370](#) [suppress](#)

The PGP signing plugin in Gradle before 6.0 relies on the SHA-1 algorithm, which might allow an attacker to replace an artifact with a different one that has the same SHA-1 message digest, a related issue to CVE-2005-4900.

CWE-327 Use of a Broken or Risky Cryptographic Algorithm

CVSSv2:

- Base Score: MEDIUM (4.3)
- Vector: /AV:N/AC:M/Au:N/C:N/I:P/A:N

CVSSv3:

- Base Score: MEDIUM (5.9)
- Vector: CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:H/A:N

References:

- MISC - [https://github.com/gradle/gradle/commit/425b2b7a50cd84106a77cdf1ab665c89c6b14d2f](#)
- MISC - [https://github.com/gradle/gradle/pull/10543](#)

Vulnerable Software & Versions:

- [cpe:2.3:a:gradle:gradle:*:*:*:*:* versions up to \(excluding\) 6.0](#)

[CVE-2021-29429](#) [suppress](#)

In Gradle before version 7.0, files created with open permissions in the system temporary directory can allow an attacker to access information downloaded by Gradle. Some builds could be vulnerable to a local information disclosure. Remote files accessed through TextResourceFactory are downloaded into the system temporary directory first. Sensitive information contained in these files can be exposed to other local users on the same system. If you do not use the 'TextResourceFactory' API, you are not vulnerable. As of Gradle 7.0, uses of the system temporary directory have been moved to the Gradle User Home directory. By default, this directory is restricted to the user running the build. As a workaround, set a more restrictive umask that removes read access to other users. When files are created in the system temporary directory, they will not be accessible to other users. If you are unable to change your system's umask, you can move the Java temporary directory by setting the System Property 'java.io.tmpdir'. The new path needs to limit permissions to the build user only.

CWE-377 Insecure Temporary File

CVSSv2:

- Base Score: LOW (1.9)
- Vector: /AV:L/AC:M/Au:N/C:P/I:N/A:N

CVSSv3:

- Base Score: MEDIUM (5.5)
- Vector: CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:N/A:N

11) Change Management during pre-commit in repositories

Pre-commit:

Step 1: Install python and GO as required for this setting from the below links.

<https://www.python.org/downloads/> - Python

<https://go.dev/doc/install> - Go

Step 2: Go to Command prompt, run the below command

- pip install pre-commit

Step 3: now do the push operations

- cd desktop
- mkdir cyber
- cd cyber
- git init
- now create one text document as demo or anything else in the folder which we created.
- Then come to cmd , run the below command
- git status -> it will show the newly added file

```

Microsoft Windows [Version 10.0.22000.1335]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Moulya N M>pip install pre-commit
Requirement already satisfied: pre-commit in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (2.21.0)
Requirement already satisfied: cfgv<=2.0.0 in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from pre-commit)
Requirement already satisfied: identify<=1.0.0 in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from pre-commit)
Requirement already satisfied: nodeenv<=0.11.1 in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from pre-commit)
Requirement already satisfied: pyyaml<=5.1 in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from pre-commit)
Requirement already satisfied: virtualenv<=20.10.0 in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from pre-commit)
Requirement already satisfied: setuptools in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from nodeenv)
Requirement already satisfied: distlib<1, >=0.3.6 in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from setuptools)
Requirement already satisfied: filelock<4, >=3.4.1 in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from setuptools)
Requirement already satisfied: platformdirs<3, >=2.4 in c:\users\moulya n m\appdata\local\programs\python\python311\lib\site-packages (from setuptools)

C:\Users\Moulya N M>cd desktop

C:\Users\Moulya N M\Desktop>mkdir cyber

C:\Users\Moulya N M\Desktop>cd cyber

C:\Users\Moulya N M\Desktop\cyber>git init
Initialized empty Git repository in C:/Users/Moulya N M/Desktop/cyber/.git/

C:\Users\Moulya N M\Desktop\cyber>git status
On branch master

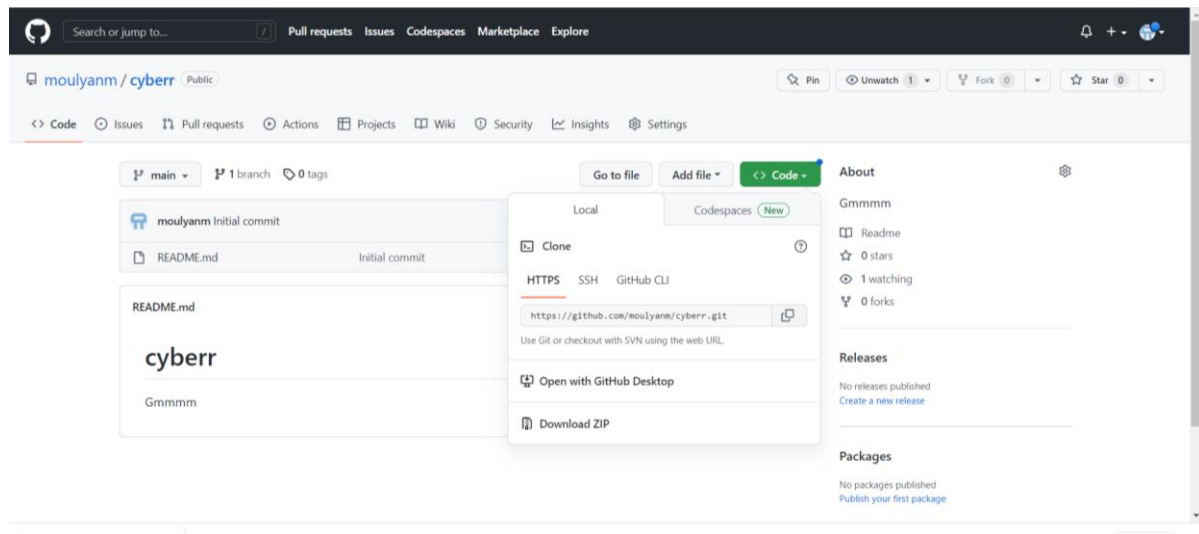
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        demo.txt

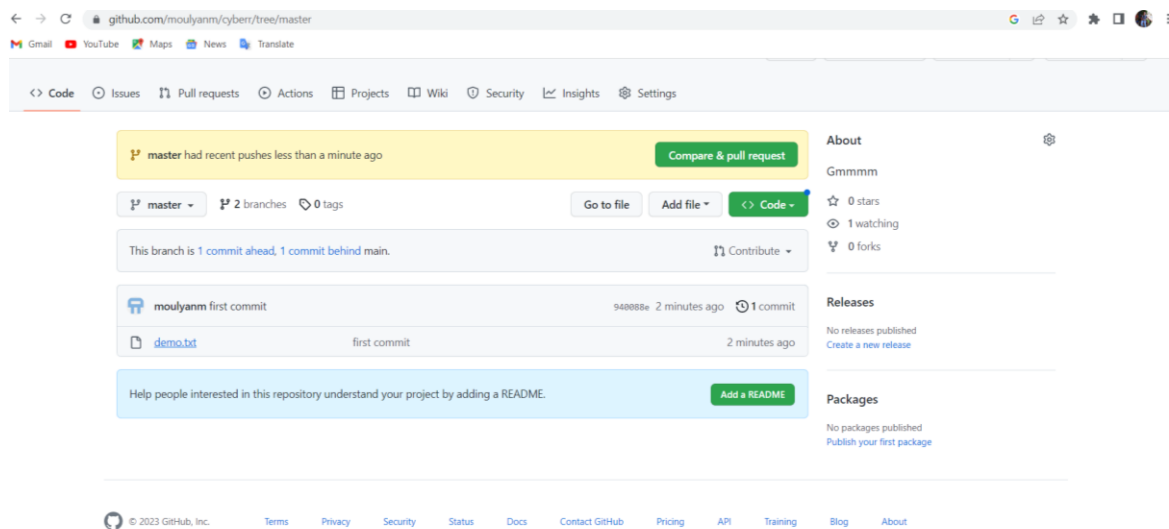
nothing added to commit but untracked files present (use "git add" to track)
  
```

- git add demo.txt
- git commit -m "first commit"
- git config --global user.moulyanm

- create new repository at git hub and copy the code



- git remote add origin <https://github.com/moulyanm/cyber.git> [pastecode]
- git push origin master



- ➔ now install the pre-commit by giving the below command
- **pre-commit install**
- ➔ Go to chrome > search git leaks > scroll down > copy the 3 lines of pre-commit code

.pre-commit-config.yaml

repos:

- repo: <https://github.com/zricethezav/gitleaks>

rev: v8.12.0

hooks:

- id: gitleaks

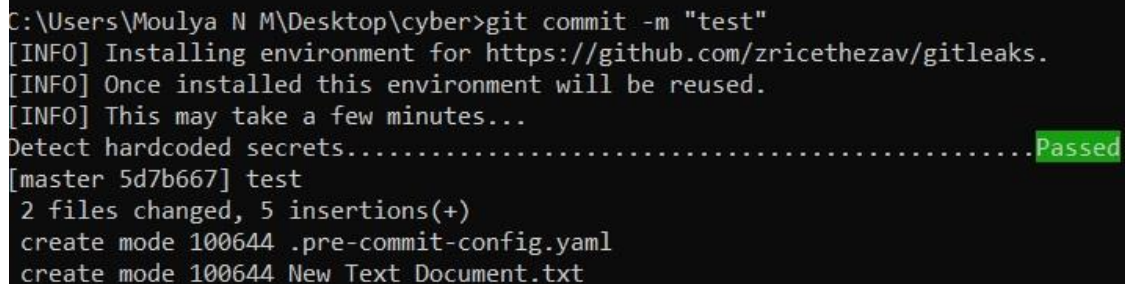
➔ Create a new text document in the folder which we created and then paste that code and save it as “. pre-commit-config.yaml”

➔ Then go cmd, run the below command

➤ **git add.**

➤ **git commit -m “test”**

➔ now it will show passed because there is no barcode in the folder



```
C:\Users\Moulya N M\Desktop\cyber>git commit -m "test"
[INFO] Installing environment for https://github.com/zricethezav/gitleaks.
[INFO] Once installed this environment will be reused.
[INFO] This may take a few minutes...
Detect hardcoded secrets.....Passed
[master 5d7b667] test
2 files changed, 5 insertions(+)
create mode 100644 .pre-commit-config.yaml
create mode 100644 New Text Document.txt
```

➔ now you can add any barcode to your folder to check and then go to cmd add and commit, it will show failed

Government Polytechnic, Tumakuru

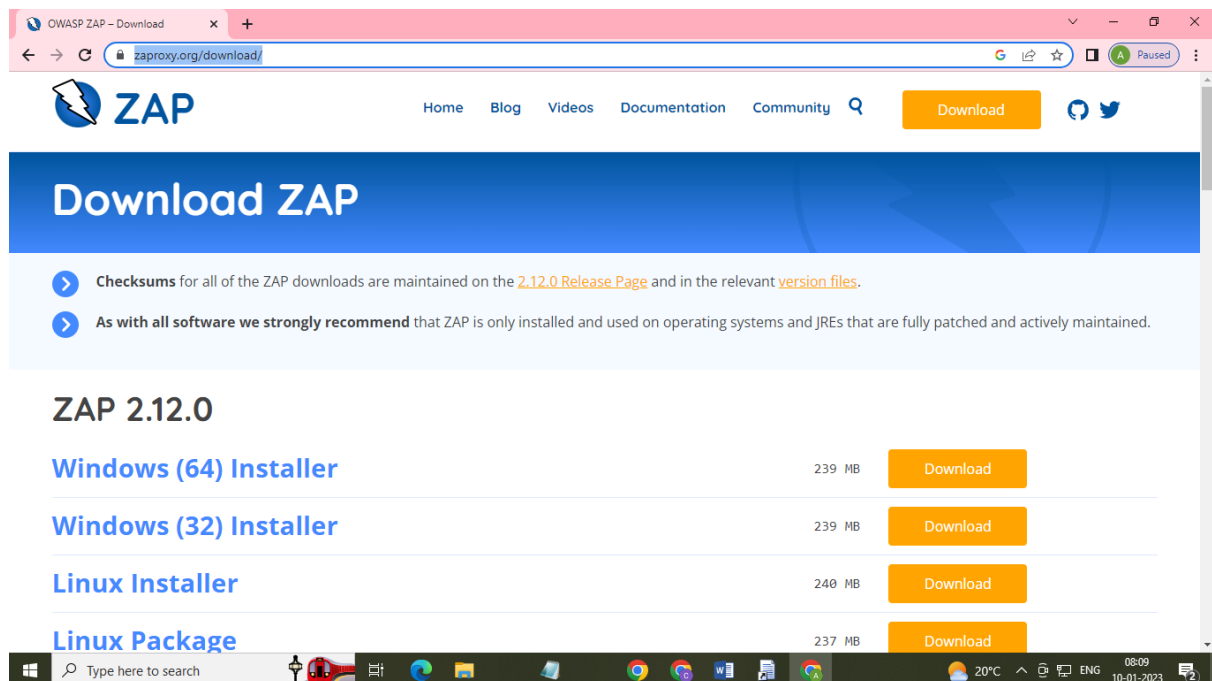
12) Conduct Penetration testing on any web site/web application and report the vulnerabilities. Explain Dynamic Analysis using an example – owasp zap.

DAST TOOL

Dynamic Application Security Testing ([DAST](#)) is the process of analyzing a web application through the front-end to find vulnerabilities through simulated attacks. This type of approach evaluates the application from the “outside in” by attacking an application like a malicious user would. After a DAST scanner performs these attacks, it looks for results that are not part of the expected result set and identifies security vulnerabilities.

AUTOMATED SCAN

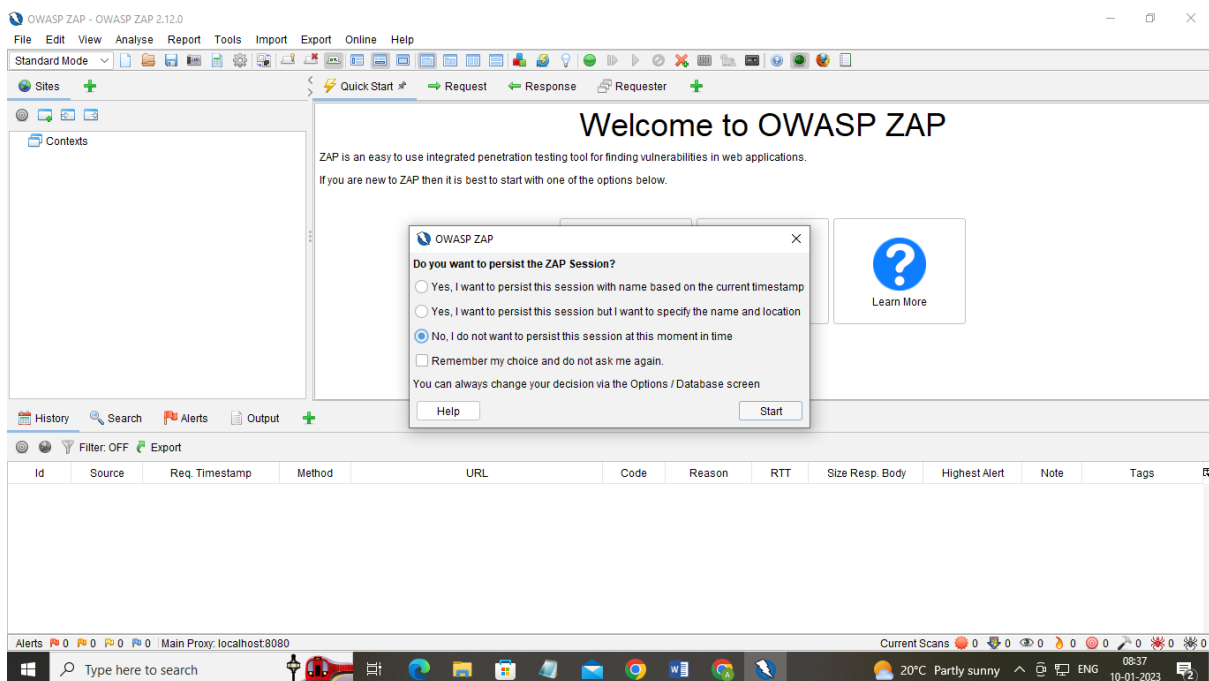
- **STEP 1** – Go to Chrome > Download Owasp Zap.
- <https://www.zaproxy.org/download/> (LINK TO DOWNLOAD OWASP ZAP).



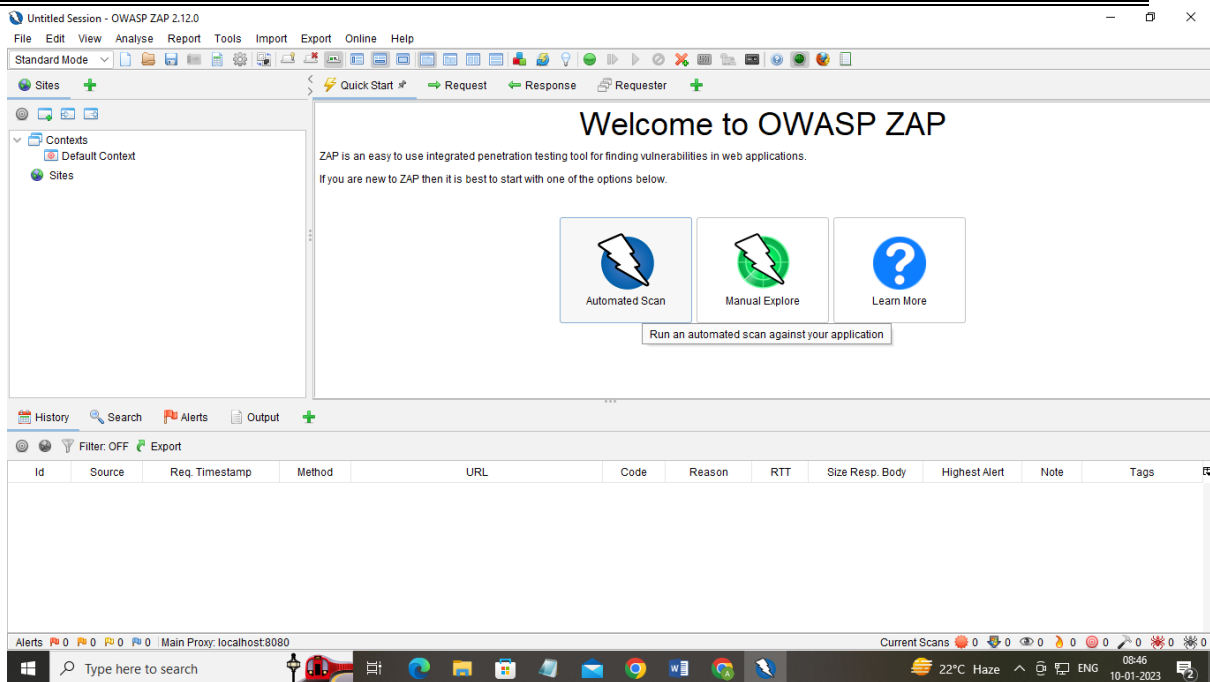
- After download complete the installation.
- **STEP 2** – Open owasp zap tool.



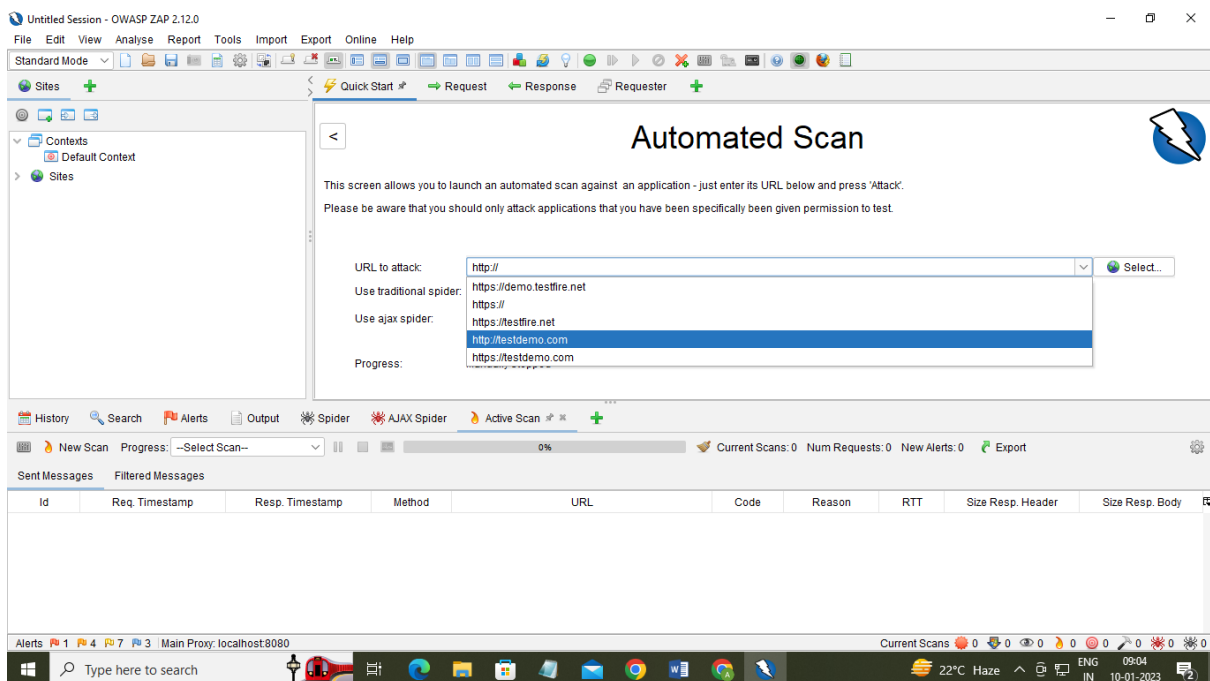
- **STEP 3** – Select 3rd option and click start.



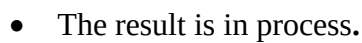
- **STEP 4** - Click on Automation Scan.

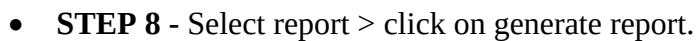


- **STEP 5** - Select the url (<http://testdemo.com>).

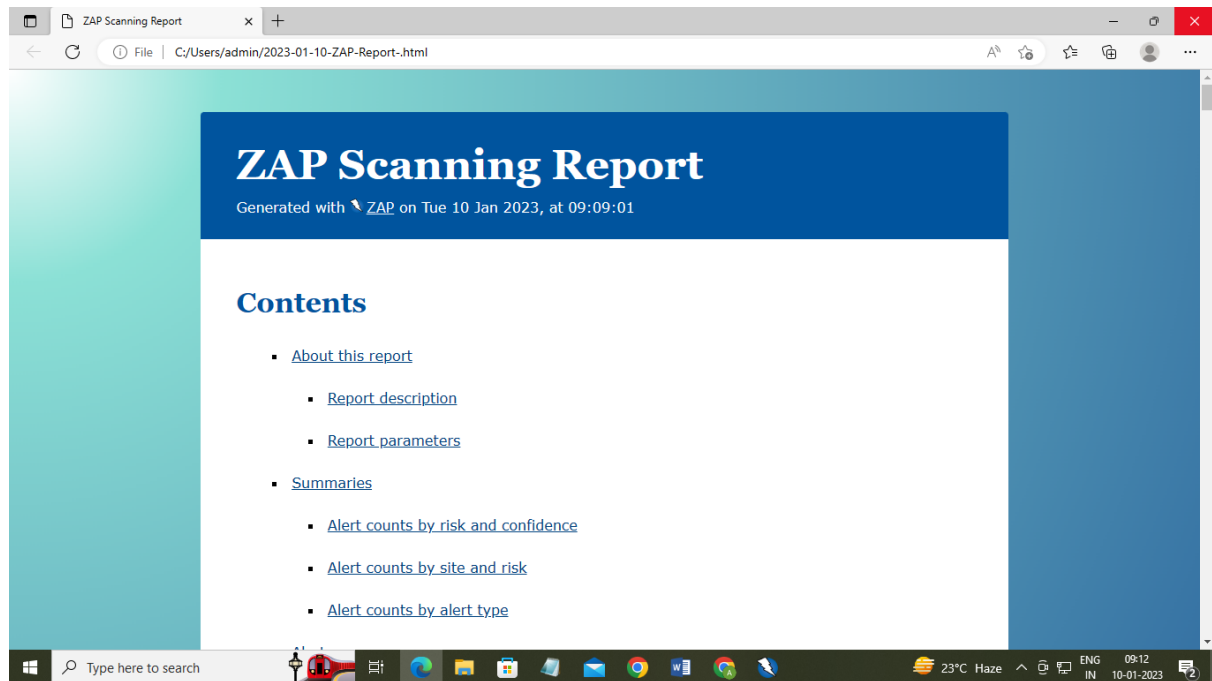


- **STEP 6** – Select chrome.





- [ZAP Scanning Report](#)

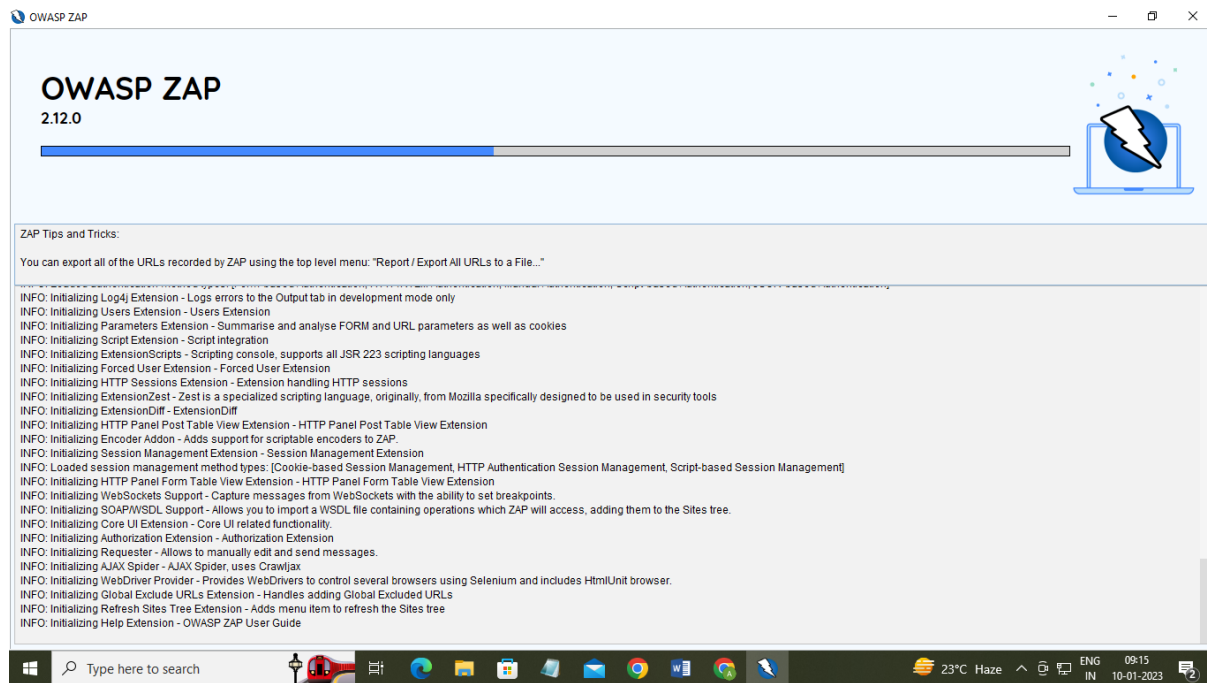


13) Conduct Penetration testing on any web site/web application and report the vulnerabilities. Exploring the application manually and Explore pages protected by login

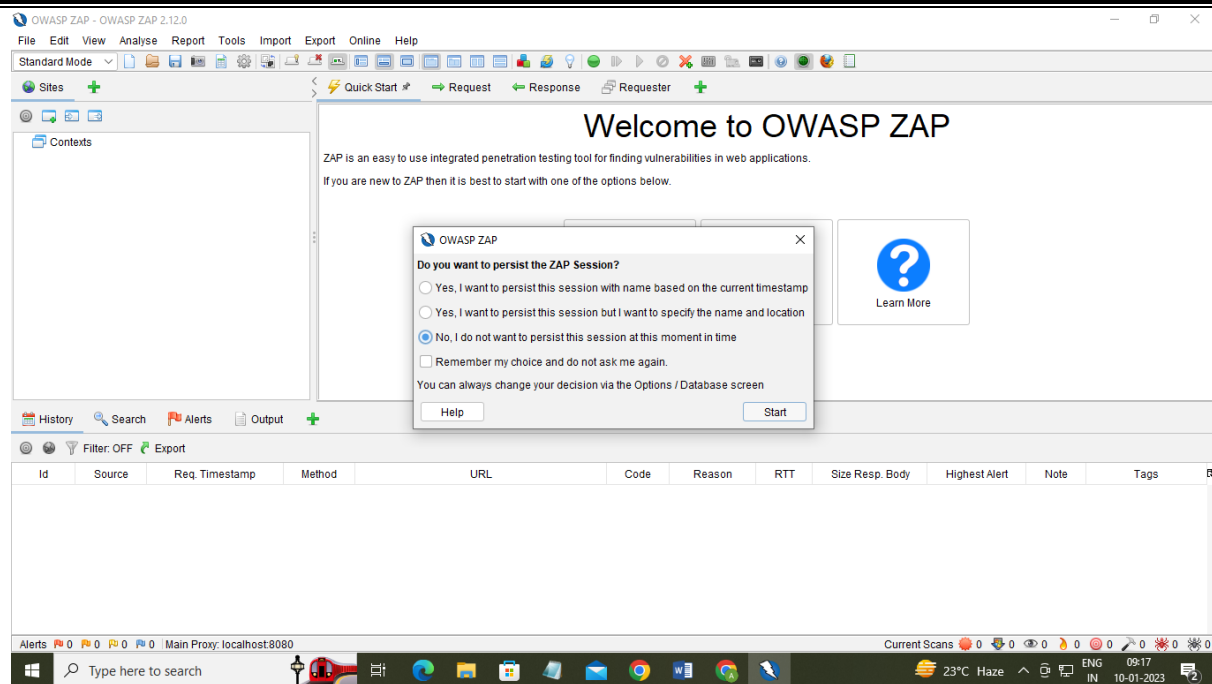
DAST TOOL

MANUAL EXPLORE

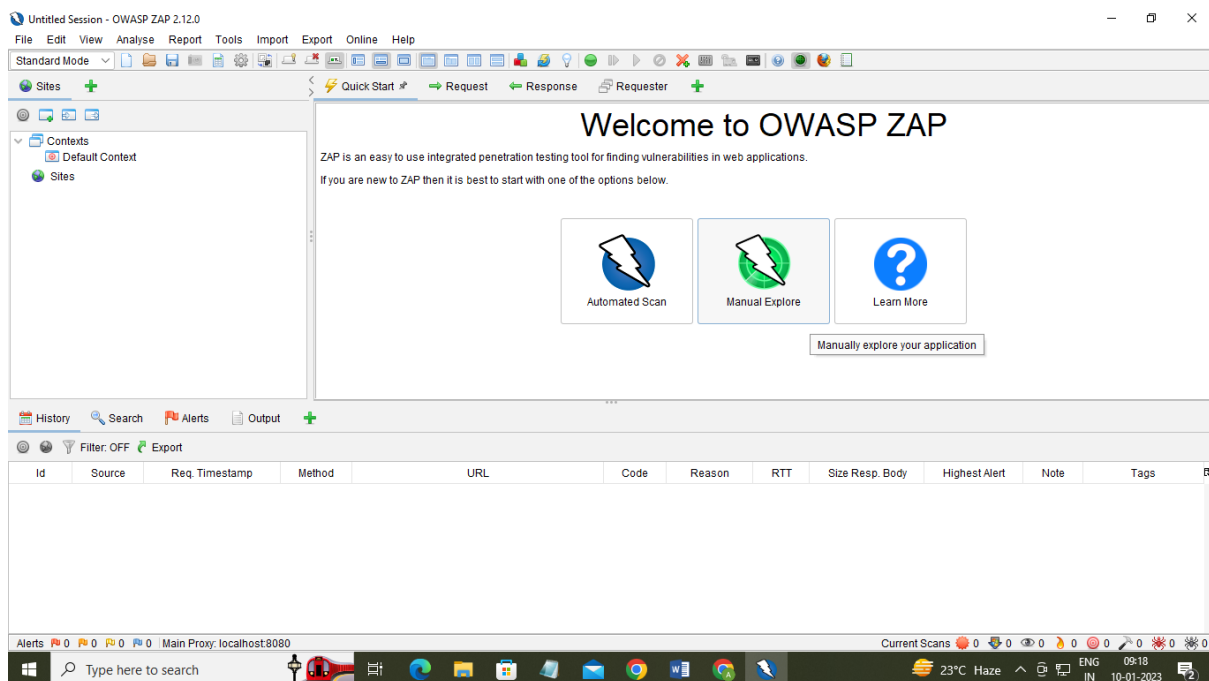
- **STEP 1** – Open owasp zap.



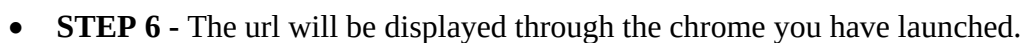
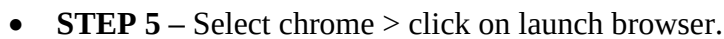
- **STEP 2** – Click on start.

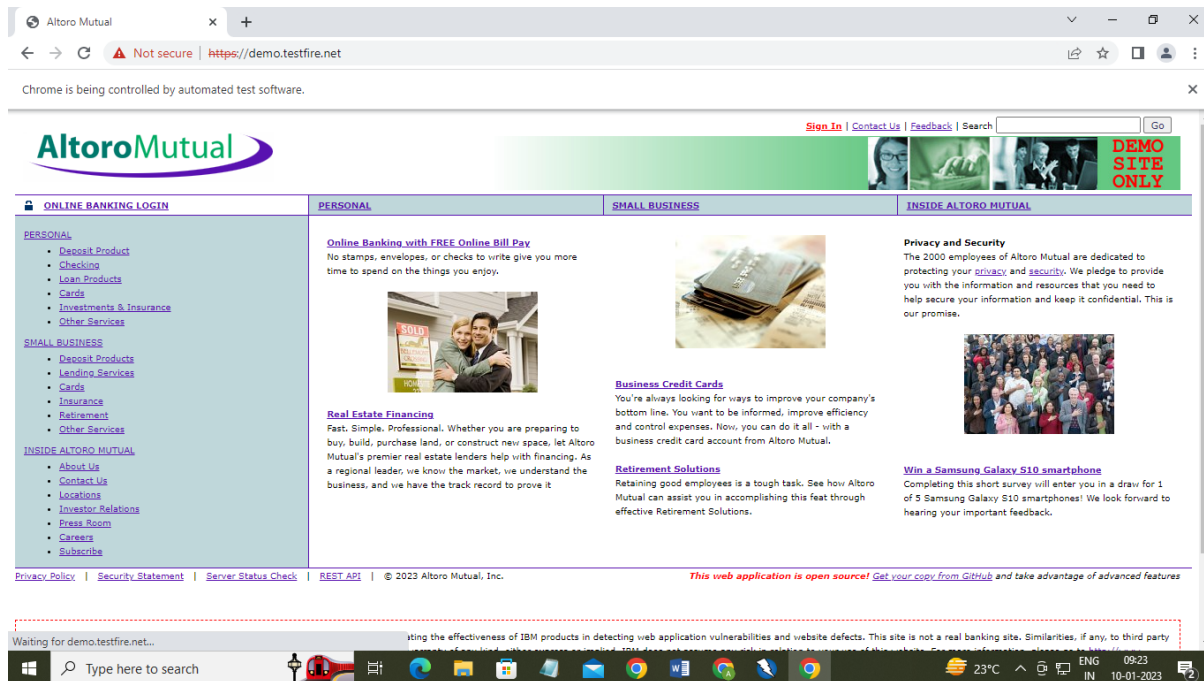


- **STEP 3 – Select manual explore**

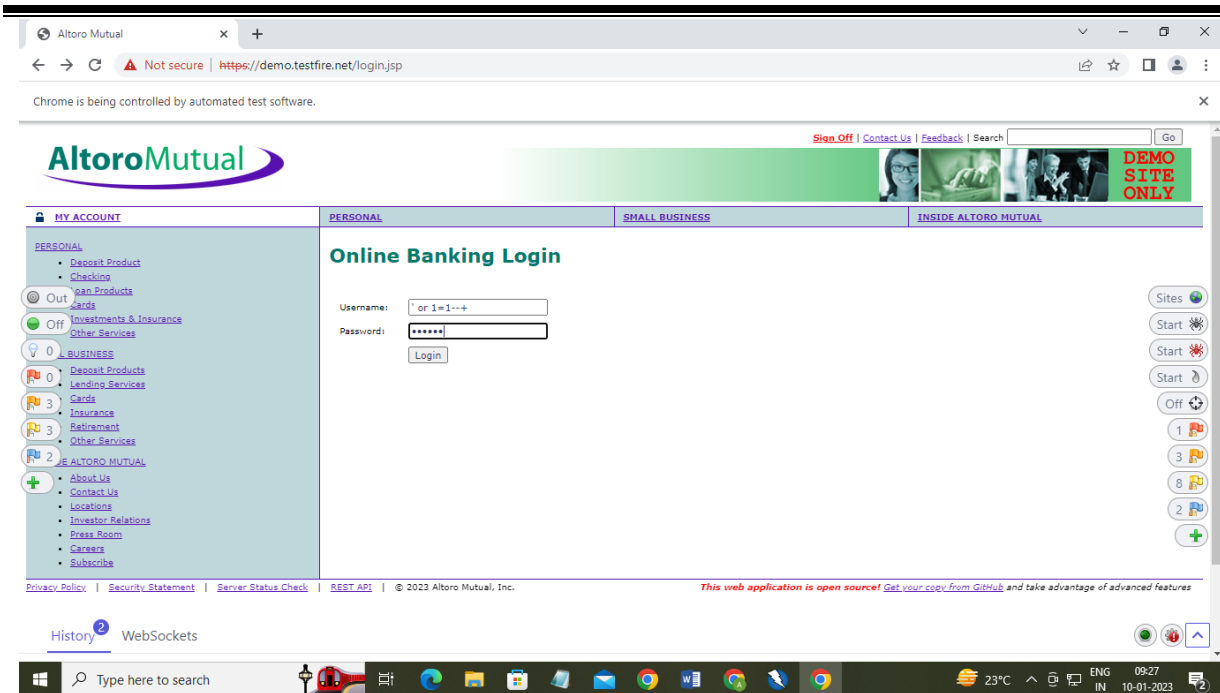


- **STEP 4 –Enter the url you need to scan (<https://demo.testfire.net>)**

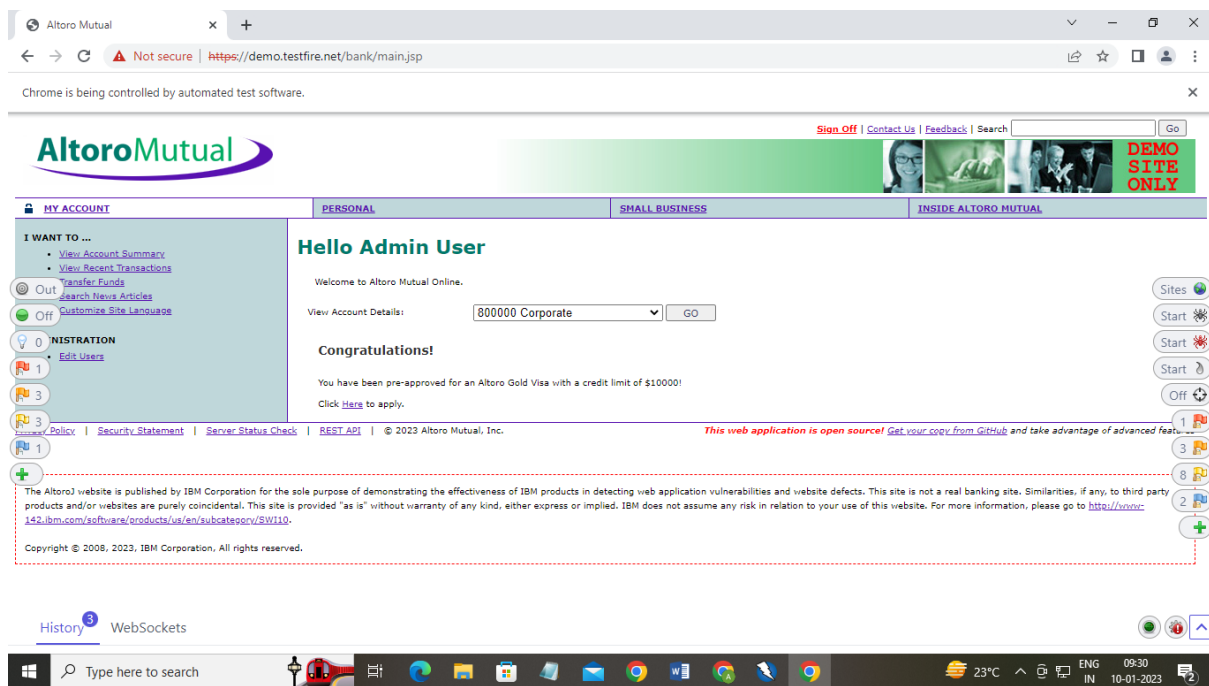




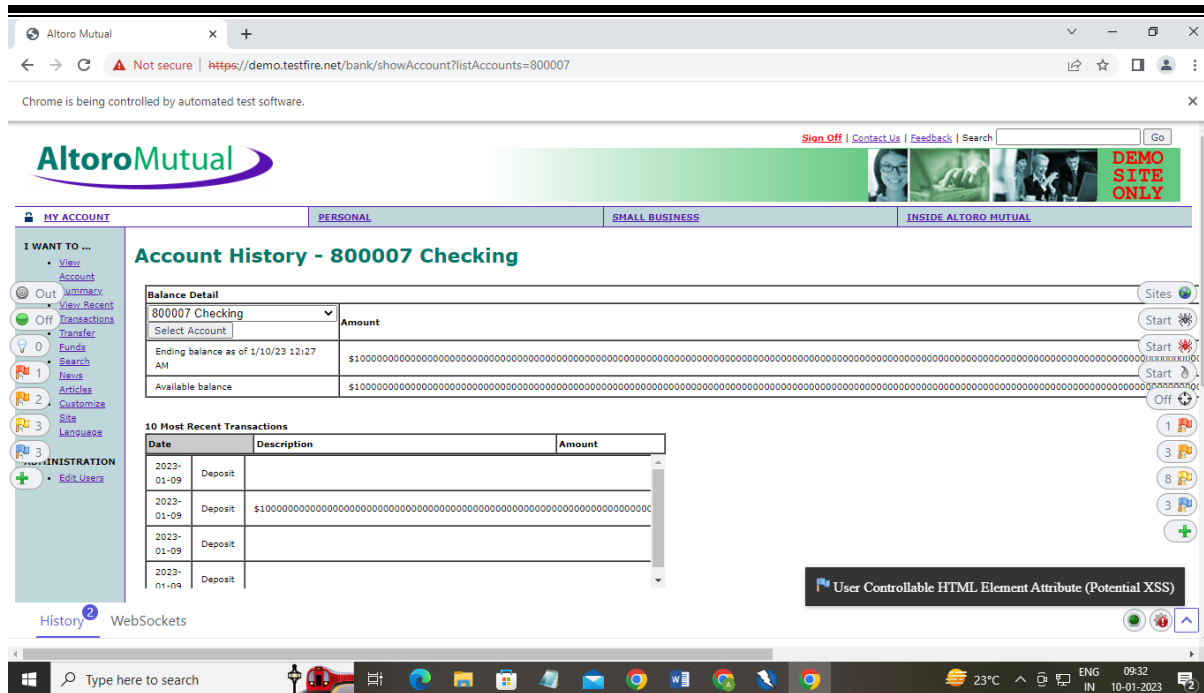
- **STEP 7** – Click on sign in.
- Perform the following operations (sql injection)
- Username - ' or 1=1--+
- Password = (give anything)
- Click on log in



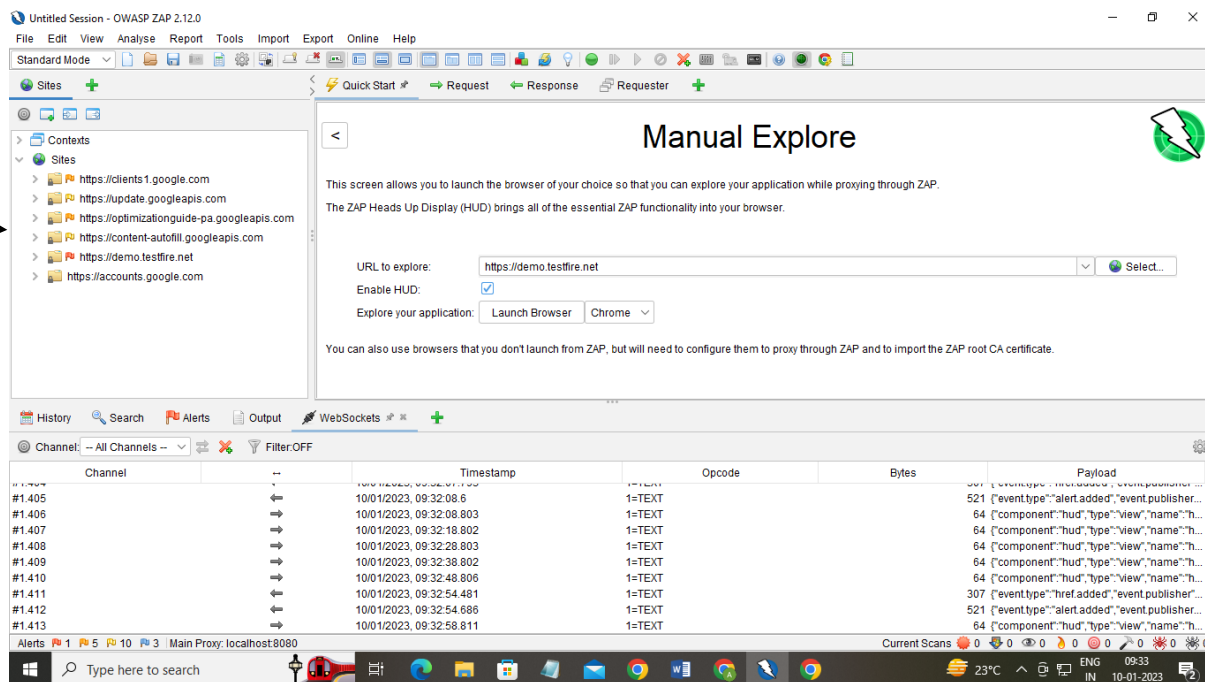
- After log in, the page will be displayed



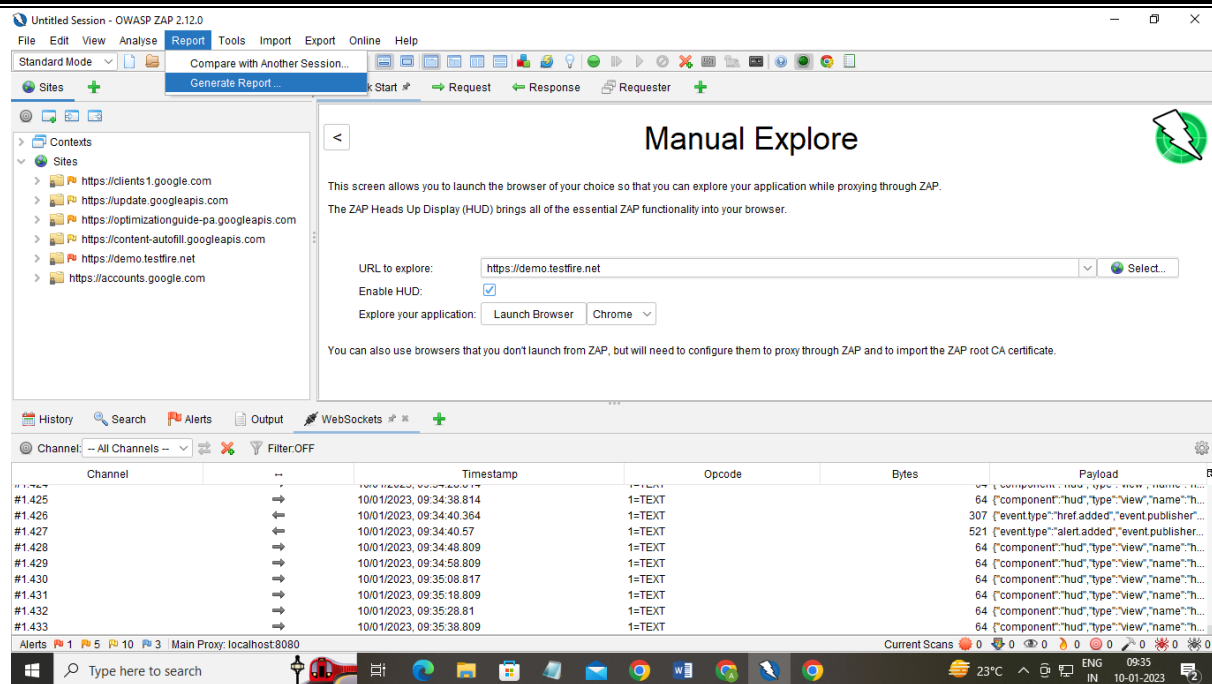
- **STEP 8** – Visit the few pages



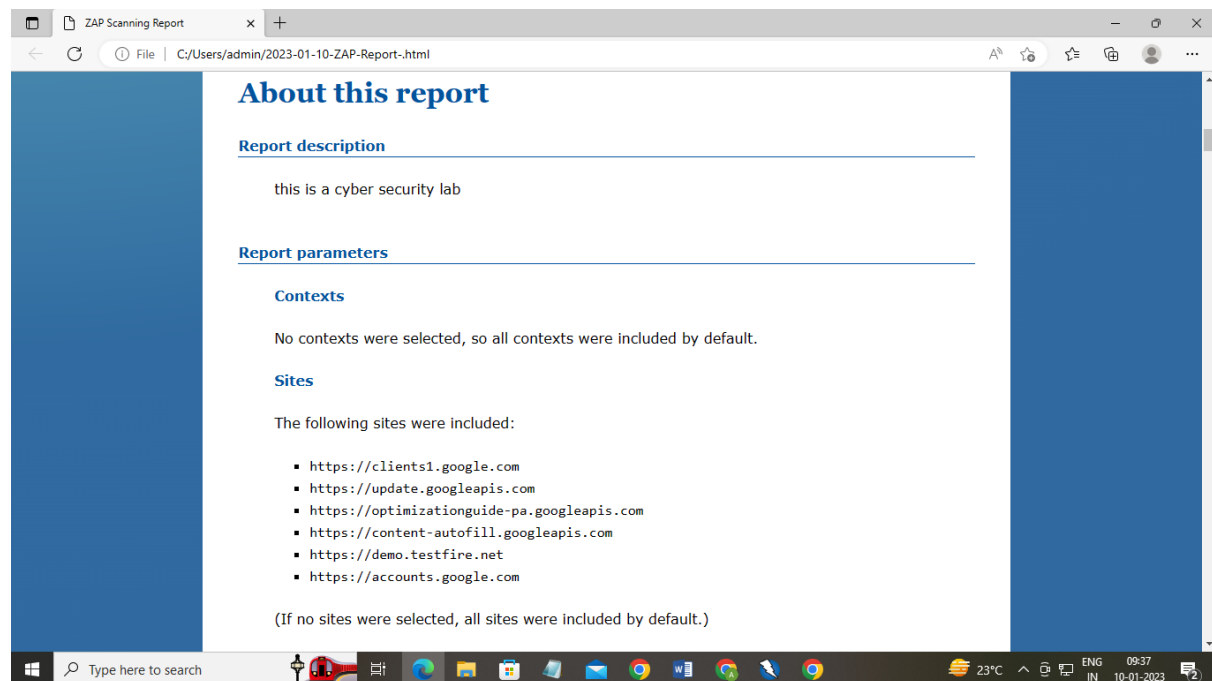
- **STEP 9** - The sites you have visited will be displayed in the left side



- **STEP 10** – Select report > click on generate report.



- **STEP 11** – In the report we can view the sites we have visited, and the vulnerabilities the site possess.



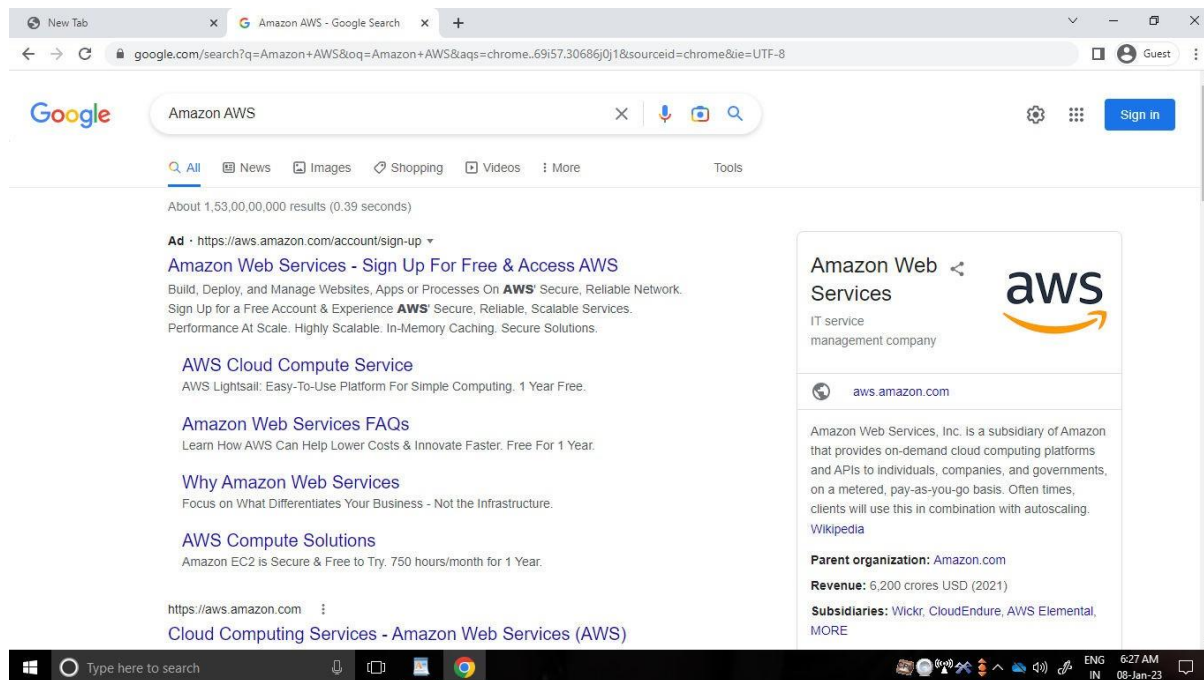
14) Create a cloud account & Setup 2Factor Authentication on account

Account Pre requisites

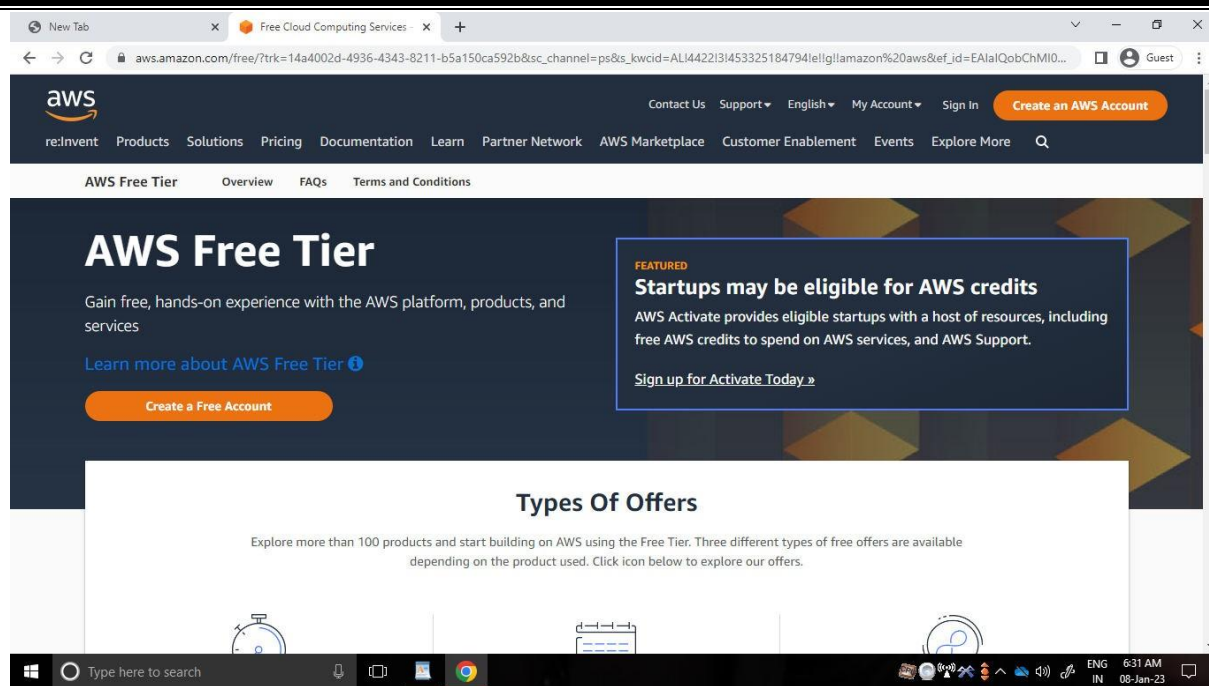
→AWS cloud account

→Google Authenticator App

Step 1: Go to browser search for Amazon AWS



Step 2: Click on the first link displayed On the Screen

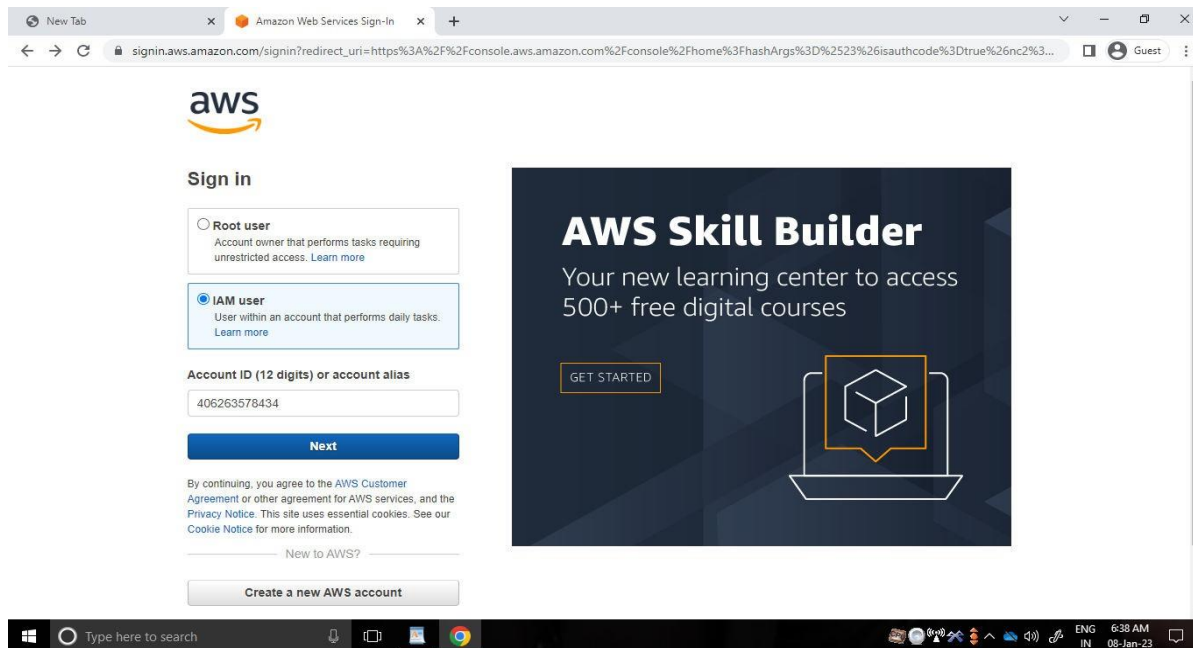


Step 3: Go to Sign in console

Select IAM user

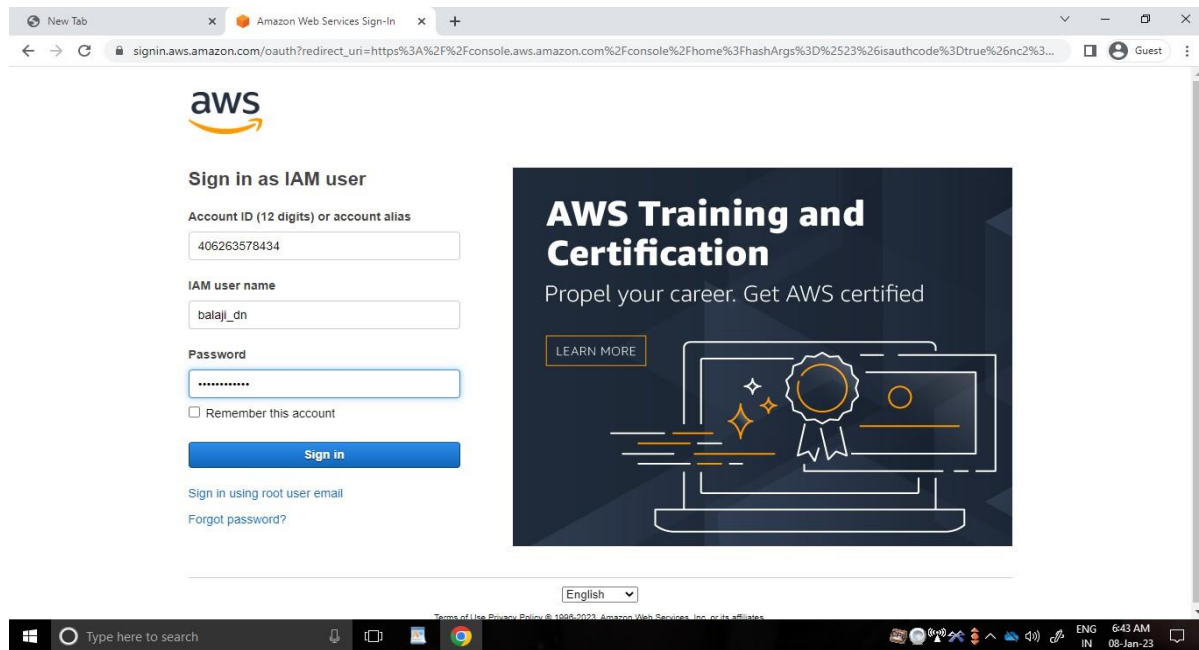
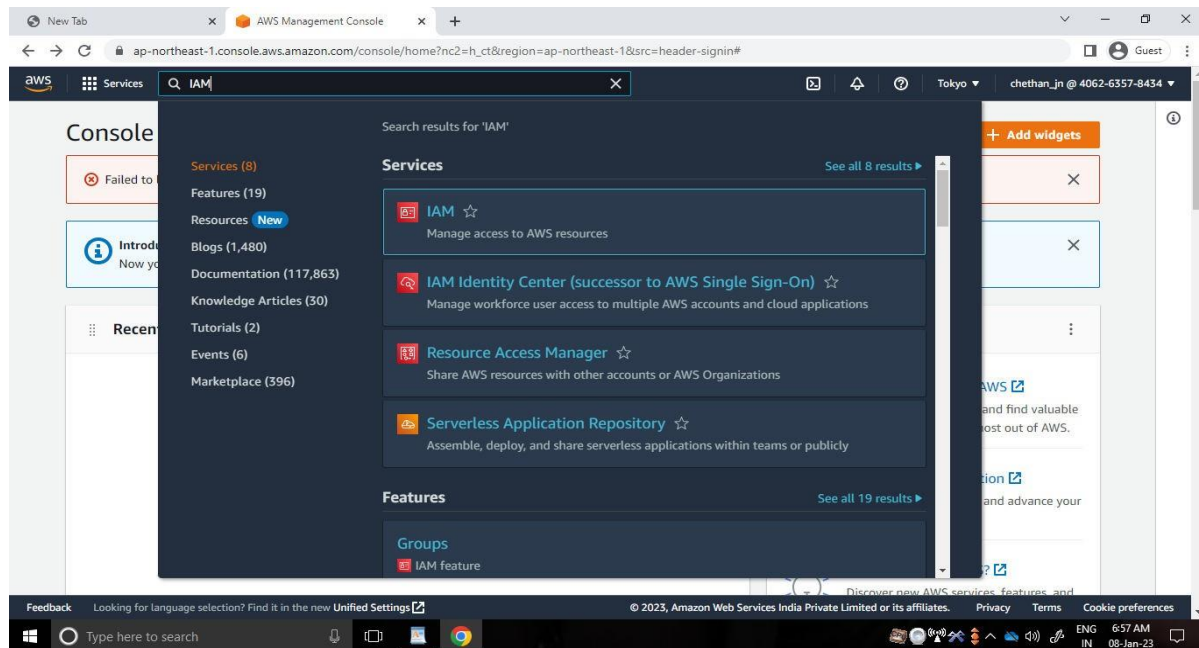
Enter 12 digits account ID

Click on next

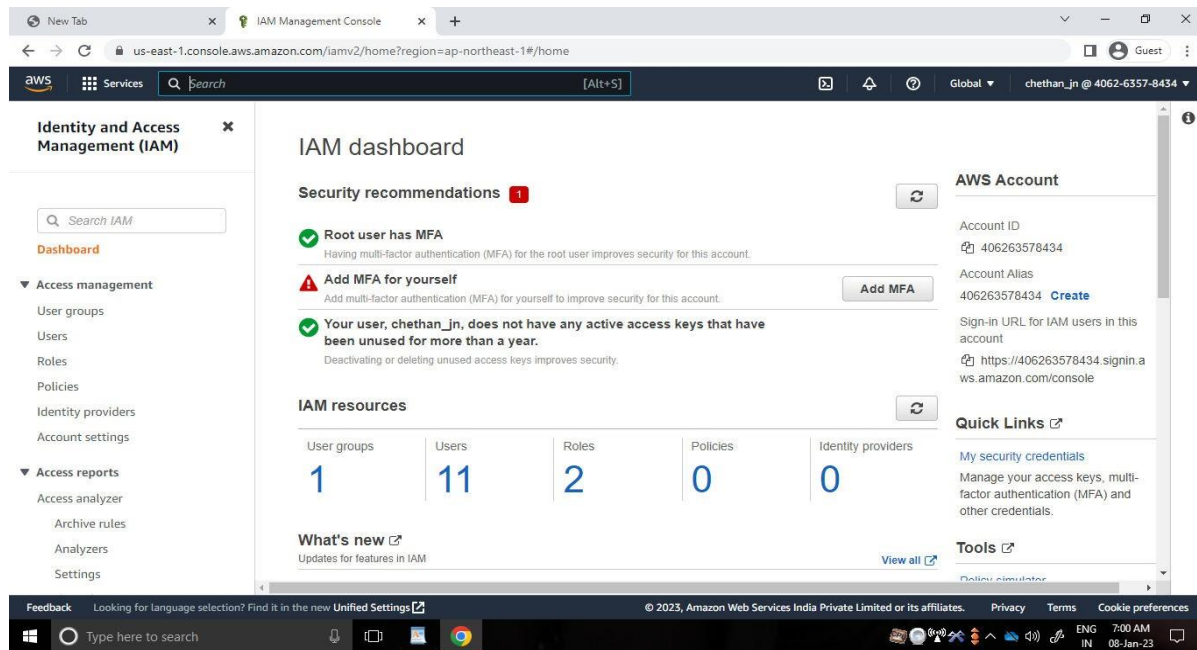


Step 4: Enter username and Password

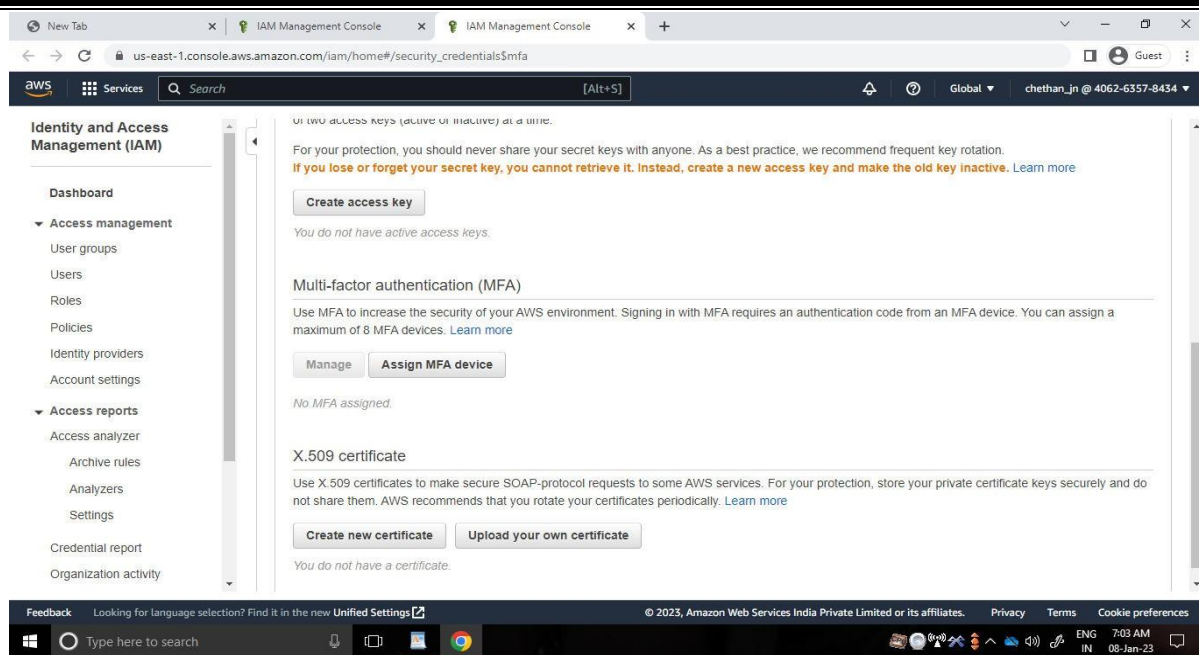
Click on sign in

**Step 5: After successful login, in search bar search for IAM and click on it**

Step 6: Click on Add MFA for yourself

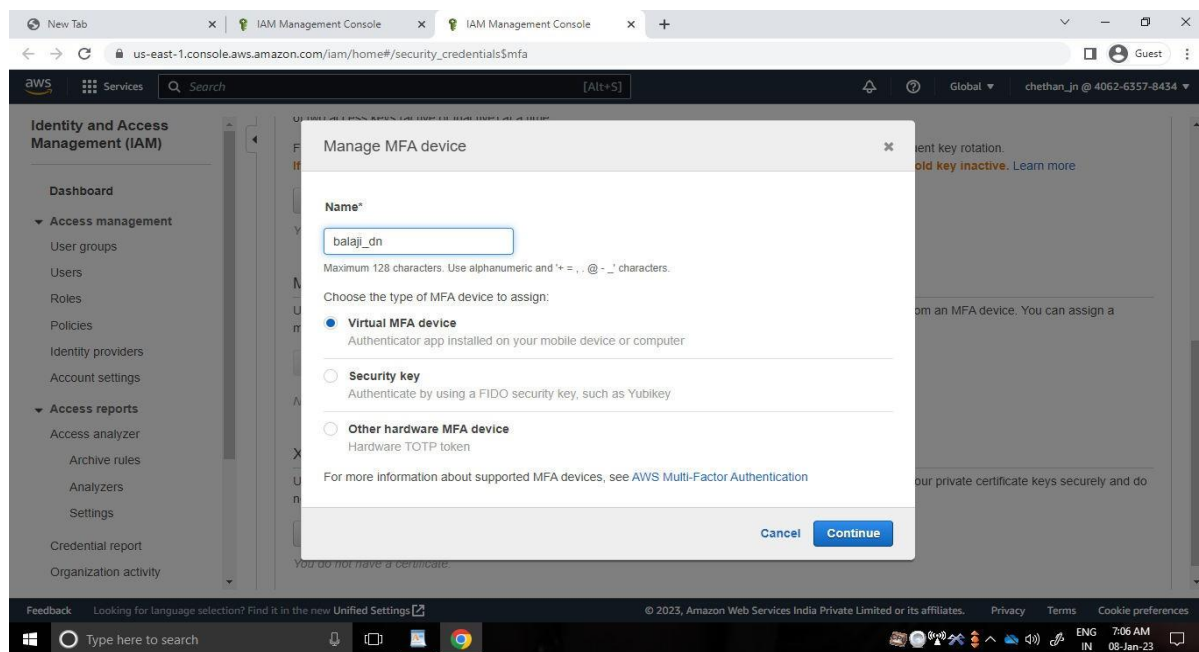


Step 7: Scroll Down and click on Assign MFA device



Step 8: Select Virtual MFA Device

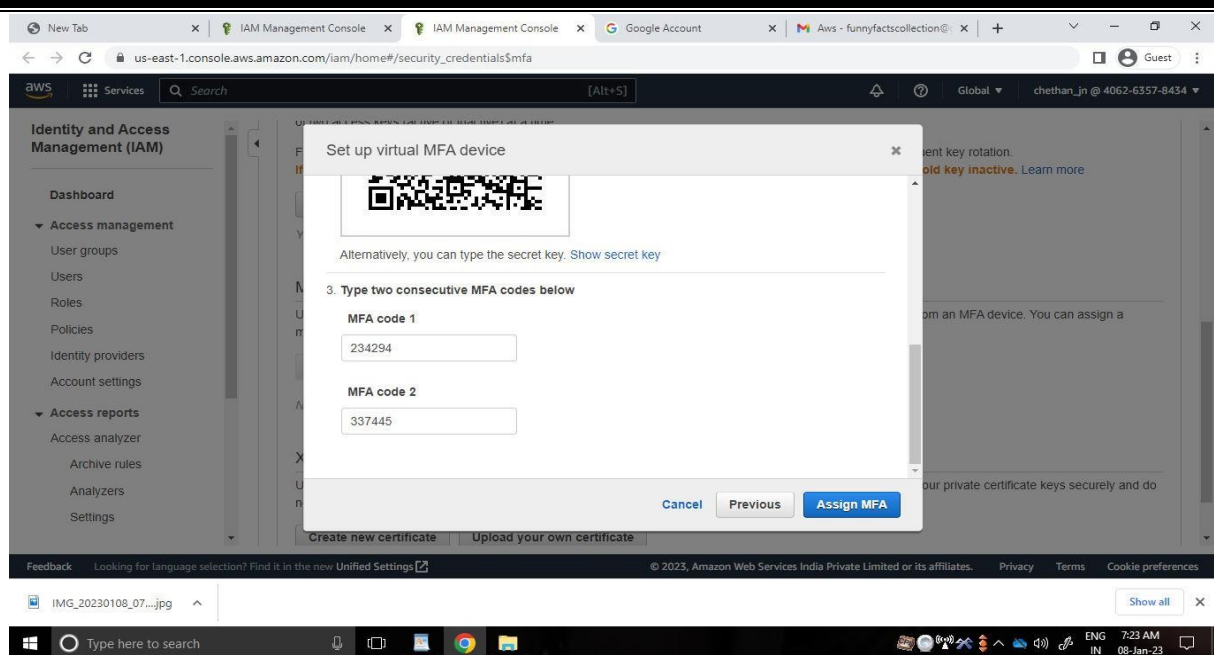
: Give a Name and Click on Continue



Step 9: Scan the Displayed QR Code on Google Authenticator App

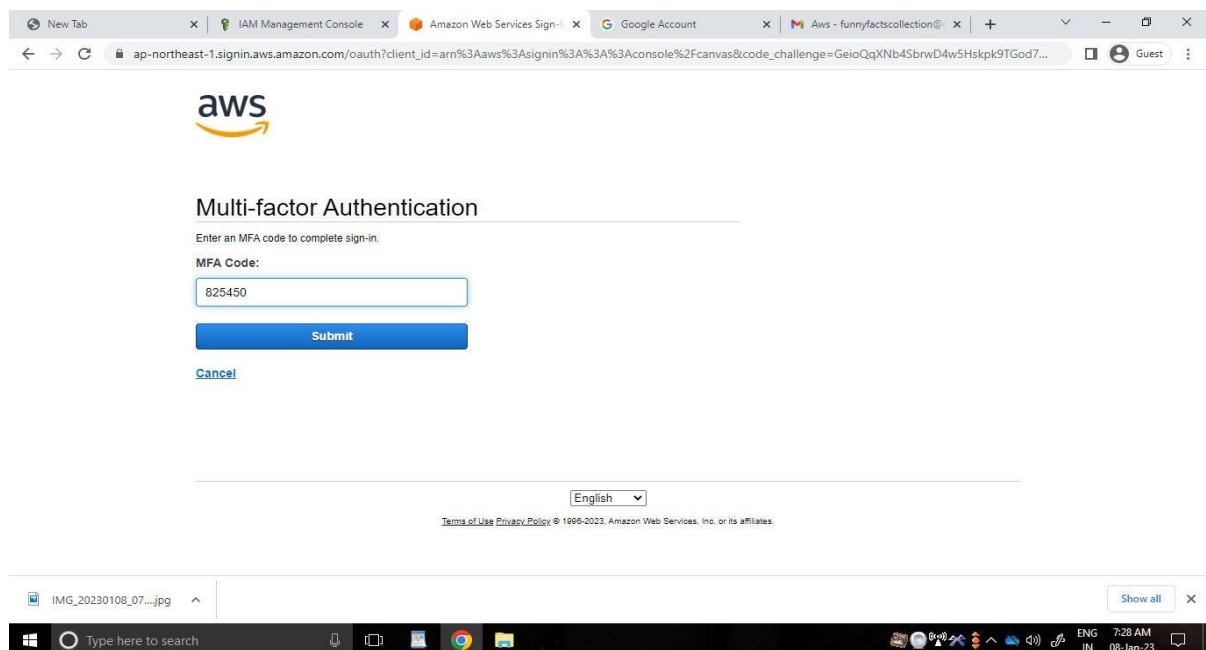
: Enter two MFA codes it generates

: click on Assign MFA



Step 10: Sign out from Account

: sign in again it will ask for for MFA Code



15) Setup Burp Suite on local machine and observe traffic of 1 website.

Intercept HTTP traffic with Burp Proxy

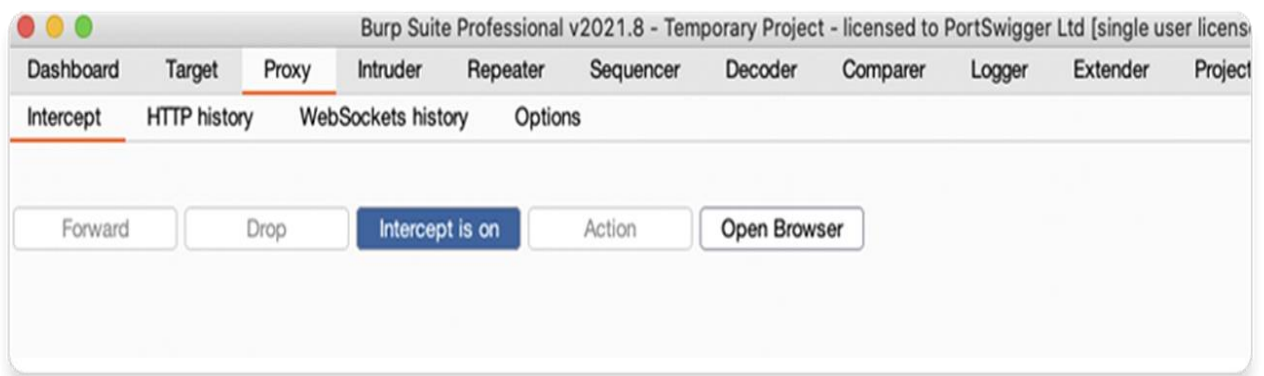
- Intercepting a request

Burp Proxy lets you intercept HTTP requests and responses sent between Burp's browser and the target server. This enables you to study how the website behaves when you perform different actions.

Step 1: Launch Burp's browser

Go to the Proxy > Intercept tab.

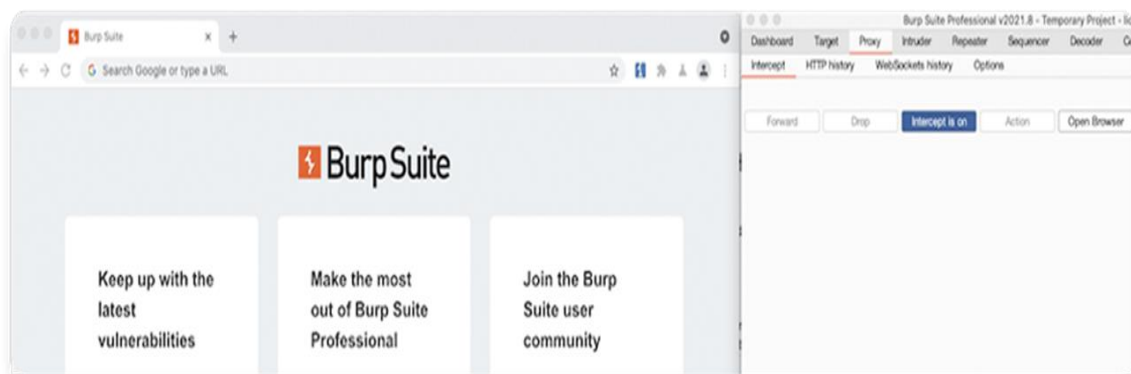
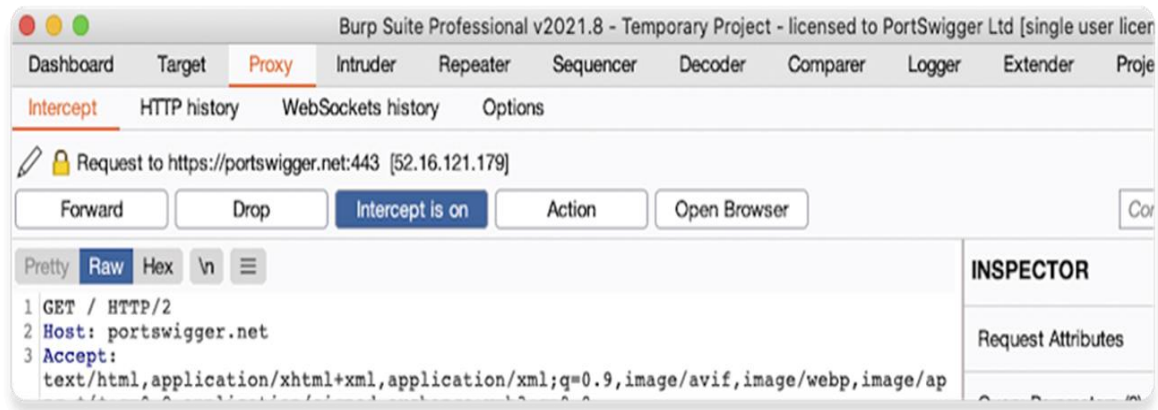
Click the Intercept is off button, so it toggles to Intercept is on.



Click Open Browser. This launches Burp's browser, which is preconfigured to work with Burp right out of the box. Position the windows so that you can see both Burp and Burp's browser.

Step 2: Intercept a request

Using Burp's browser, try to visit <https://portswigger.net> and observe that the site doesn't load. Burp Proxy has intercepted the HTTP request that was issued by the browser before it could reach the server. You can see this intercepted request on the Proxy > Intercept tab.



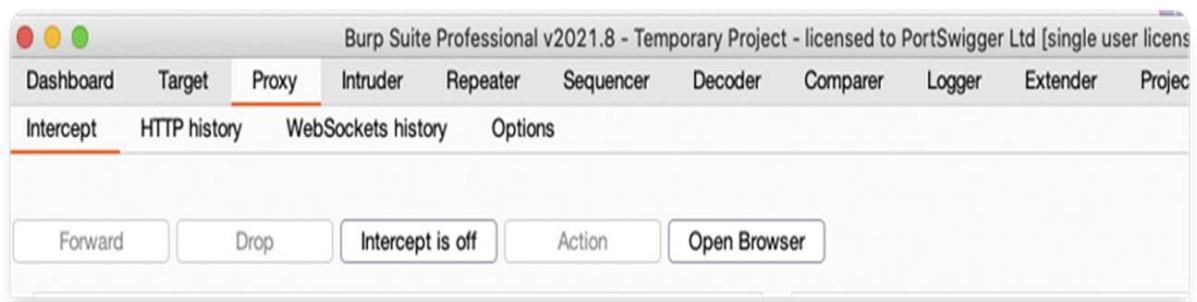
The request is held here so that you can study it, and even modify it, before forwarding it to the target server.

Step 3: Forward the request

Click the Forward button several times to send the intercepted request, and any subsequent ones, until the page loads in Burp's browser.

Step 4: Switch off interception

Due to the number of requests browsers typically send, you often won't want to intercept every single one of them. Click the Intercept is on button so that it now says Intercept is off.



Go back to the browser and confirm that you can now interact with the site as normal.

Step 5: View the HTTP history

In Burp, go to the Proxy > HTTP history tab. Here, you can see the history of all HTTP traffic that has passed through Burp Proxy, even while interception was switched off.

Click on any entry in the history to view the raw HTTP request, along with the corresponding response from the server.

Dashboard
Target
Proxy
Intruder
Repeater
Sequencer
Decoder
Comparer
Logger
Extender
Project options

Intercept
HTTP history
WebSockets history
Options

Filter: Hiding CSS, image and general binary content

#	Host	Method	URL	Params	Edited	Status	Length	MIME type	Extens
25	https://0ac9003503634ff7c01d...	GET	/resources/labheader/images/logoAca...			200	8930	XML	svg
24	https://0ac9003503634ff7c01d...	GET	/academyLabHeader			101	147		
23	https://0ac9003503634ff7c01d...	GET	/resources/labheader/images/ps-lab-...			200	934	XML	svg
22	https://0ac9003503634ff7c01d...	GET	/resources/images/shop.svg			200	7250	XML	svg
2	https://0ac9003503634ff7c01d...	GET	/resources/labheader/js/labHeader.js			200	867	script	js
1	https://0ac9003503634ff7c01d...	GET	/			200	8319	HTML	

Request
Raw
Hex

1 GET /academyLabHeader HTTP/1.1
2 Host: 0ac9003503634ff7c01d5eb4003d0076.web-security-academy.net
3 Connection: Upgrade
4 Pragma: no-cache

Response
Raw
Hex
Render

1 HTTP/1.1 101 Switching Protocol
2 Connection: Upgrade
3 Upgrade: websocket
4 Sec-WebSocket-Accept: urFasr0py7aAmDQCaISVxmkaV54=
5 Content-Length: 0

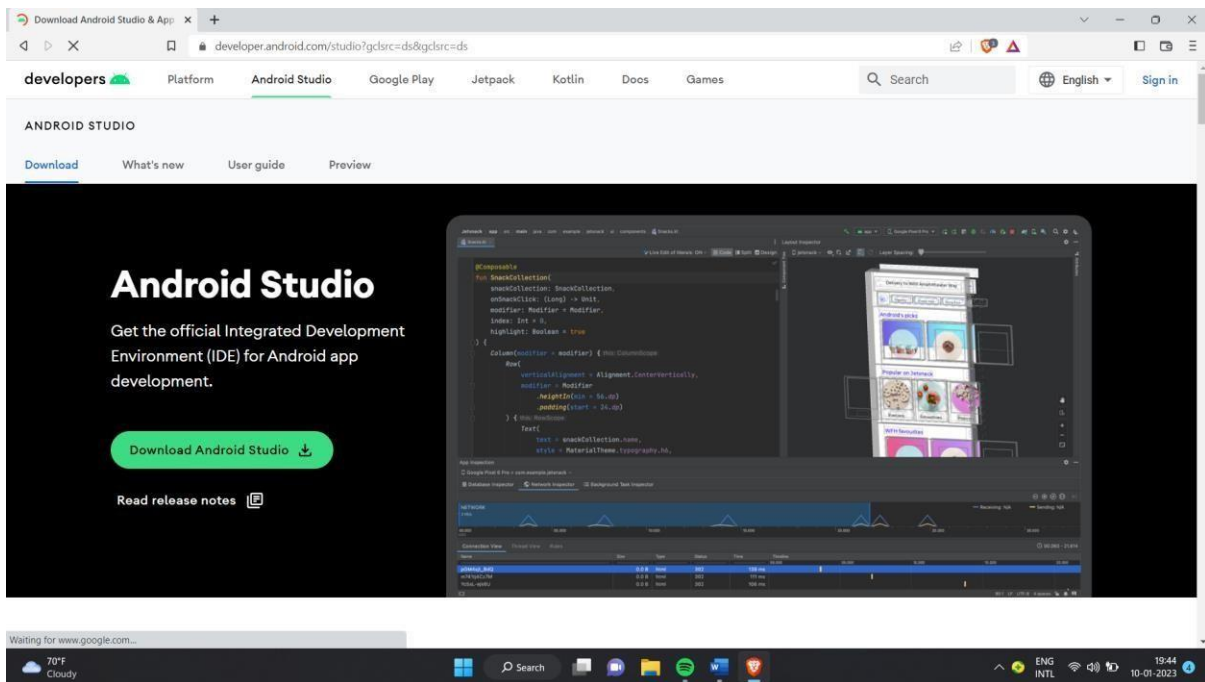
This lets you explore the website as normal and study the interactions between Burp's browser and the server afterward, which is more convenient in many cases.

16) Setting up the environment:

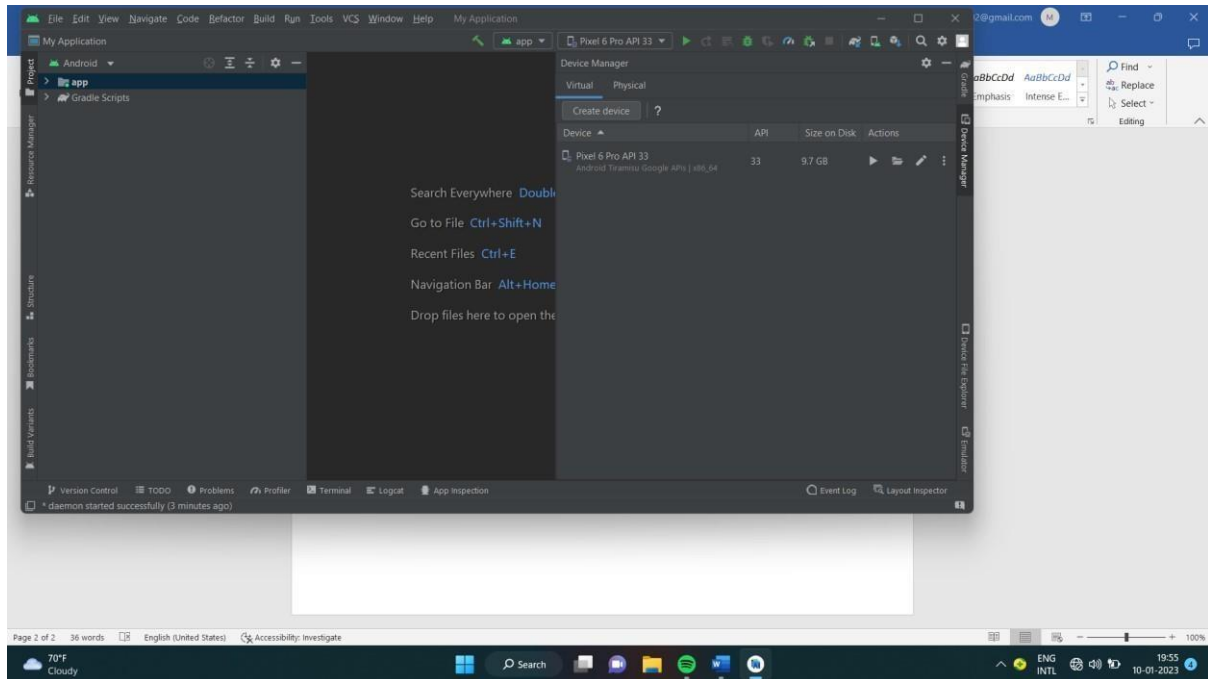
- **Installing Android Studio and Creating Android Virtual Devices**
- **Using Android Debug Bridge (ADB) to interact with the Android Virtual Devices (AVD)**
- **Transferring files between Host machine and AVD using ADB**

Installing Android Studio

→Go to chrome search for android studio for windows

**Create Android Virtual Devices**

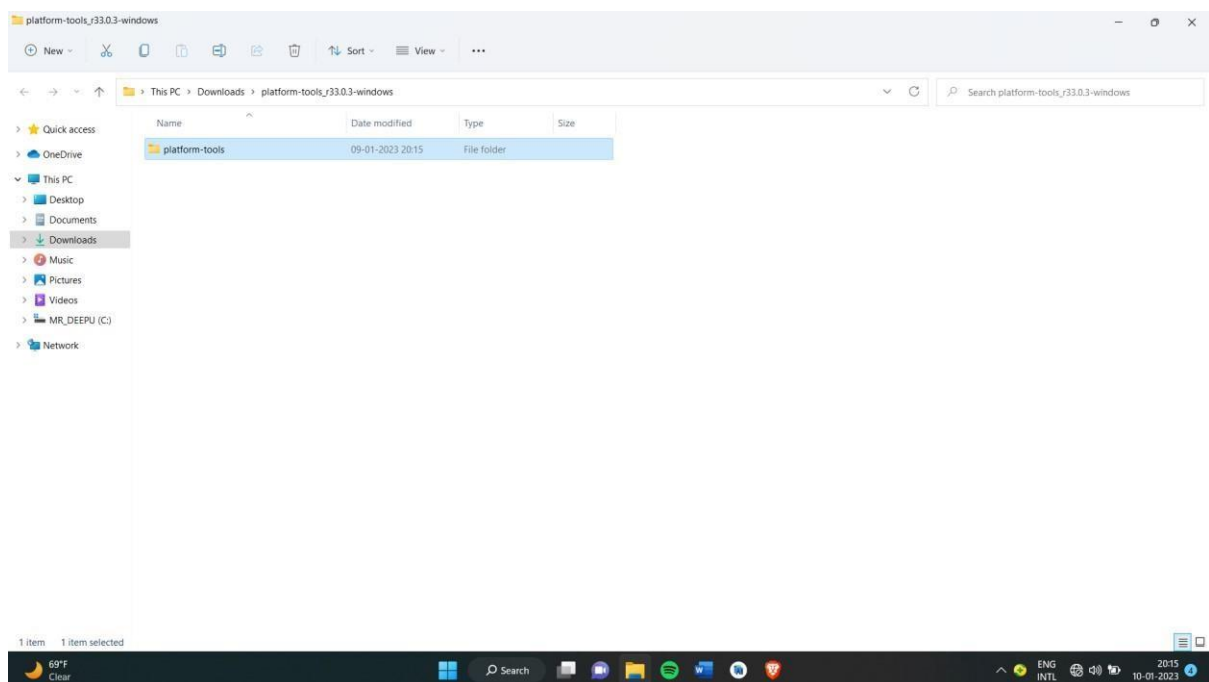
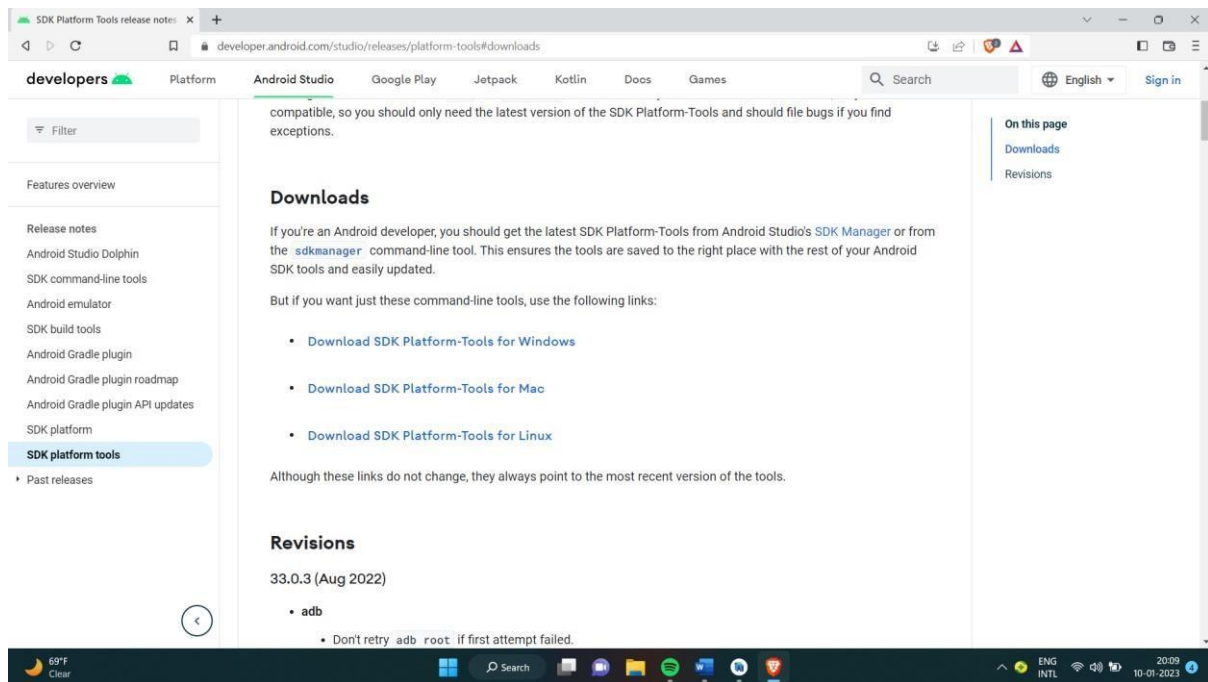
- Go to Android Studio
- Select tool and click on Device manager



- Select virtual
- Click on create virtual device
- Select a device
- Select specification and finish it.

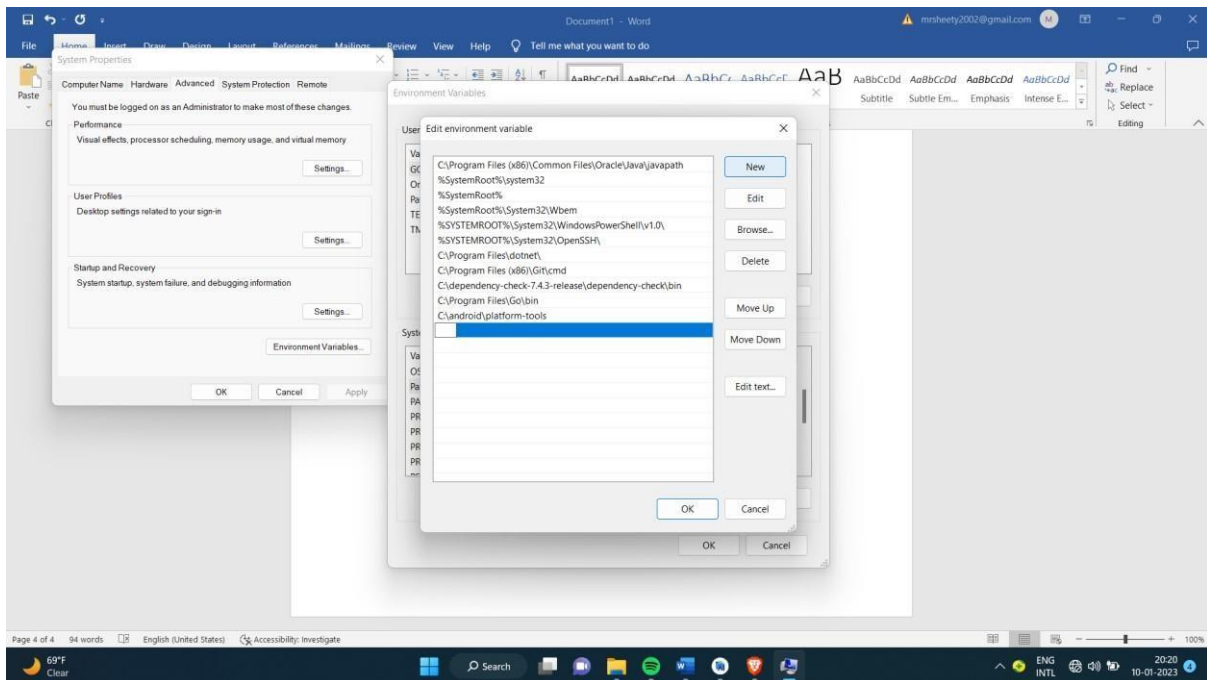
Setup a ADB environment on host system

- Go to chrome
- Search for ADB Platform tools
- Extract files
- copy the platform tools path



- Edit Environmental Variables for your system
- Select environmental variables
- Select System variable and Path
- Click on new

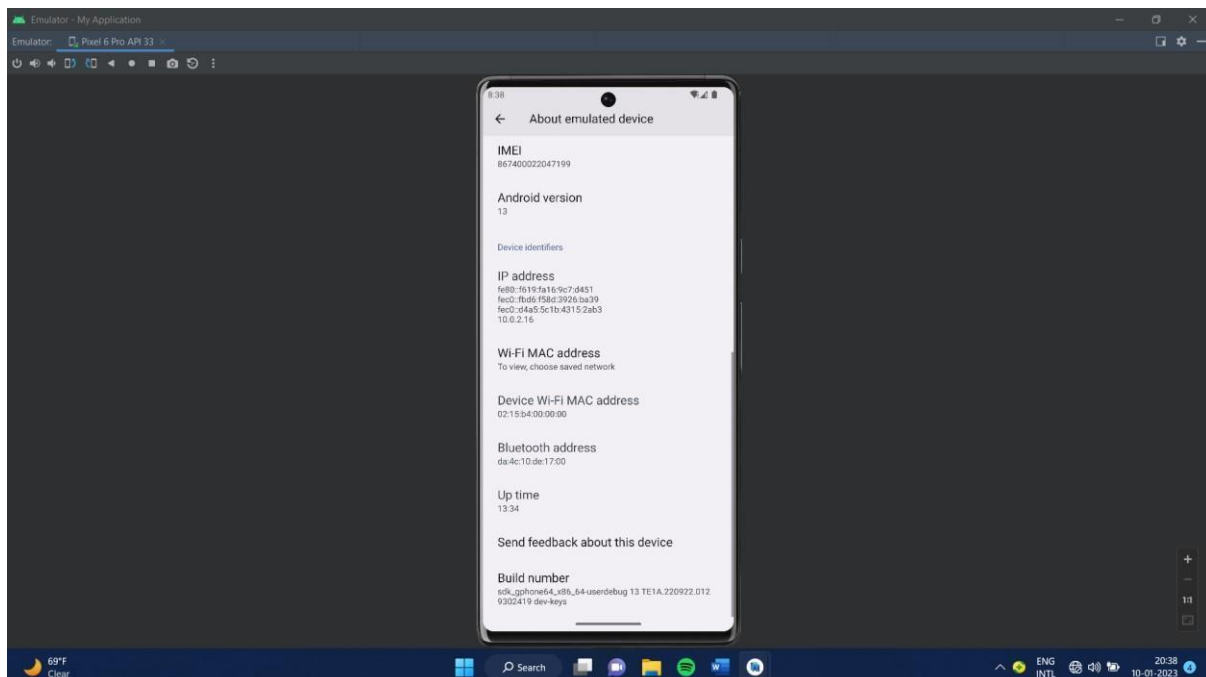
→ Paste the path click on ok



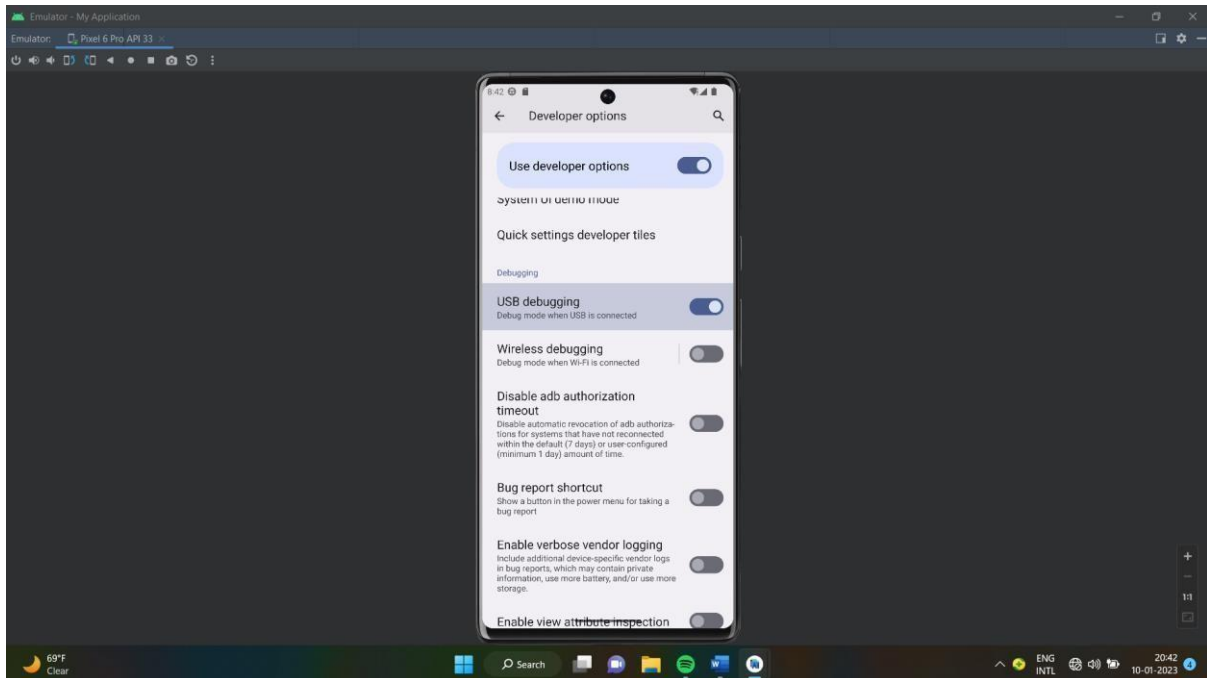
Transferring Files between host Machine and AVD using ADB

→ Open Virtual Device

→ Turn on developer option by clicking 7 times on Build number



→Turn on USB Debugging



→Go to Command Prompt

→Type the below Commands

\$adb (to confirm the adb installed successful on your system)

\$adb devices

\$adb shell

\$ls

\$cd sdcard

\$touch cyber.txt

→Check the created file in Virtual Device

→Go to file manager

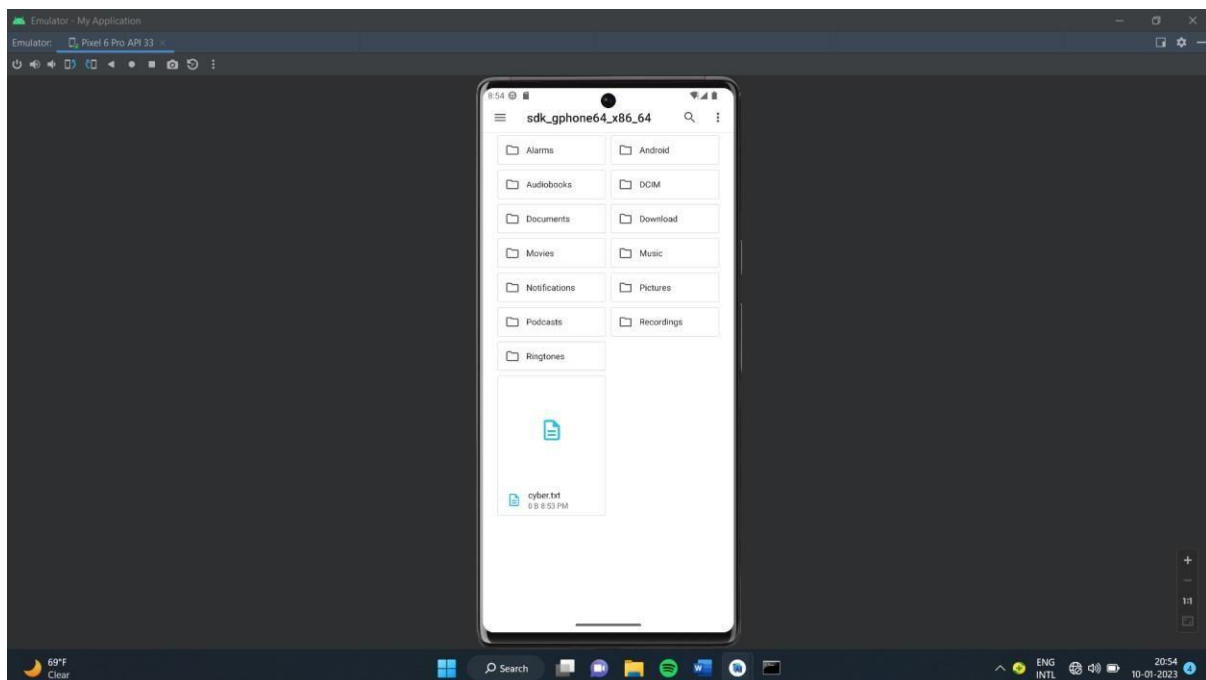
→Select emulator path on left side options

```
Command Prompt - adb shell
Microsoft Windows [Version 10.0.22000.1335]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Wp_Deepu>adb devices
List of devices attached
emulator-5554    device

C:\Users\Wp_Deepu>adb shell
emulator-5554 / $ ls
acct      cache      debug_ramdisk  linkerconfig  oem_dkms      sdcard        system_dkms
adb_keys  config     dev            lost+found    oem           second_stage_resources  system_ext
apex      data      etc            metadata      postinstall   storage        system
bin       data_mirror  init           mnt           proc          sys            vendor
bugreports  data_mirror  init.environ.rc  oda           product       system         vendor_dkms
emulator-5554 / $ cd sdcard
emulator-5554 /sdcard $ touch cyber.txt
emulator-5554 /sdcard $
```

Transferring Files between host machine and AVD using ADB isSuccessful



17) Setup the following tools onto your machine and reverse the application.

- Apktool
- Dex2Jar
- JDGUI

Reverse engineering

STEP 01 :- Install the dex2.jar, ju-gui and apktool. Links are below :

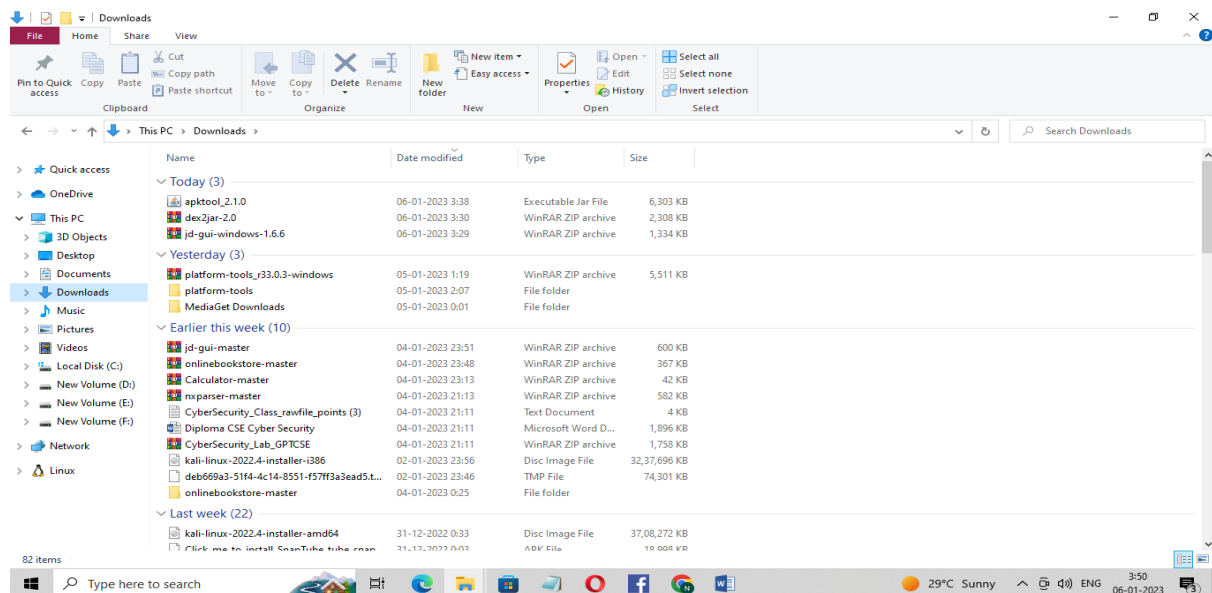
<https://www.filecroco.com/download-jd-gui/download/> : **ju-gui**

<https://sourceforge.net/projects/dex2jar/> : **dex2.jar**

<https://ibotpeaches.github.io/Apktool/> : **apktool**

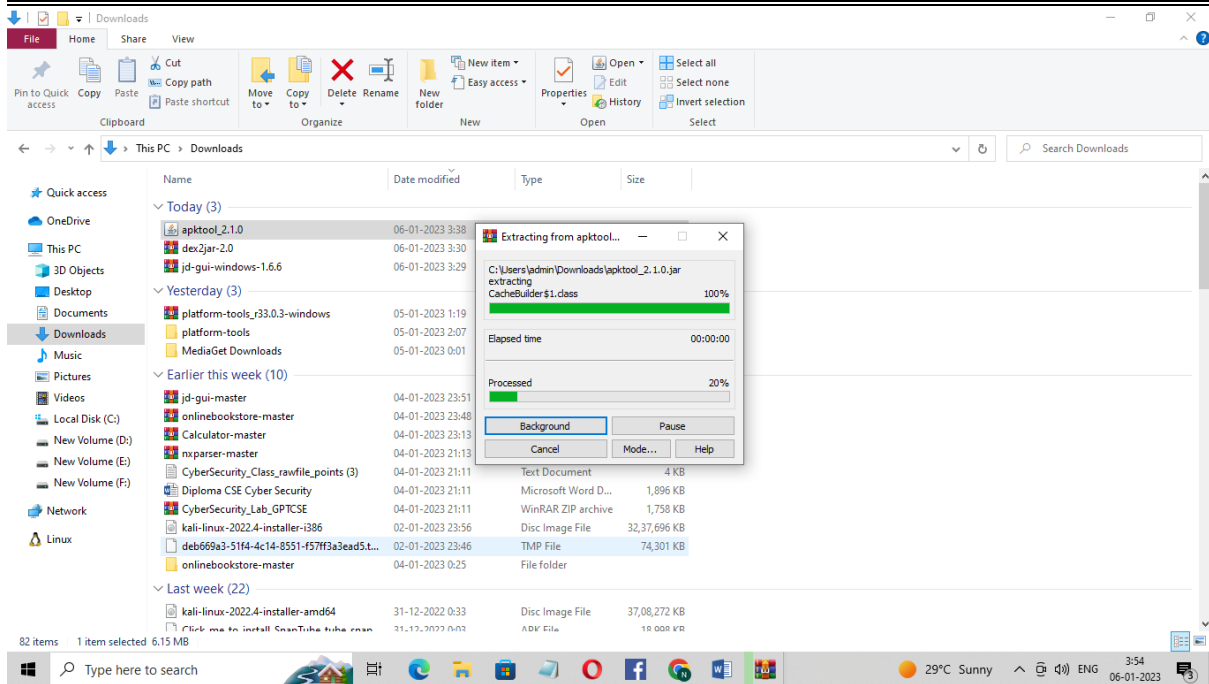
STEP 02 :- Now go to This PC

STEP 03 :- Click on DOWNLOAD

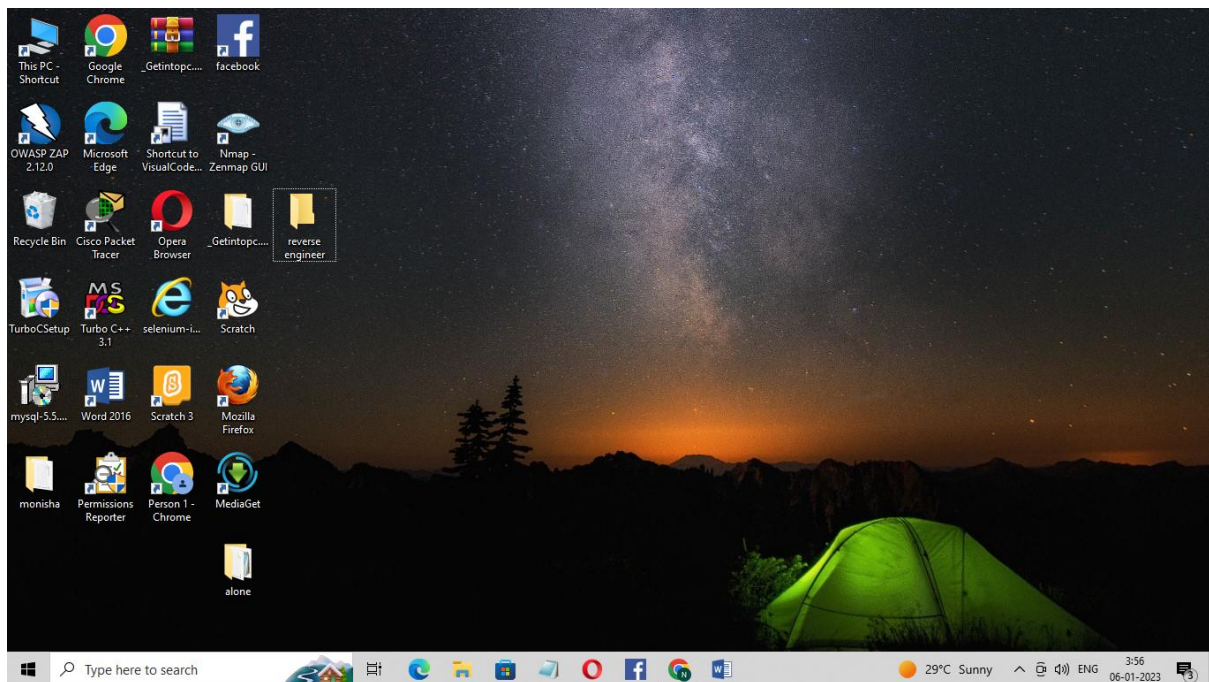


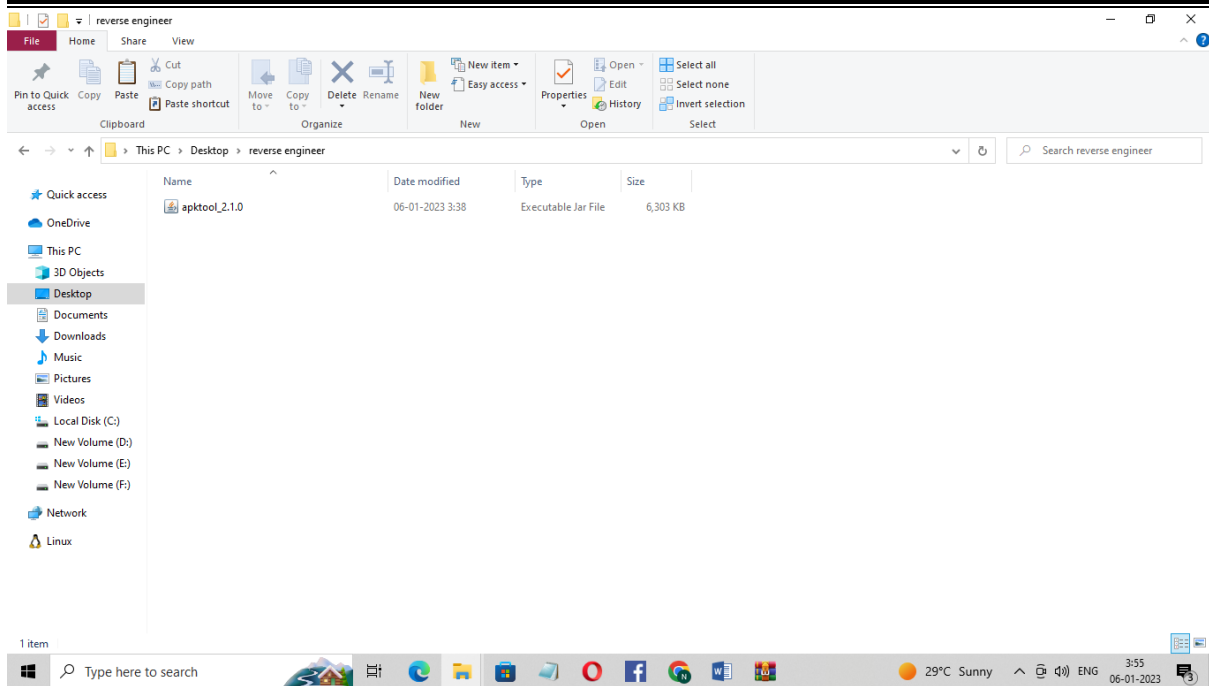
STEP 04 :- Now extract the APTTOOL to C:/ DRIVE

STEP 05 :- Go to *C:/ DRIVE * And copy the apktool



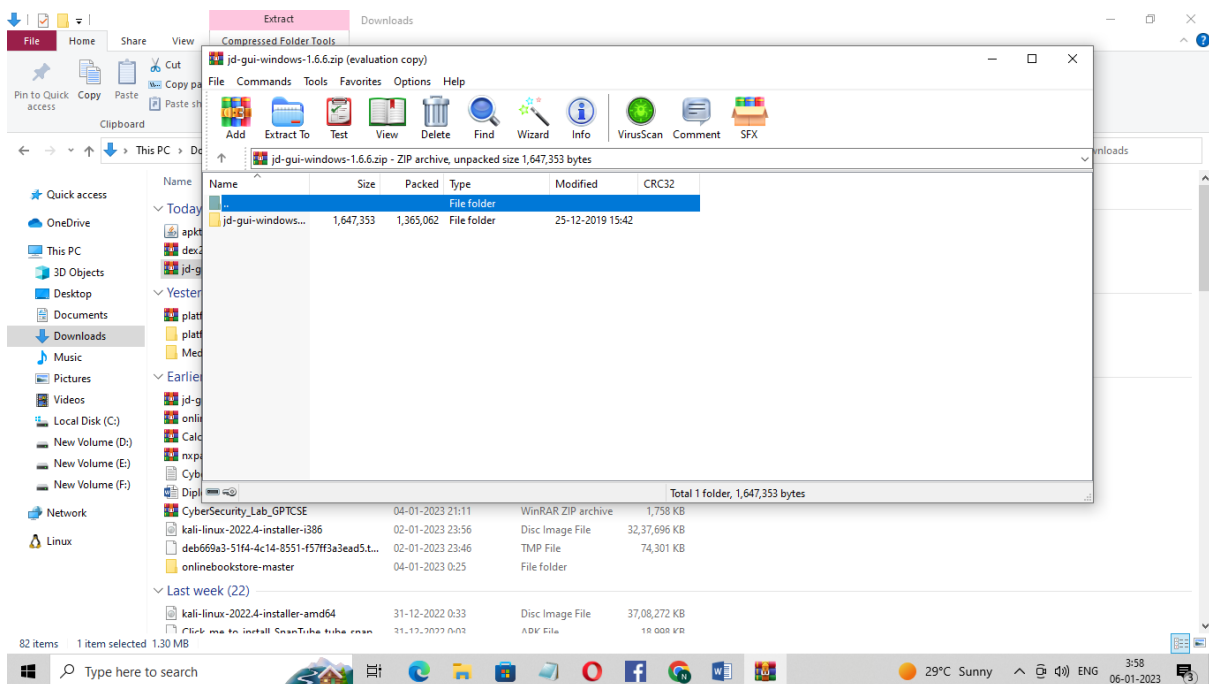
STEP 06 :- Then go to DESKTOP And create a NEW FOLDER Name as reverse engineer and In inside paste the copied APKTOOL and SAVE



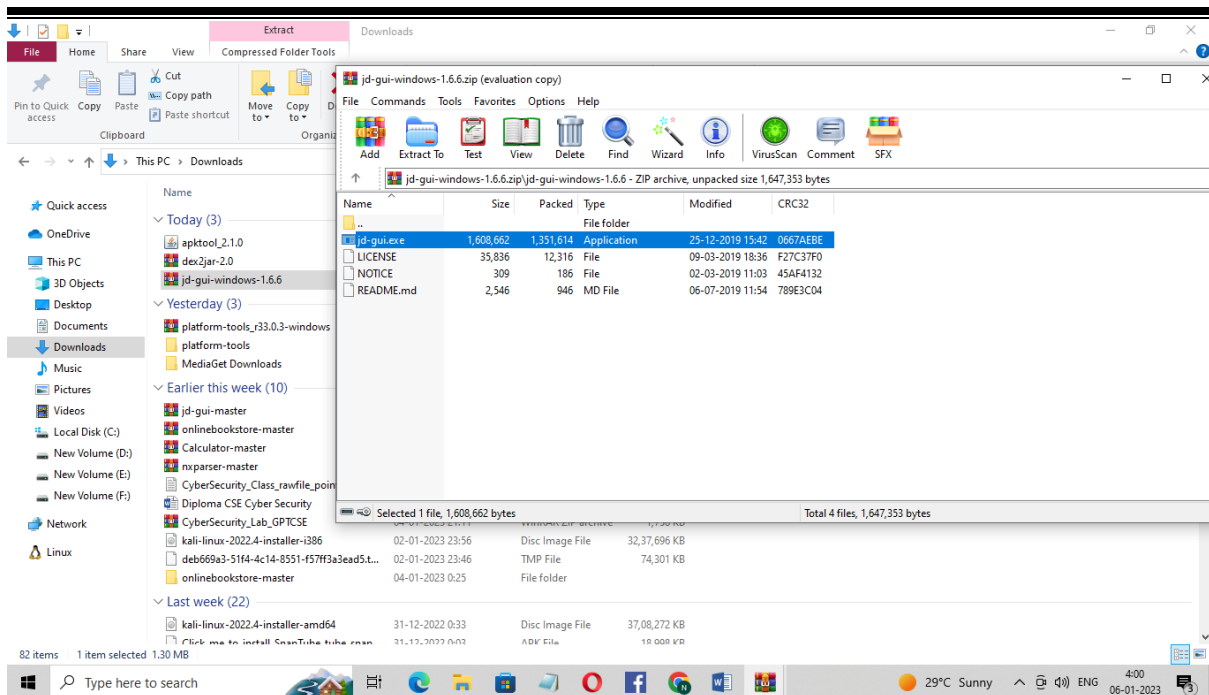


STEP 07 : Then go to DOWNLOADS

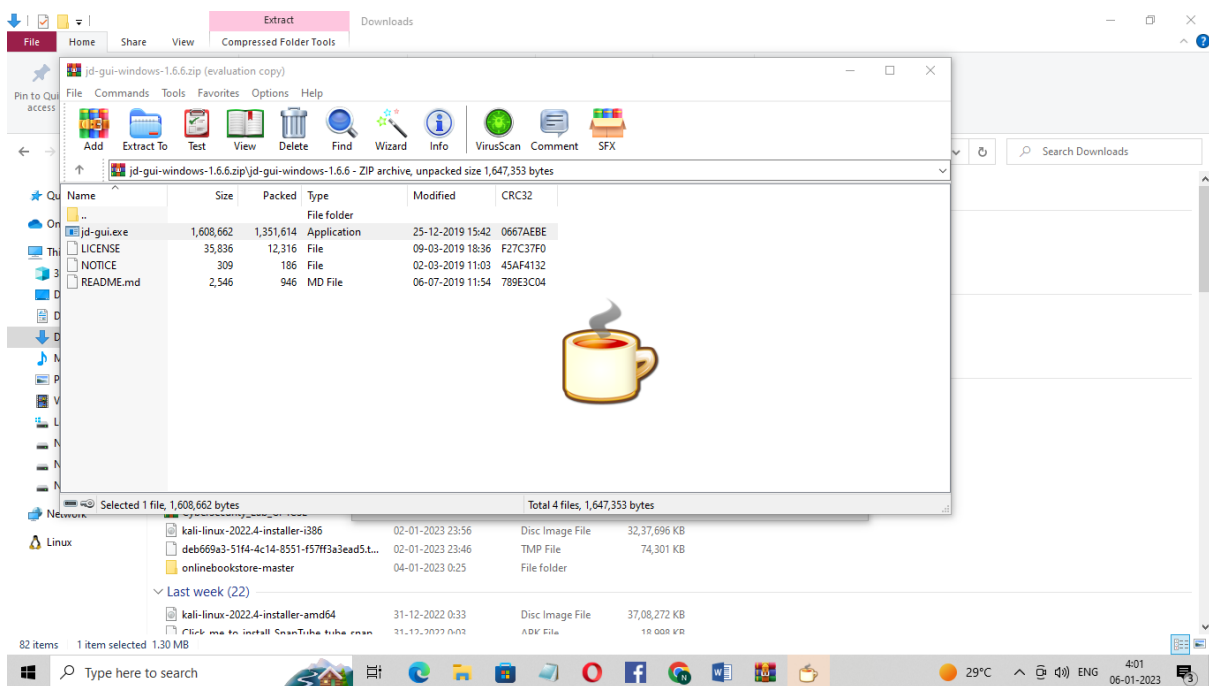
STEP 08 :- Click on JD-GUI TOOL



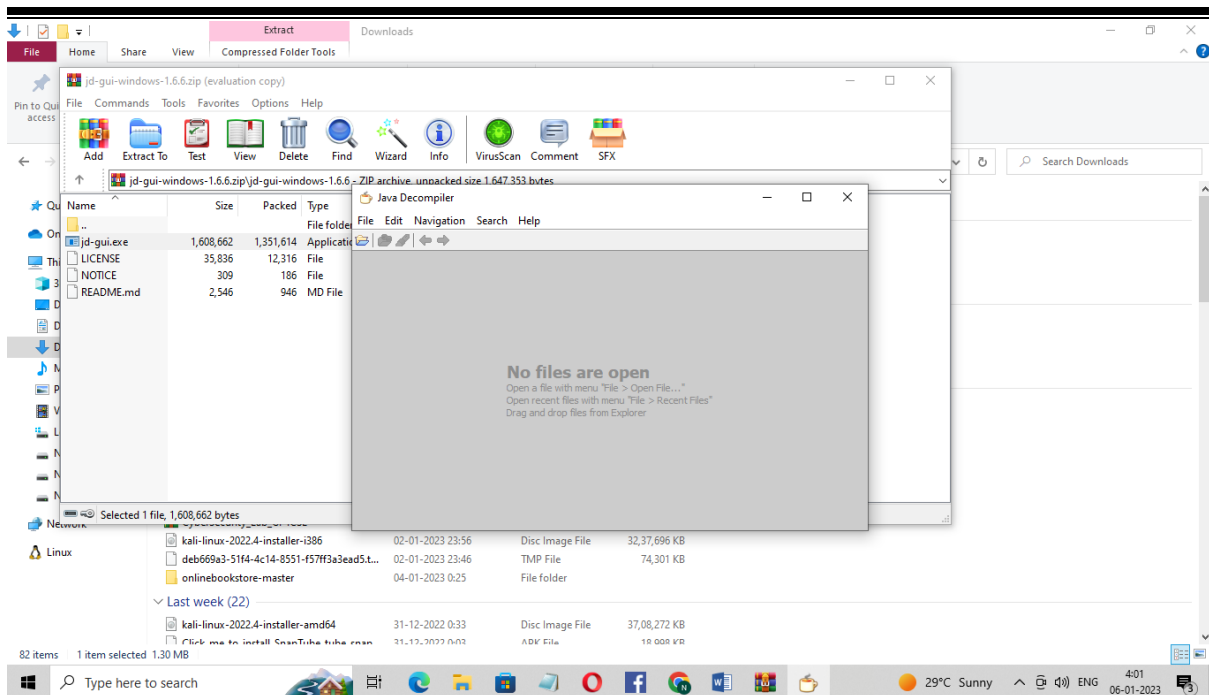
STEP 09 :- Now Select the JD-GUI APPLICATION AND CLICK ON IT



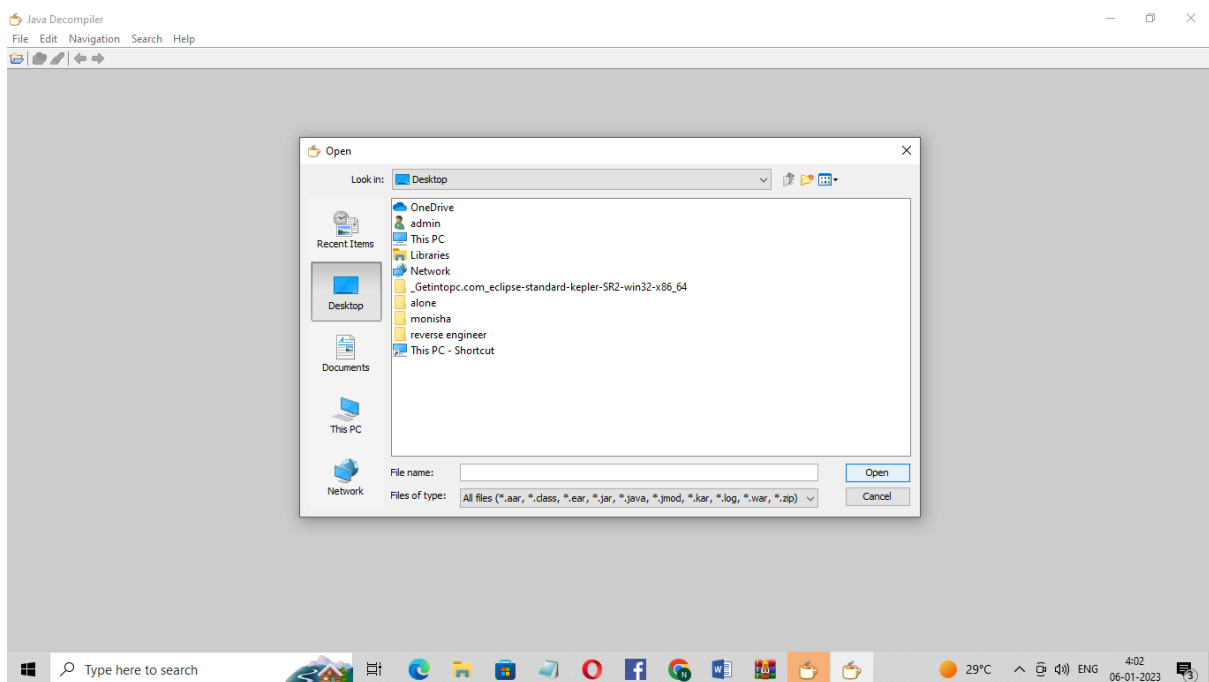
STEP 10 :- Now Select the JD-GUI.exe and click on it



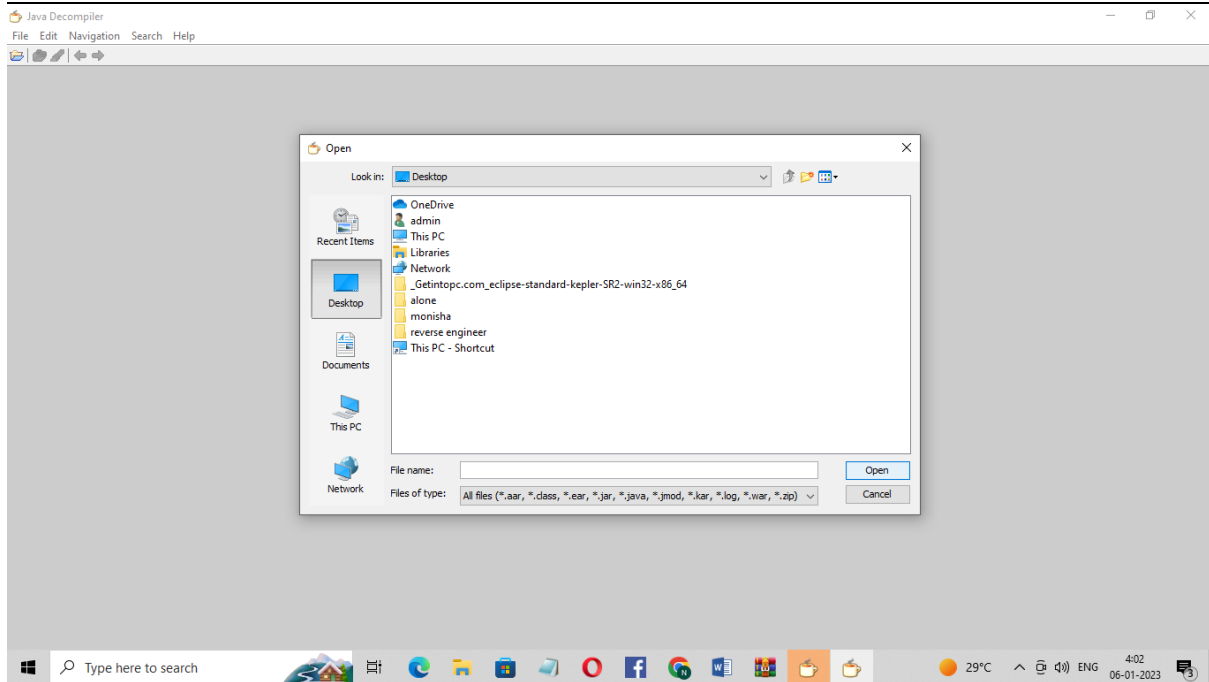
STEP 10 :- Now Create a NEW FILE. Click on OPEN FILE



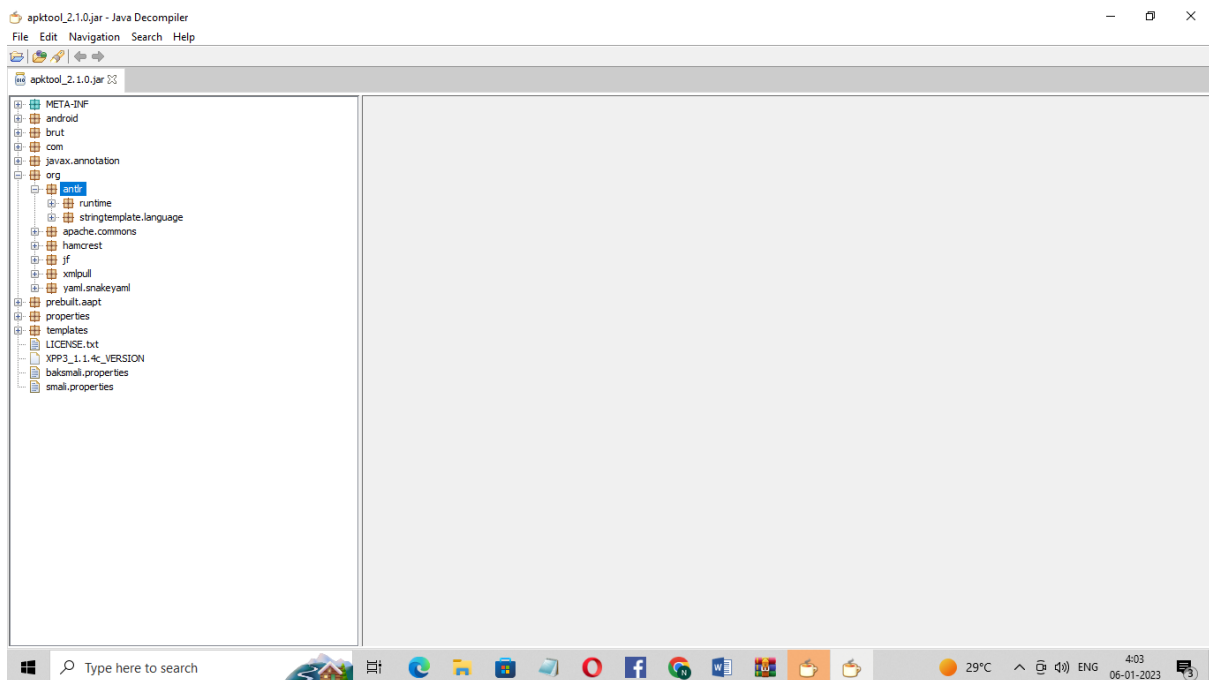
STEP 11 :- Now Select the DESKTOP then Select your FOLDER Reverse engineer



STEP 12 :- Now inside the Reverse engineer file Select the APKTOOL and CLICK ON IT



STEP 13 :- After SELECT the APKTOOL.2.1.0.jar will be OPENED like this



STEP 14 :- Now select the ORG and click on the RUNTIME in APKTOOL.2.1.0.jar now the java source code programs will be generated automatically.

